Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

PickleVerse – Multiplayer Pickleball Skill-Training and Match Game with AI Coaching

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I. Record of Changes

Date	A* M, D	In charge	Change Description
27/10/2026	A	Nguyễn Gia Huy	v0.1 – Feature list F01–F18, actors and workflows prepared for the Review 2 commitment (BM04).
06/11/2026	A	Nguyễn Văn Thành Đạt	v0.2 – First SRS on the official template: 23 use cases, FR-01 to FR-23, 10 NFRs.
07/11/2026	M	Nguyễn Văn Thành Đạt	v0.3 – Linked the Review 2 feature list to the requirements; added business rules and API messages.
11/11/2026	M	Nguyễn Gia Huy	v1.0 – Rewritten after the internal review: corrected context diagram; six actors; 68 use cases in eight UML use case diagrams; screen flows from the real scenes; screen authorization matrix; conceptual ERD; 33 functional requirements with real screenshots, data fields, validation and exception flows; gameplay, economy and authentication rules; measurable NFRs; message list in the product language.
11/11/2026	M	Nguyễn Gia Huy	v1.1 – Appendix A: workflow and BM04 owner added for each Review 2 feature.
12/11/2026	M	Nguyễn Gia Huy	v1.2 – Revised after the second internal review: time-outs removed (FR-12, BR-15, Bot Match screen); quick doubles added to UC-19, FR-14 and the non-screen functions; reconnection rule BR-28 and FR-17 aligned with the code; accuracy history, streak rewards and phone controls described; Menu, Online Lobby and Bot Match screens and their screenshots updated; screenshot references clarified.
13/11/2026	M	Nguyễn Gia Huy	v1.3 – Communication interfaces: the staging environment serves the Web API and the admin web over HTTPS since 13/11/2026.

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

PickleVerse is a new 3D pickleball game that teaches beginners the rules and core techniques of pickleball and lets them practise against bots and other players online. Release 1.0 consists of five deployable parts that together form the system under specification: the Unity game client for Windows and Android, the Unity dedicated game server that runs online matches with server

authority, the ASP.NET Core Web API with its PostgreSQL database and Redis cache, the Python AI coaching service, and the React administration web application. The Web API hosts the background workers for matchmaking, dedicated-server leases, AI analysis and moderation expiry.

Figure 1 shows the context of release 1.0. The system exchanges data with two human actors and three external systems. Photon Cloud provides the real-time session and relay service used by the game client and the dedicated server; Gmail SMTP delivers one-time passwords; S3-compatible object storage keeps match replays. Real-money payment, social login and push notifications are outside the boundary of this release.

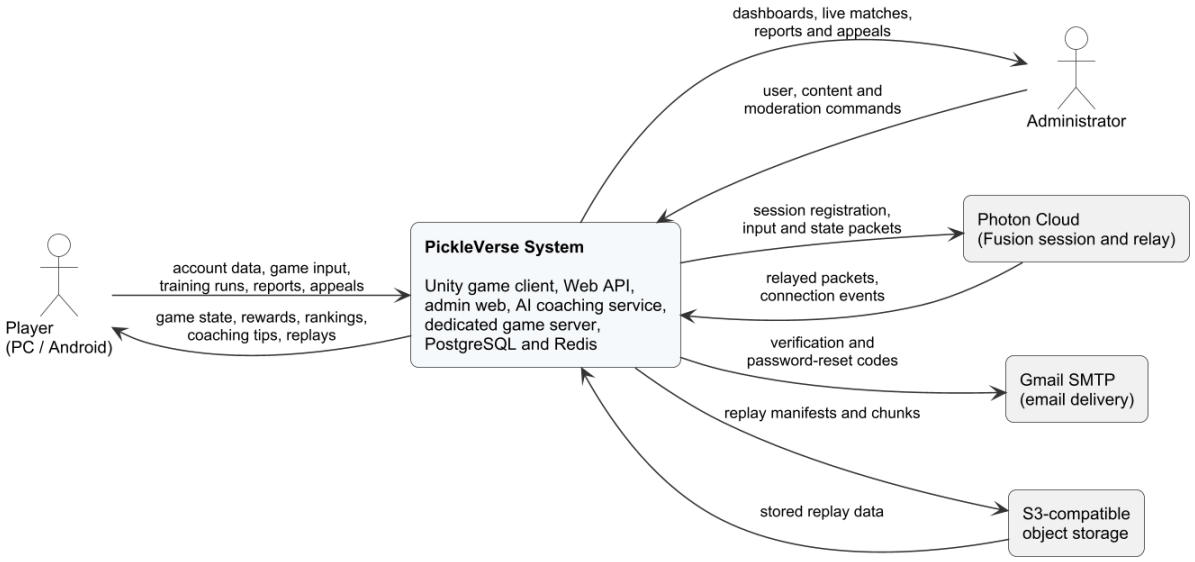


Figure 1. Context diagram of PickleVerse release 1.0

Table 1. External entities of the context diagram

External entity	Type	Data exchanged with PickleVerse
Player	Human (Windows PC or Android phone)	Sends account data, game input, training attempts, reports and appeals; receives game state, rewards, rankings, coaching tips and replays
Administrator	Human (web browser)	Sends user, content, match and moderation commands; receives dashboards, live match status, reports, appeals and analytics
Photon Cloud	External real-time service	Receives session registration and input/state packets from the client and the dedicated server; returns relayed packets and connection events
Gmail SMTP	External e-mail service	Receives verification and password-reset messages to deliver to the player's mailbox
S3-compatible object storage	Storage service	Stores replay manifests and compressed replay chunks; returns them for the replay viewer and moderation

2. User Requirements

2.1 Actors

Table 2. Actors

#	Actor	Description
1	Visitor	A person who uses the game client without being signed in. A visitor can register, verify an e-mail address, sign in, reset a forgotten password and, on a device that has a cached profile, start offline training.

#	Actor	Description
2	Player	A signed-in user with the Player role and an active account. A player inherits every use case of a visitor and can learn, train, play bot and online matches, view progress, use the shop and the social features, and report other players.
3	Restricted Player	A player whose account is suspended or banned. After signing in, a restricted player is limited to viewing the sanction decision and submitting one appeal for it.
4	Administrator	A user with the Admin role who signs in to the administration web application to manage accounts, content, matches, moderation and analytics, and to label coaching sessions as an expert reviewer.
5	Email Service	Secondary actor (Gmail SMTP) that delivers verification and password-reset codes.
6	Photon Cloud	Secondary actor that hosts the real-time sessions through which the game client, other players and the dedicated server exchange match packets.

2.2 Use Cases

2.2.1 Diagram(s)

The 68 use cases are grouped into eight packages. Figure 2 gives the package overview; Figures 3 to 11 show each package with its actors and the «include» and «extend» relationships (package P4 is drawn in two parts for readability). Player and Restricted Player are specialisations of Visitor, so they inherit the visitor use cases.

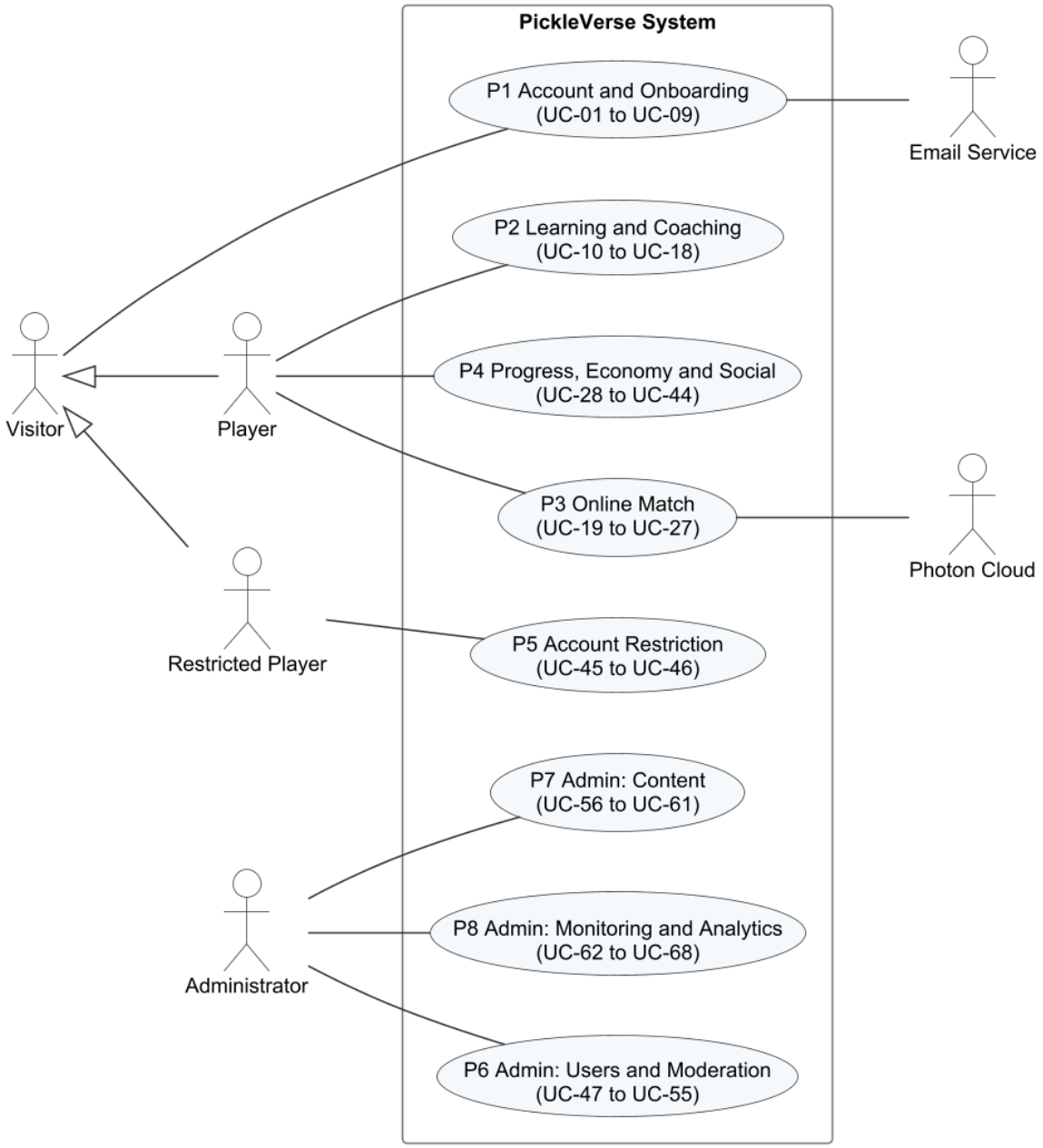


Figure 2. Use case overview by package

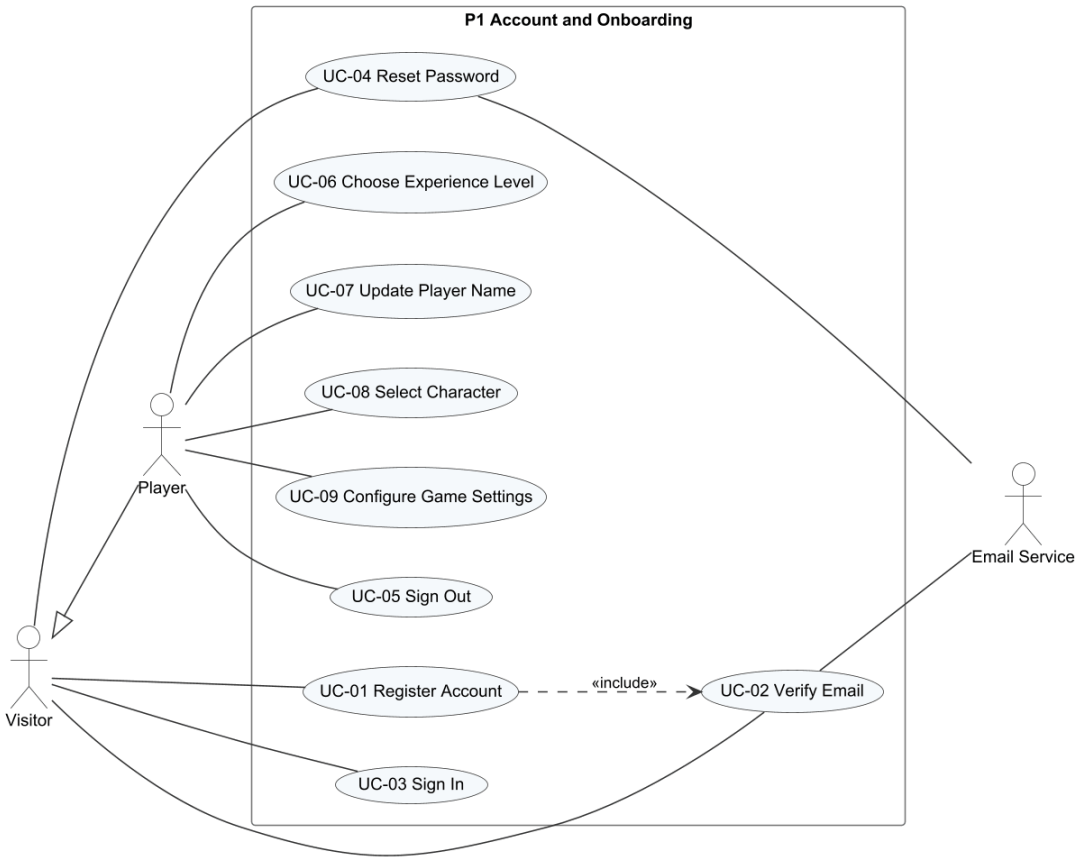


Figure 3. Use case diagram P1 – Account and Onboarding

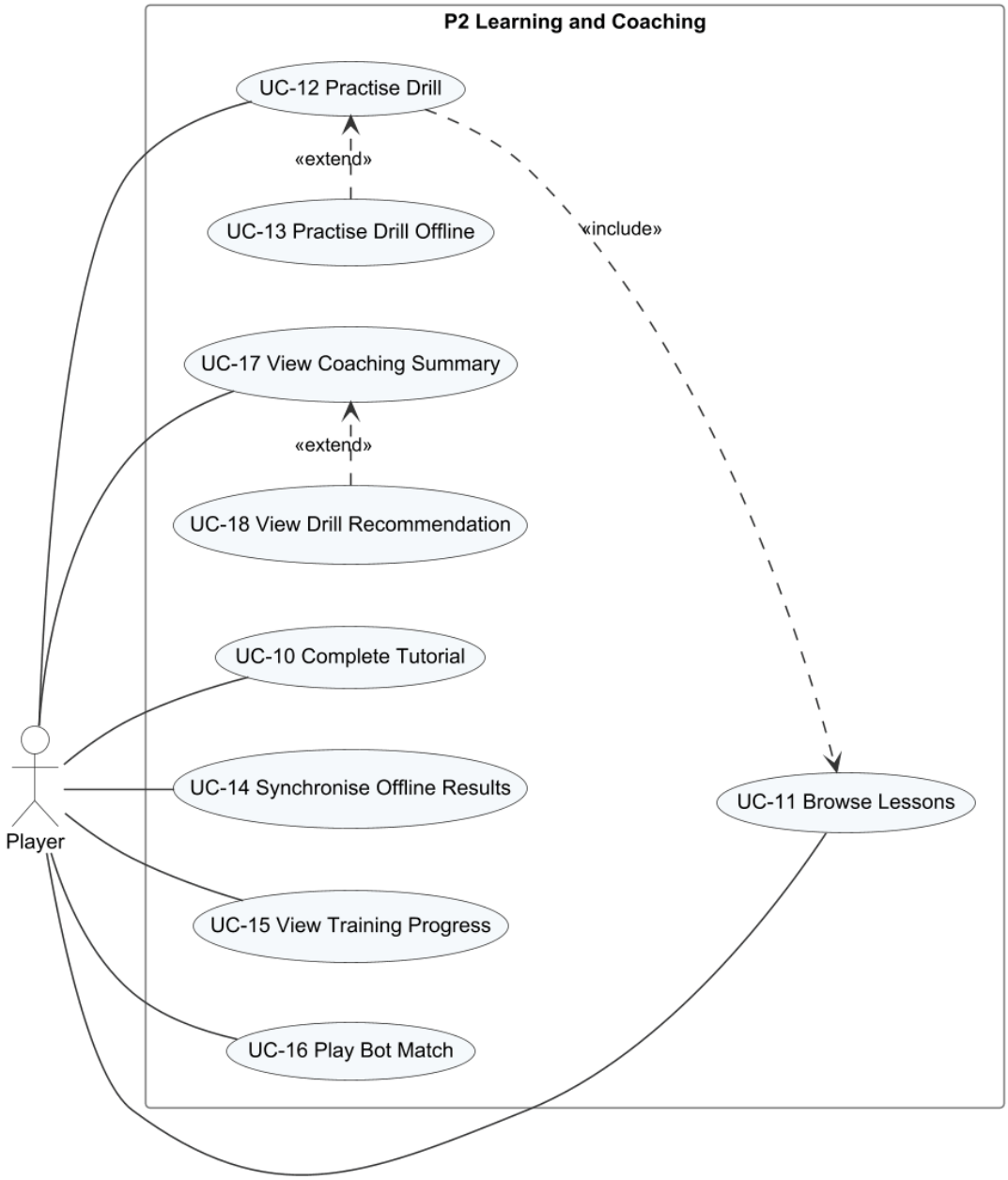


Figure 4. Use case diagram P2 – Learning and Coaching

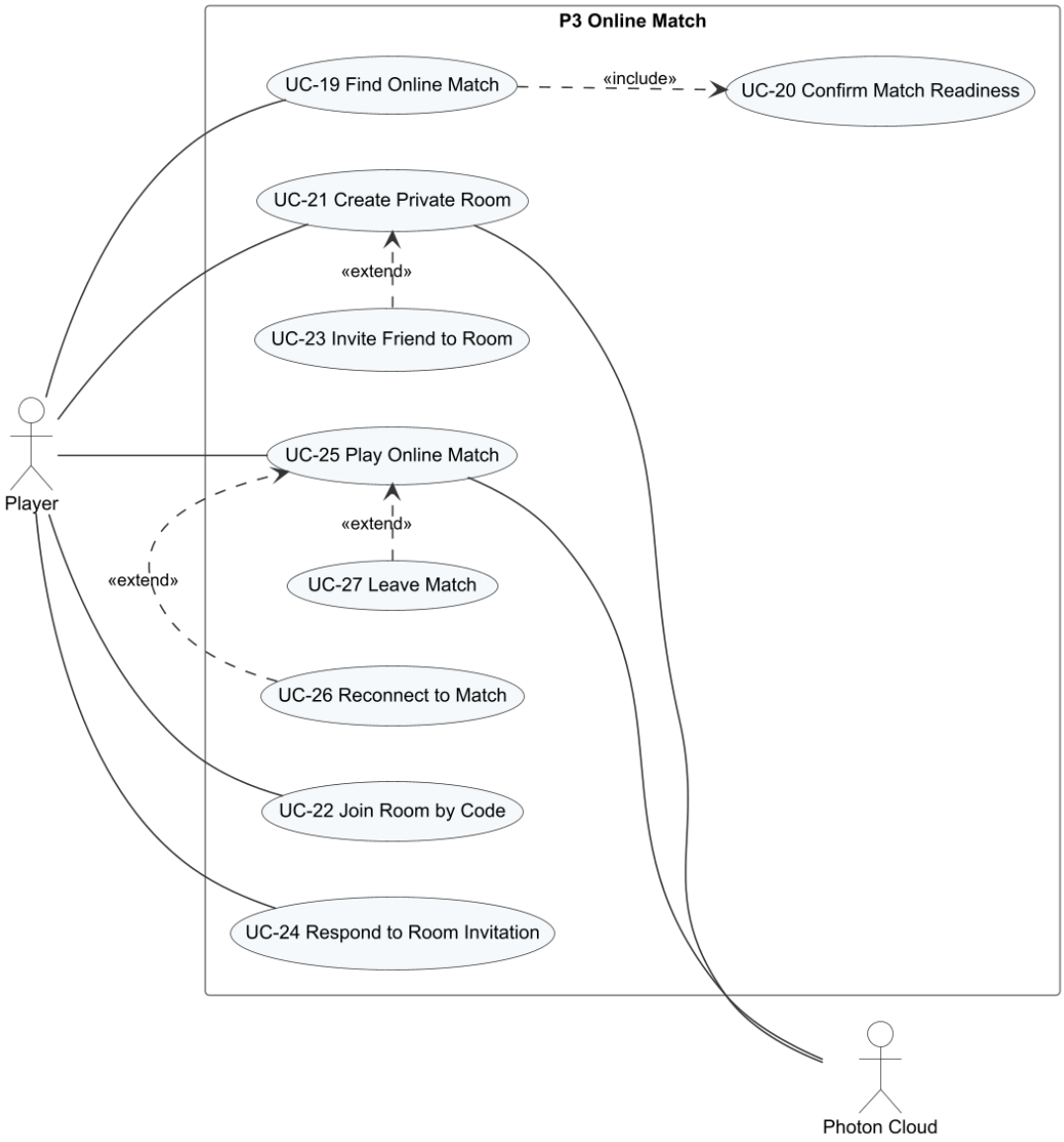


Figure 5. Use case diagram P3 – Online Match

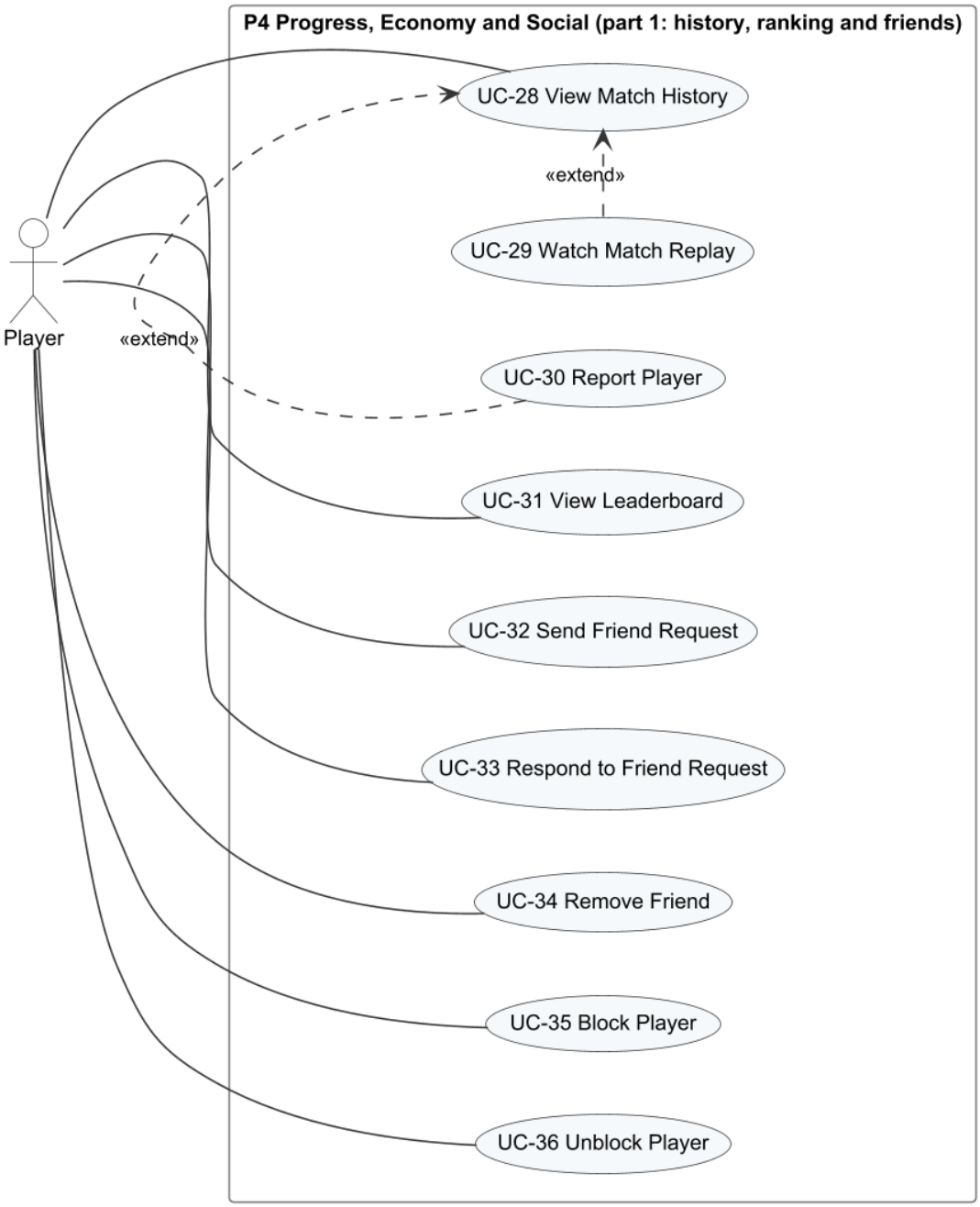


Figure 6. Use case diagram P4 (part 1) – History, Ranking and Friends

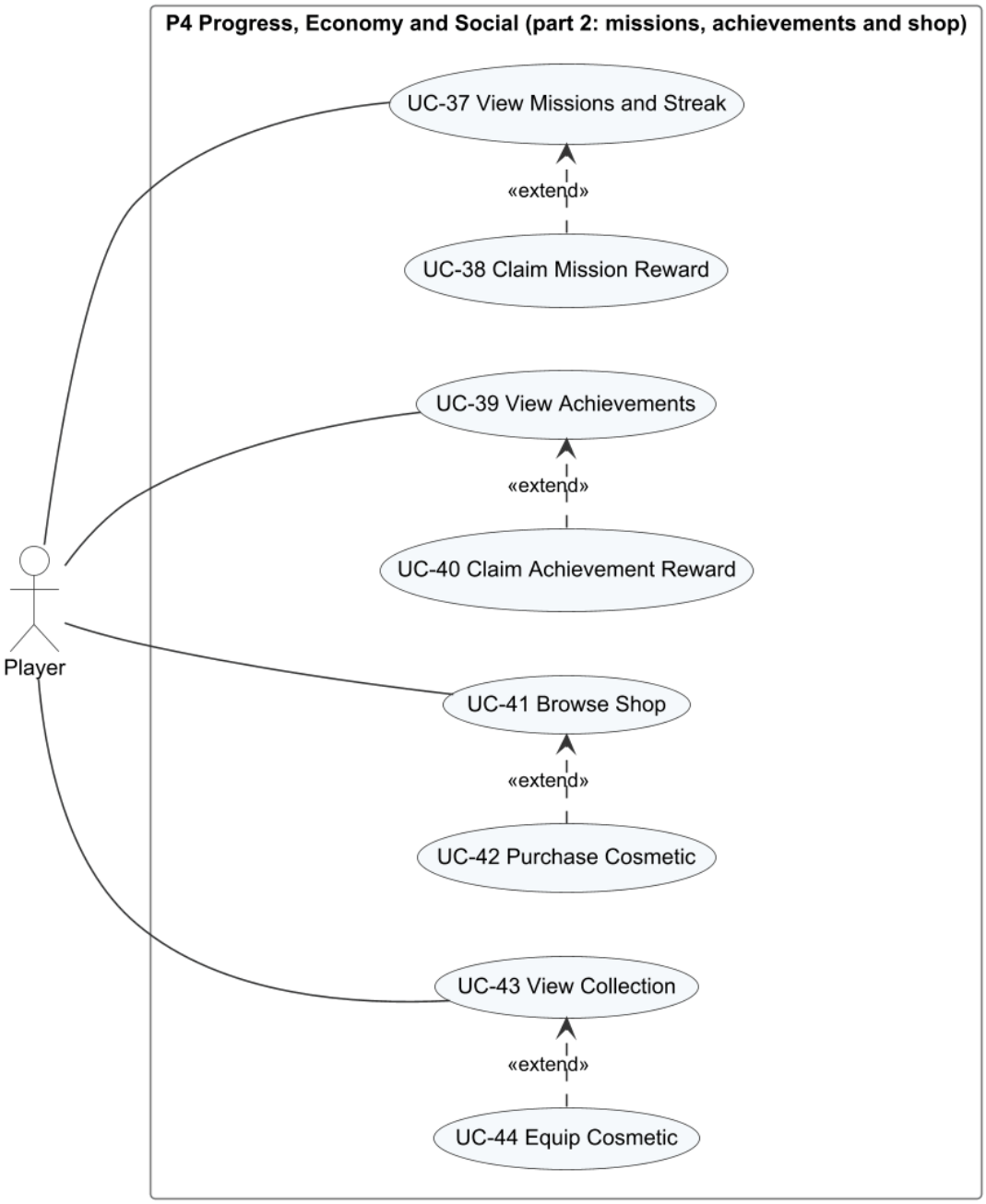


Figure 7. Use case diagram P4 (part 2) – Missions, Achievements and Shop

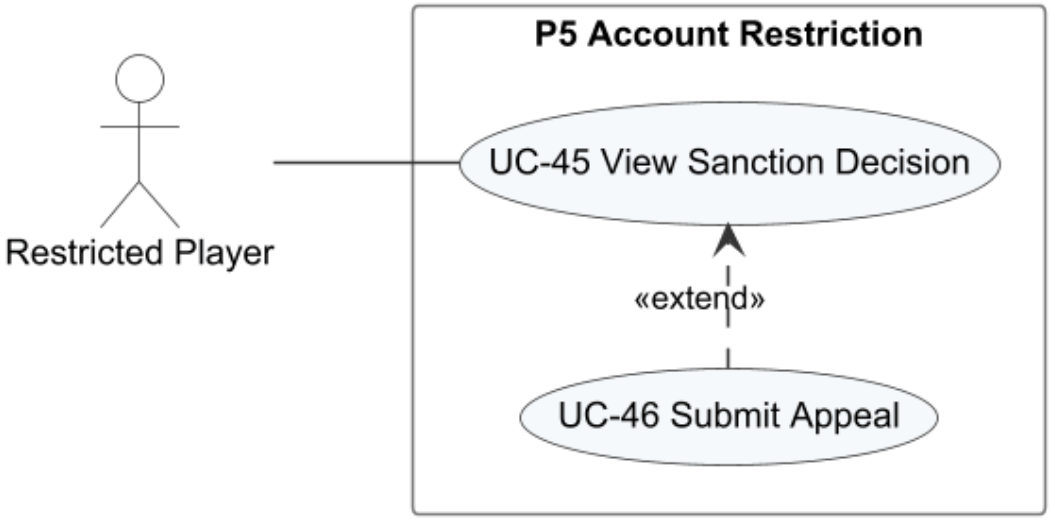


Figure 8. Use case diagram P5 – Account Restriction

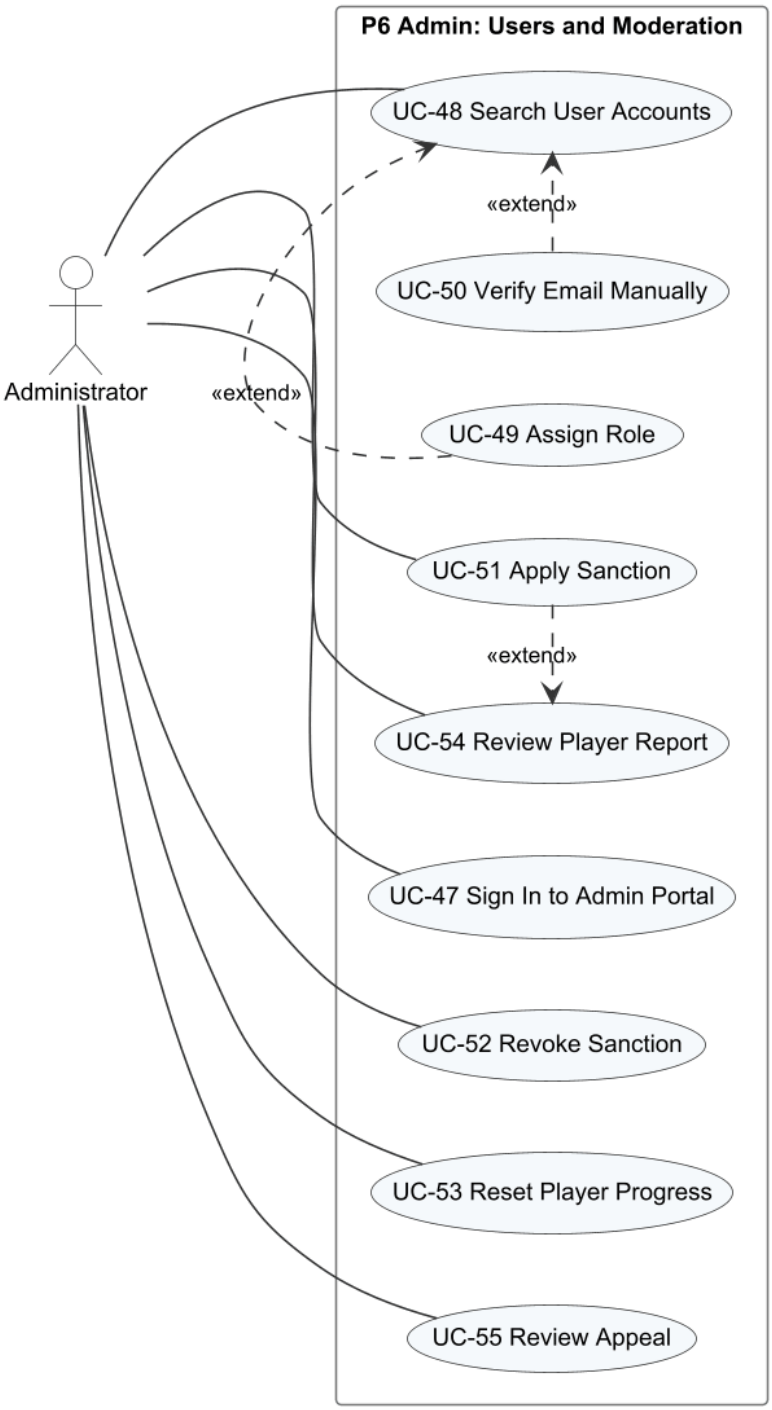


Figure 9. Use case diagram P6 – Admin: Users and Moderation

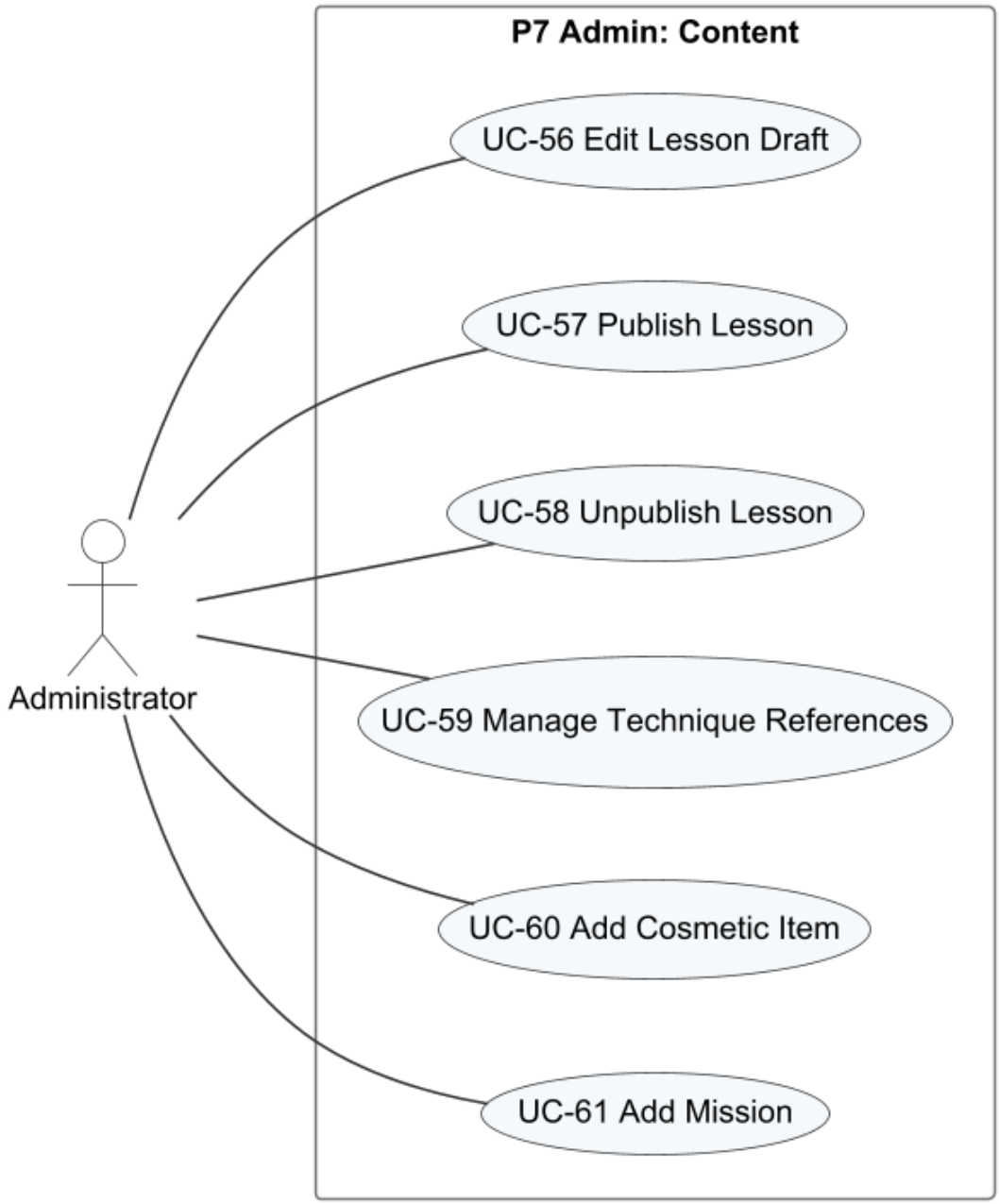


Figure 10. Use case diagram P7 – Admin: Content

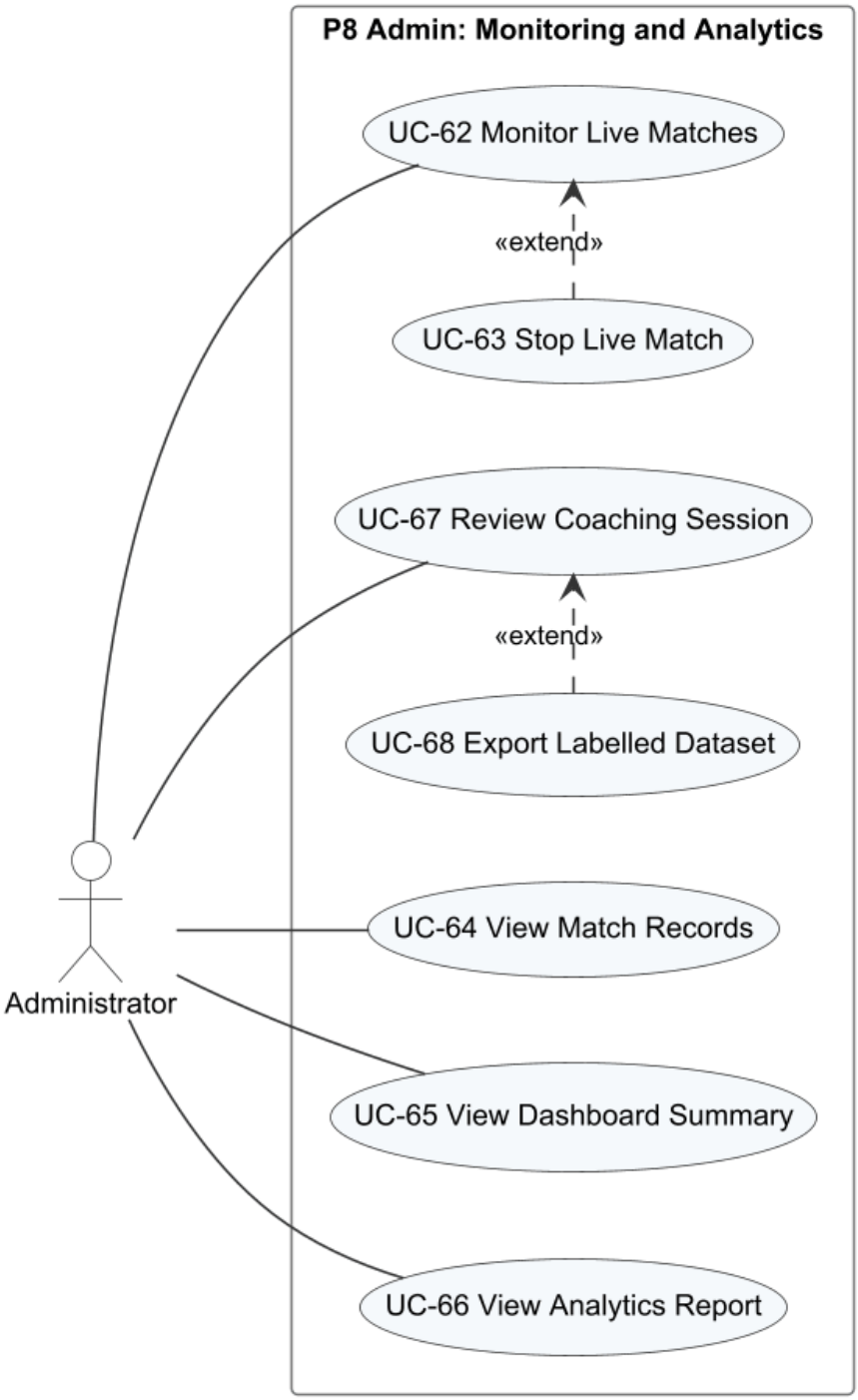


Figure 11. Use case diagram P8 – Admin: Monitoring and Analytics

2.2.2 Descriptions

Table 3. Use case descriptions

ID	Use Case	Actors	Use Case Description	FR
UC-01	Register Account	Visitor	The visitor creates an account with an e-mail address and a password; the system sends a six-digit code to that address.	FR-01
UC-02	Verify Email	Visitor, Email Service	The visitor enters the code received by e-mail to activate the account, or asks for a new code.	FR-01
UC-03	Sign In	Visitor	The visitor signs in with e-mail and password and is taken to the menu with the saved profile.	FR-02

ID	Use Case	Actors	Use Case Description	FR
UC-04	Reset Password	Visitor, Email Service	The visitor receives a reset code by e-mail and sets a new password; all existing sessions are revoked.	FR-03
UC-05	Sign Out	Player	The player ends the session on the device.	FR-02
UC-06	Choose Experience Level	Player	At the first sign-in the player states whether they are new to pickleball, which decides whether the tutorial starts.	FR-04
UC-07	Update Player Name	Player	The player changes the display name shown to other players.	FR-04
UC-08	Select Character	Player	The player chooses the male or female character used in matches.	FR-04
UC-09	Configure Game Settings	Player	The player adjusts audio, graphics, key bindings and the touch layout.	FR-05
UC-10	Complete Tutorial	Player	The player follows the 17-step tutorial on the court, rules and controls.	FR-06
UC-11	Browse Lessons	Player	The player views the lesson catalogue by group, with stars and lock status.	FR-07
UC-12	Practise Drill	Player	The player plays the drill of an unlocked lesson and receives a score of 0 to 3 stars.	FR-08
UC-13	Practise Drill Offline	Player	Without a connection, the player practises a cached drill; the result is stored for later synchronisation.	FR-09
UC-14	Synchronise Offline Results	Player	When online again, the stored offline results are sent and applied exactly once.	FR-09
UC-15	View Training Progress	Player	The player reviews passed lessons, stars, best and recent accuracy of each lesson, practice time, common errors and post-session feedback.	FR-10
UC-16	Play Bot Match	Player	The player configures and plays a singles match against a computer opponent.	FR-11, FR-12, FR-13
UC-17	View Coaching Summary	Player	After a drill or match the player reads the analysis of their shots and the main errors.	FR-19
UC-18	View Drill Recommendation	Player	The player opens a recommended drill that targets a detected weakness.	FR-19
UC-19	Find Online Match	Player	The player searches a ranked singles match (opponent of similar Elo, or a bot after 180 seconds) or a quick doubles match (unrated 2v2, alone or with a party of two, human players only).	FR-14
UC-20	Confirm Match Readiness	Player	Each paired player accepts the match within 10 seconds before it starts.	FR-14
UC-21	Create Private Room	Player, Photon Cloud	The player creates a singles or doubles room identified by a room code.	FR-15
UC-22	Join Room by Code	Player	The player joins a friend's room by typing its code.	FR-15
UC-23	Invite Friend to Room	Player	The host invites an online friend into the room.	FR-15
UC-24	Respond to Room Invitation	Player	The invited player accepts or declines the invitation.	FR-15
UC-25	Play Online Match	Player, Photon Cloud	The players play a match whose rules and score are decided by the dedicated server.	FR-16
UC-26	Reconnect to Match	Player	After a disconnection the player rejoins the same match within the reconnection window.	FR-17
UC-27	Leave Match	Player	The player leaves a match before it ends and accepts the consequences.	FR-17
UC-28	View Match History	Player	The player views the last 20 matches with result, score and Elo change.	FR-24
UC-29	Watch Match Replay	Player	The player watches the replay of one of their own matches with pause, seek and speed control.	FR-24
UC-30	Report Player	Player	The player reports an opponent from a finished match with a reason.	FR-27
UC-31	View Leaderboard	Player	The player views players ordered by Elo and their own position.	FR-26
UC-32	Send Friend Request	Player	The player searches by name or ID and sends a friend request.	FR-25
UC-33	Respond to Friend Request	Player	The player accepts or declines a received request.	FR-25
UC-34	Remove Friend	Player	The player removes a friend from the friend list.	FR-25
UC-35	Block Player	Player	The player blocks another player, which removes friendship and prevents requests and invitations.	FR-25

ID	Use Case	Actors	Use Case Description	FR
UC-36	Unblock Player	Player	The player removes a block.	FR-25
UC-37	View Missions and Streak	Player	The player views daily and weekly missions, their progress and the daily streak.	FR-21
UC-38	Claim Mission Reward	Player	The player claims the reward of a completed mission or mission set once.	FR-21
UC-39	View Achievements	Player	The player views achievement progress.	FR-21
UC-40	Claim Achievement Reward	Player	The player claims the reward of a completed achievement once.	FR-21
UC-41	Browse Shop	Player	The player views cosmetic items with their prices and currencies.	FR-22
UC-42	Purchase Cosmetic	Player	The player buys an item with in-game currency.	FR-22
UC-43	View Collection	Player	The player views owned cosmetic items.	FR-23
UC-44	Equip Cosmetic	Player	The player equips an owned paddle, which is then shown in matches.	FR-23
UC-45	View Sanction Decision	Restricted Player	The restricted player reads the restriction type, reason and expiry.	FR-28
UC-46	Submit Appeal	Restricted Player	The restricted player submits one written appeal against a sanction.	FR-28
UC-47	Sign In to Admin Portal	Administrator	The administrator signs in to the web application; only the Admin role is accepted.	FR-29
UC-48	Search User Accounts	Administrator	The administrator searches and filters accounts by text and status.	FR-29
UC-49	Assign Role	Administrator	The administrator changes the role of an account between Player and Admin.	FR-29
UC-50	Verify Email Manually	Administrator	The administrator activates an account whose verification e-mail could not be delivered.	FR-29
UC-51	Apply Sanction	Administrator	The administrator suspends or bans an account with a reason, optionally from a report.	FR-29, FR-30
UC-52	Revoke Sanction	Administrator	The administrator lifts an active sanction with a reason.	FR-29
UC-53	Reset Player Progress	Administrator	The administrator previews and resets selected progress groups of a player.	FR-29
UC-54	Review Player Report	Administrator	The administrator examines a report with match evidence and accepts or declines it.	FR-30
UC-55	Review Appeal	Administrator	The administrator accepts or declines an appeal; acceptance revokes the related sanction.	FR-30
UC-56	Edit Lesson Draft	Administrator	The administrator creates or edits a lesson and saves it as a draft.	FR-31
UC-57	Publish Lesson	Administrator	The administrator publishes the saved draft so that new attempts use it.	FR-31
UC-58	Unpublish Lesson	Administrator	The administrator hides a published lesson from players.	FR-31
UC-59	Manage Technique References	Administrator	The administrator maintains the reference values used to judge each technique.	FR-31
UC-60	Add Cosmetic Item	Administrator	The administrator adds an item to the shop with type, price and currency.	FR-31
UC-61	Add Mission	Administrator	The administrator creates a daily or weekly mission with target and reward.	FR-31
UC-62	Monitor Live Matches	Administrator	The administrator watches active matches with score and connection status refreshed every 5 seconds.	FR-32
UC-63	Stop Live Match	Administrator	The administrator stops a dedicated-server match with a reason.	FR-32
UC-64	View Match Records	Administrator	The administrator browses finished and interrupted matches.	FR-32
UC-65	View Dashboard Summary	Administrator	The administrator sees today's key figures and items that need action.	FR-32
UC-66	View Analytics Report	Administrator	The administrator views active users, retention, lesson pass rates, technique errors and matchmaking health for a date range.	FR-32
UC-67	Review Coaching Session	Administrator	Acting as expert, the administrator inspects shot evidence of a training session and labels the player level and errors.	FR-33
UC-68	Export Labelled Dataset	Administrator	The administrator exports the labelled sessions used to evaluate the AI coaching quality.	FR-33

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screens Flow

Figure 12 shows the screens of the game client and how a player moves between them; Figure 13 shows the administration web application. Rounded white boxes marked [popup], [dialog], [drawer] or [overlay] are windows opened above another screen; the other boxes are full scenes or pages.

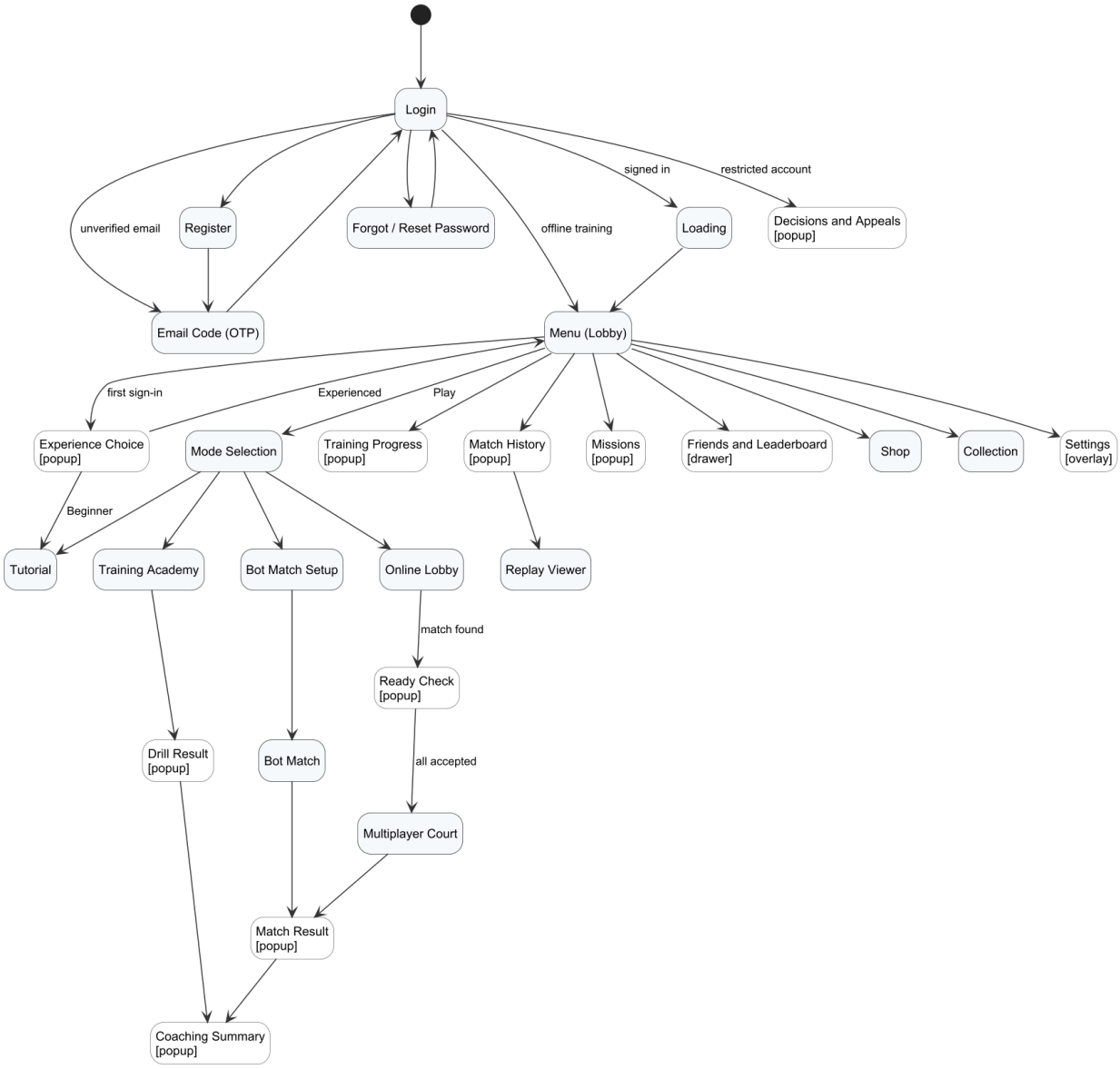


Figure 12. Screen flow of the game client

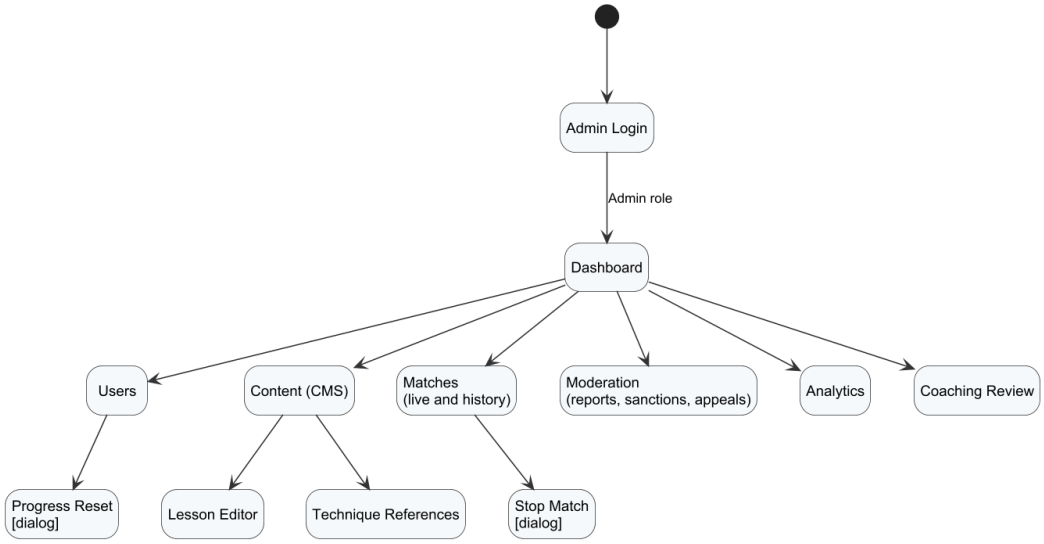


Figure 13. Screen flow of the administration web application

3.1.2 Screen Descriptions

Table 4. Screen descriptions

#	Feature	Screen	Description
1	FE-01	Login	E-mail and password fields with a show/hide control, the Sign in button, links to e-mail verification, password recovery and registration, and an Offline training button when a profile is cached on the device.
2	FE-01	Register	E-mail, password and password confirmation fields with inline validation and the Create account button.
3	FE-01	Email Verification (OTP)	Six-digit code field, Verify and Resend code buttons, and a link back to sign-in.
4	FE-01	Forgot / Reset Password	Step 1 asks for the e-mail address; step 2 asks for the reset code, the new password and its confirmation.
5	FE-01	Loading	Animated artwork while the profile, wallet, content catalogue and settings are loaded.
6	FE-01	Experience Choice [popup]	Asked once at the first sign-in: "Bạn đã quen với pickleball chưa?" with Beginner and Experienced options.
7	FE-01	Menu (Lobby)	3D court with the player's character; name tag with level, rank and Elo; Tokenpad and Elite Token balances; male/female selector; day/night toggle; music player; Play button (Enter); one row of seven tiles – Progress, Decisions and Appeals ("!"), History, Missions, Collection, Shop and Settings; friend drawer on the left edge.
8	FE-01	Settings [overlay]	Tabs Overview (with Sign out), Audio, Graphics and Controls (key binding and touch layout); Restore defaults, Undo and Apply and save buttons.
9	FE-02	Tutorial	Seventeen steps on the court; each step has a title, an explanation and a demonstration, with Continue, Replay and Later buttons.
10	FE-02	Mode Selection	Four cards – Bot match, Online match, Training and Create room – and a Tutorial strip.
11	FE-02	Training Academy	Lesson catalogue with group tabs (All, Basic, Control, Advanced, Doubles); lesson briefing with target, reward and star rules; practice HUD with shot counter, timing feedback and key hints.
12	FE-02	Drill Result [popup]	Stars, accuracy (hits in target / shots), mean timing error, practice time, most frequent error with a tip, and Retry, Next lesson and Lesson list buttons.
13	FE-02	Training Progress [popup]	Passed lessons out of 15, next lesson, post-session feedback, practice time, most frequent error, and for each lesson the stars, best accuracy and the four most recent accuracy results ("Gần đây") with a Retry button.
14	FE-03	Bot Match Setup	Difficulty (Easy, Medium, Hard, Adaptive – shown as "Tự động"), time of day (day, night) and court (professional, outdoor), with a summary line and the Enter court button.
15	FE-03	Bot Match	Court seen from behind the player's baseline with both players; score panel with serving side and serve timer; rule hint; fault banners; shot cue under the player with the key combination (touch buttons on phones); Settings (Esc) and help (F1).

#	Feature	Screen	Description
16	FE-03	Match Result [popup]	Winner, game scores, Tokenpad and XP earned, Elo change for ranked matches, and Play again, Mode selection and View analysis buttons.
17	FE-05	Coaching Summary [popup]	Counts of shots, contacts and serves; main errors with their frequency; up to three recommended drills with a Practise now link.
18	FE-04	Online Lobby	Singles 1v1 and Doubles 2v2 tabs, room code with Copy, Join by code, roster slots with Invite buttons, a collapsible friends column listing online friends who can be invited, and the Find match / Cancel button with an elapsed-time counter.
19	FE-04	Ready Check [popup]	Match found message with Accept and Decline buttons and a 10-second countdown.
20	FE-04	Multiplayer Court	Online match on the dedicated server with score, connection state, reconnecting overlay and leave control.
21	FE-07	Match History [popup]	Level card and the last 20 matches with opponent, mode, Vietnam time, result, game score, Elo change, and Report and Replay buttons.
22	FE-07	Replay Viewer	Playback of a recorded match with play/pause, a seek bar and speeds from 0.5× to 2×.
23	FE-06	Missions [popup]	Daily and weekly tabs, mission list with progress and reward, set bonus, daily streak, reset countdown, and Claim buttons.
24	FE-07	Friends and Leaderboard [drawer]	Friends tab with search by name or ID and sub-tabs Friends, Find friends and Blocked; Leaderboard tab with own position.
25	FE-06	Shop	Featured and all items, item preview, price in Tokenpad and/or Elite Token, wallet balances and the Buy button.
26	FE-06	Collection	Owned paddles with preview and the Equip button.
27	FE-08	Decisions and Appeals [popup]	List of sanctions on the account with type, reason and expiry; appeal form and appeal status.
28	FE-08	Report Player [dialog]	Opened from a history entry: reported player, match, reason field and Send button.
29	FE-09	Admin Login	Account or e-mail and password fields and the Authenticate button; only Admin accounts are accepted.
30	FE-09	Dashboard	Total users, daily active users, matches played, online time, pending reports, technique-error heat map and quick links.
31	FE-09	Users	Search and status filter; table of player and e-mail, role, status, Elo, play time and creation date; actions to assign a role, sanction, verify e-mail and reset progress.
32	FE-09	Progress Reset [dialog]	Selection of LEARNING, RANKING, MISSIONS and ECONOMY groups, an impact preview and the Reset button.
33	FE-09	Content (CMS)	Lesson manager (draft, publish, unpublish), drill form, shop item form, mission form and technique references.
34	FE-09	Matches	Live matches refreshed every 5 seconds with score and connection state, match history, and a Stop match dialog that requires a reason.
35	FE-09	Moderation	Reports, sanctions and appeals with match evidence and replay download, decision form and decision reason.
36	FE-09	Analytics	Date range in Vietnam time; daily active players, retention cohorts, popular lessons with pass rate, technique-error heat map and matchmaking health.
37	FE-10	Expert Review	Training sessions with shot-by-shot evidence, level and error labels, data group (training, calibration, holdout) and export.

3.1.3 Screen Authorization

X means that the role may open the screen or perform the activity. Visitor is a user who has not signed in; Restricted Player is a signed-in player with a suspended or banned account. The Web API enforces the same rules: a missing or expired token returns HTTP 401, and a valid token without the required role or scope returns HTTP 403.

Table 5. Screen authorization

Screen / Activity	Visitor	Player	Restricted Player	Administrator
Login, Register, Email Verification, Forgot Password	X			
Start offline training from a cached profile	X	X		
Menu, Settings, Mode Selection, Tutorial		X		
Update name, select character, sign out		X		

Screen / Activity	Visitor	Player	Restricted Player	Administrator
Training Academy, Drill Result, Training Progress		X		
Bot Match Setup, Bot Match, Match Result, Coaching Summary		X		
Online Lobby, Ready Check, Multiplayer Court		X		
Create room, invite friend, start a custom match (host only)		X		
Match History, Replay Viewer		X		
Query own matches and replays only		X		
Report a player of an own finished match		X		
Missions, Shop, Collection		X		
Claim rewards, purchase and equip own items		X		
Friends and Leaderboard		X		
Decisions and Appeals		X	X	
Submit an appeal for an own sanction			X	
Admin Login				X
Dashboard, Analytics				X
Users				X
Query all accounts				X
Assign role, verify e-mail, apply and revoke sanctions (not own account)				X
Preview and reset player progress				X
Content (CMS)				X
Add and update lessons, references, items and missions; publish and unpublish				X
Matches				X
Stop a live dedicated-server match				X
Moderation				X
Decide reports and appeals				X
Expert Review				X
Save expert labels and export the dataset				X

3.1.4 Non-Screen Functions

Table 6. Non-screen functions

#	Feature	System Function	Description
1	FE-01	Send verification and reset e-mails	Generates a six-digit code, stores only its HMAC, and sends the message through Gmail SMTP (STARTTLS) from a background mail queue.
2	FE-01	Refresh access token	Exchanges a valid refresh token for a new access token (2 hours) and a rotated refresh token (14 days); revoked or reused tokens are rejected.
3	FE-01	Record activity days	Records each player's login day and active day (Vietnam time) and online presence in Redis for analytics and streaks.
4	FE-02	Apply training results	Validates each training attempt from its telemetry, applies stars and first-pass rewards, and returns a receipt APPLIED, DUPLICATE or REJECTED for each request ID.
5	FE-04	Matchmaking	Keeps search tickets in a Redis queue. Ranked singles: pairs tickets within the Elo window and assigns a ranked bot after 180 seconds. Quick doubles: fills two teams of two from solo players and parties of two in the same region and protocol version, without Elo or bot fallback. Both create the ready check.
6	FE-04	Dedicated server allocation	Issues HMAC-signed match tickets, leases the dedicated server slot, receives heartbeats every 8 seconds and releases expired leases (DedicatedLeaseMonitor).
7	FE-04	Settle match result	Saves the game scores once, applies Elo for ranked matches, grants rewards and updates missions and achievements in one transaction.
8	FE-04	Store replay	Uploads the replay manifest and compressed chunks to object storage and links them to the match.
9	FE-05	Ingest telemetry	Receives shot telemetry parts from the client (or the dedicated server for online matches), stores them and creates an analysis job.

#	Feature	System Function	Description
10	FE-05	Hybrid AI worker	Sends pending analysis jobs to the AI coaching service, stores the coaching result and retries failed jobs (HybridAiWorker).
11	FE-05	AI coaching service	POST /v1/analyze-session: rule-based findings, summary, at most three drill recommendations and, when the gated model is approved, a tier estimate.
12	FE-06	Mission cycle	Opens new daily missions at 07:00 Vietnam time each day and new weekly missions each week, and updates the daily streak.
13	FE-07	Leaderboard cache	Caches the Elo leaderboard in Redis for 15 seconds.
14	FE-08	Sanction expiry	Restores suspended accounts whose sanction has expired (ModerationExpiryWorker).
15	FE-09	Analytics projection	Aggregates daily active players, retention cohorts, lesson statistics and matchmaking events (AnalyticsProjectionWorker).
16	FE-09	Admin audit	Writes an audit record for every administrative change with actor, action, target and time.

3.1.5 Entity Relationship Diagram

The conceptual ERD is drawn in Chen notation and split into six subject areas so that every diagram stays readable. Entities are business concepts rather than database tables: junction tables appear as relationships with their own attributes (for example PARTICIPATES carries Team and Outcome), and purely technical tables such as refresh tokens and telemetry parts are omitted. The logical design of all 51 tables is given in Report 4, §2.

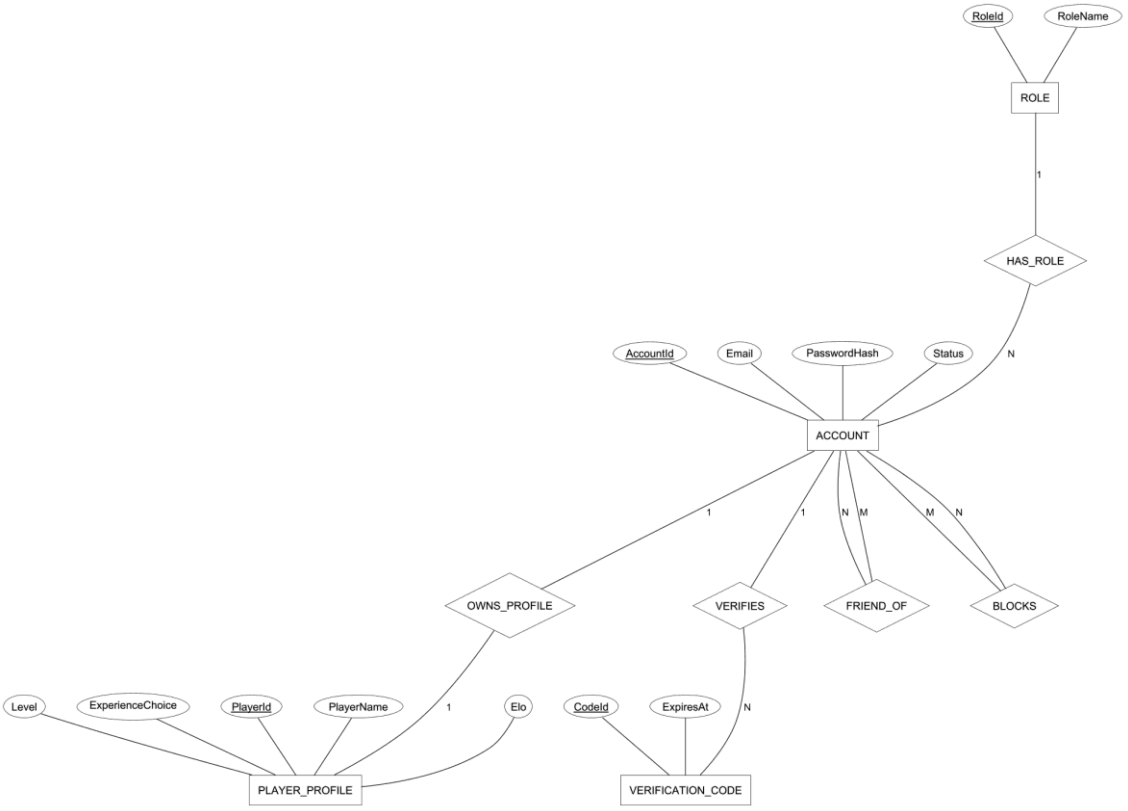


Figure 14. Conceptual ERD – accounts, profiles and social relations

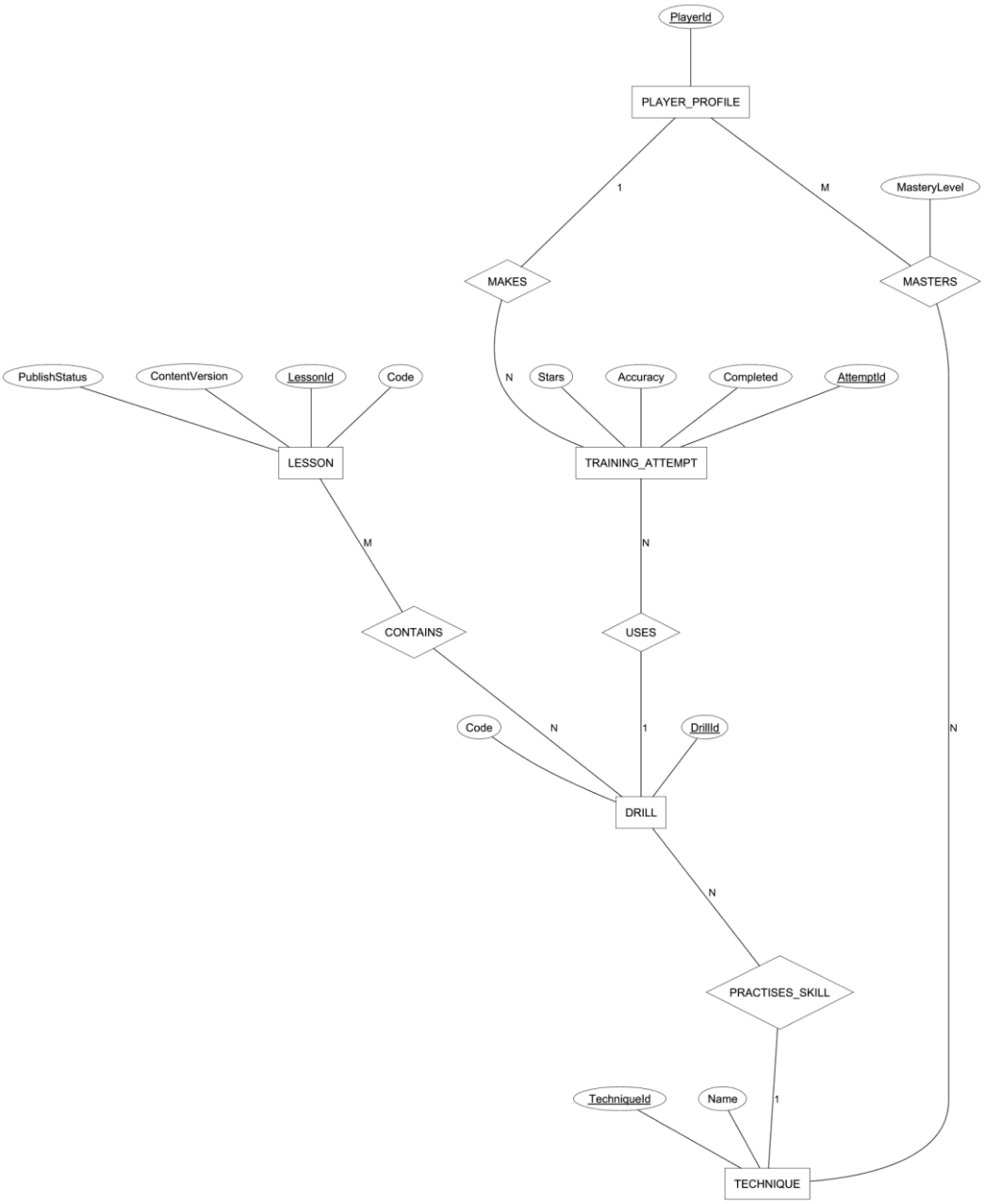


Figure 15. Conceptual ERD – techniques, lessons and training

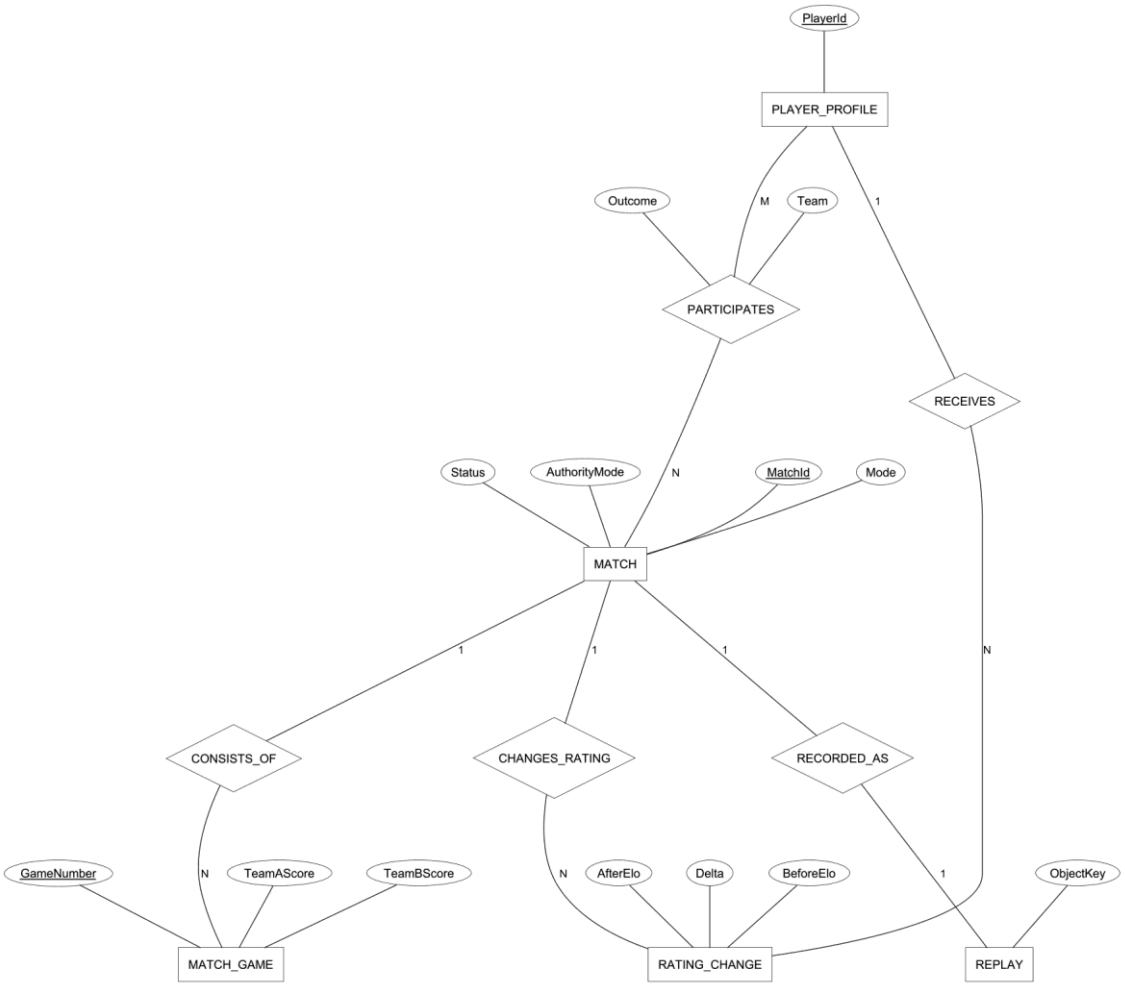


Figure 16. Conceptual ERD – matches, ratings and replays

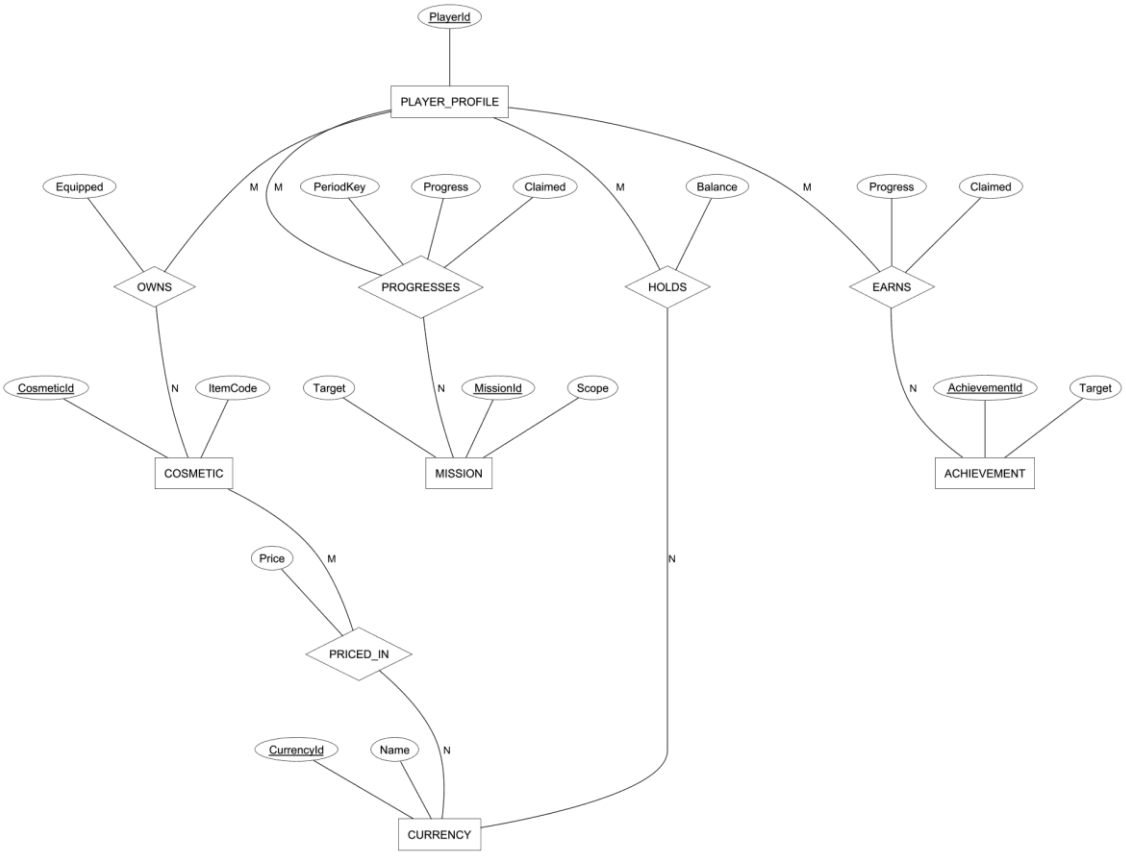


Figure 17. Conceptual ERD – currencies, cosmetics, missions and achievements

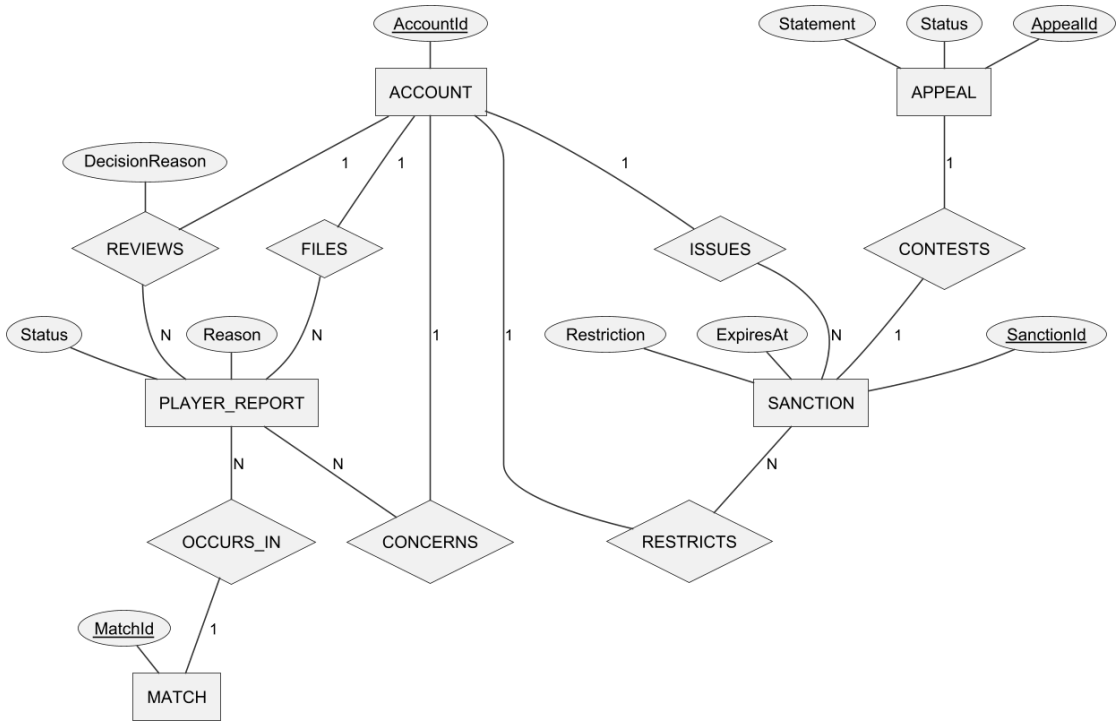


Figure 18. Conceptual ERD – reports, sanctions and appeals

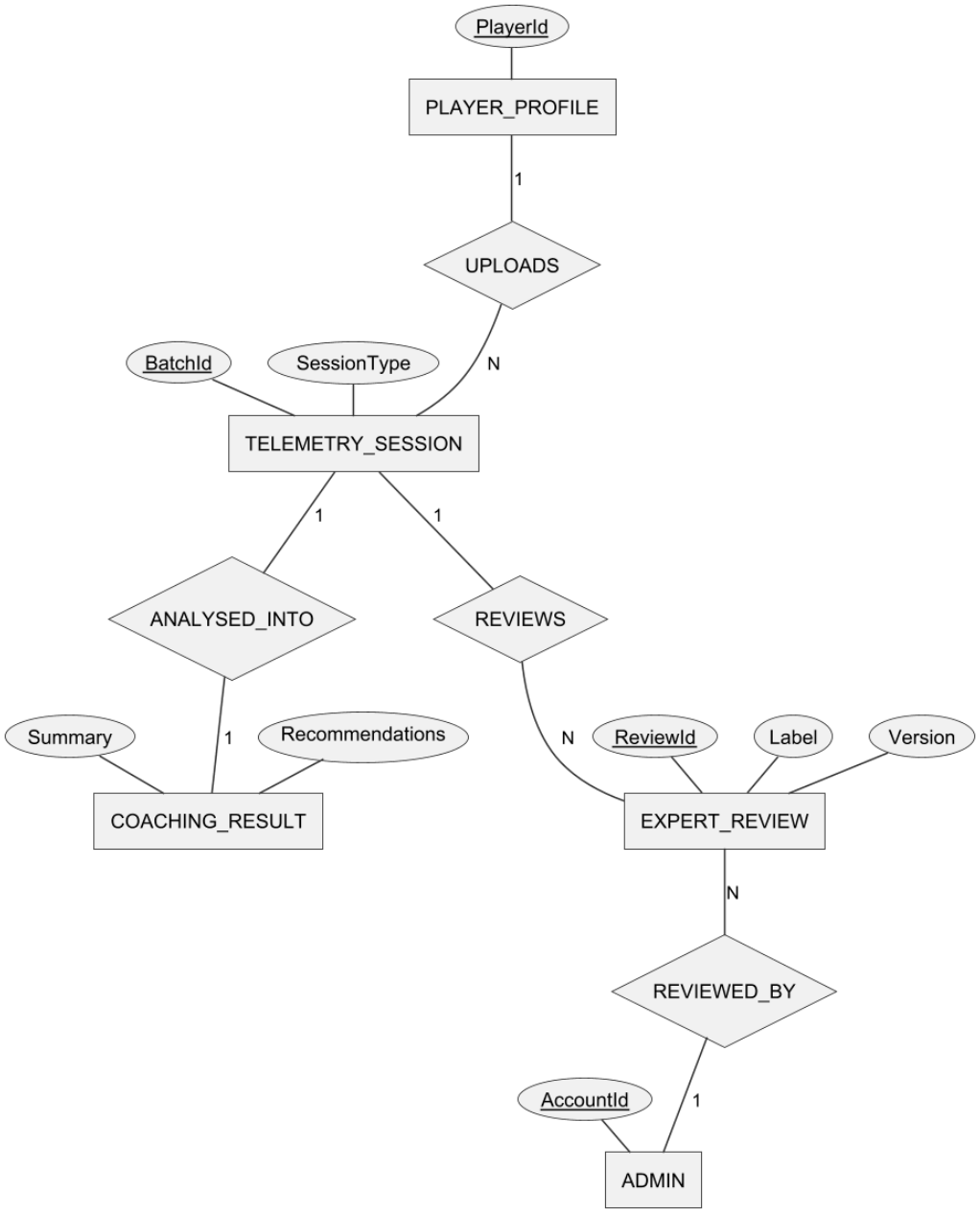


Figure 19. Conceptual ERD – telemetry, coaching and expert review

Entities Description

Table 7. Entities description

#	Entity	Description
1	ACCOUNT	A registered user identified by e-mail; holds the password hash, status (ACTIVE, SUSPENDED, BANNED) and verification state.
2	ROLE	Player or Admin; each account has exactly one role.
3	VERIFICATION_CODE	A one-time code for e-mail verification or password reset, with expiry and remaining attempts.
4	PLAYER_PROFILE	Game identity of an account: display name, character, level, XP, Elo, onboarding choice and statistics.
5	TECHNIQUE	A pickleball technique to be learnt (serve, drive, dink, volley, drop, reset, smash, footwork, approach, doubles coverage and transition, modifiers).
6	LESSON	A unit of the curriculum with code, group, order, target, reward, prerequisite and publication status.
7	DRILL	An exercise that practises one technique; a lesson contains one or more drills.

#	Entity	Description
8	TRAINING_ATTEMPT	One run of a drill by a player, with shots, hits, timing, stars and synchronisation receipt.
9	MATCH	A bot, ranked, matched doubles or custom match with mode, kind, authority mode and status.
10	MATCH_GAME	One game (1 to 3) of a match with the score of each team; a weak entity identified by match and game number.
11	RATING_CHANGE	The Elo before, change and after of one player caused by one ranked match.
12	REPLAY	The stored recording of a match (object key of its manifest).
13	PLAYER_REPORT	A complaint by one player about another player in a given match, with reason, status and decision.
14	SANCTION	A suspension (with expiry) or ban applied to an account, with reason, actor and revocation data.
15	APPEAL	A restricted player's written objection to one sanction, with status and decision.
16	CURRENCY	In-game currency: Tokenpad (earned by playing) and Elite Token.
17	COSMETIC	A purchasable or granted item (paddle, outfit, hat) with price and currency.
18	MISSION	A daily or weekly goal with target and reward.
19	ACHIEVEMENT	A long-term goal (wins, matches, points, streaks) with a one-time reward.
20	TELEMETRY_SESSION	The shot-level data captured during a drill or match and uploaded for analysis.
21	COACHING_RESULT	The AI analysis of a telemetry session: findings, summary, recommendations and optional tier.
22	EXPERT_REVIEW	An expert's labels for a telemetry session (level, errors, data group) used to evaluate the AI.
23	ADMIN	An account with the Admin role, shown as a separate box in the coaching diagram for readability.

3.2 Account and Onboarding (FE-01)

FR-01 Register Account and Verify Email

Related use cases: UC-01 Register Account, UC-02 Verify Email

Function trigger: Login screen → link "Đăng ký" (Register). The verification page also opens after a successful registration, or from the link "Bạn cần xác thực email?".

Function description: A visitor creates an account with an e-mail address and a password. The client validates the form, the Web API creates the account in the unverified state, stores a hash of a six-digit one-time password (OTP) and queues an e-mail through Gmail SMTP. The visitor enters the OTP to activate the account; after activation the account can sign in.

Screen layout

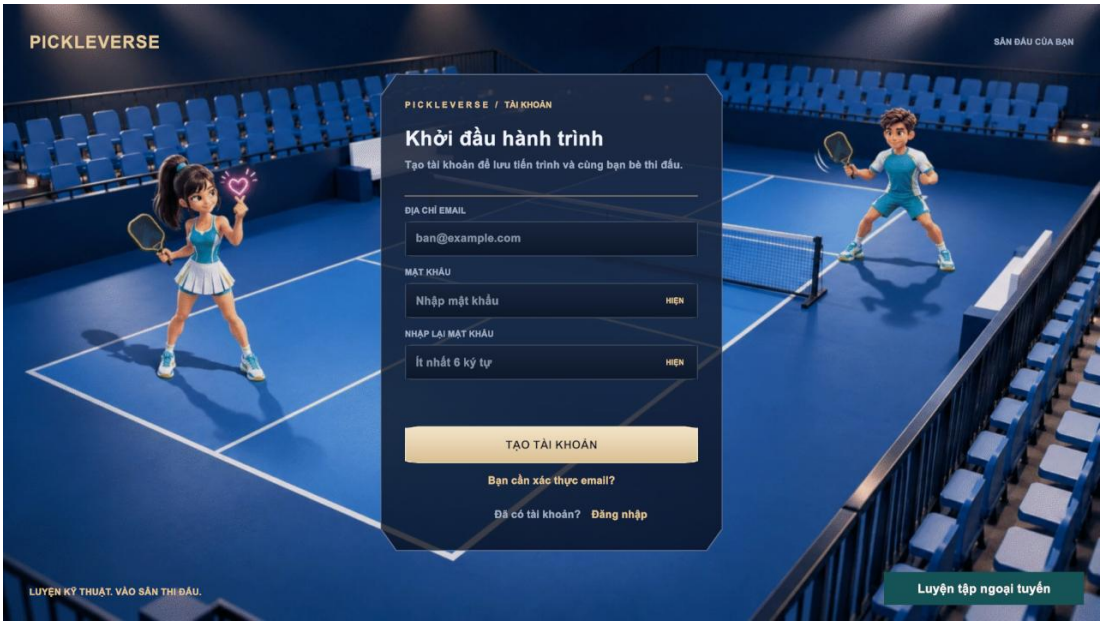


Figure 20. Register screen

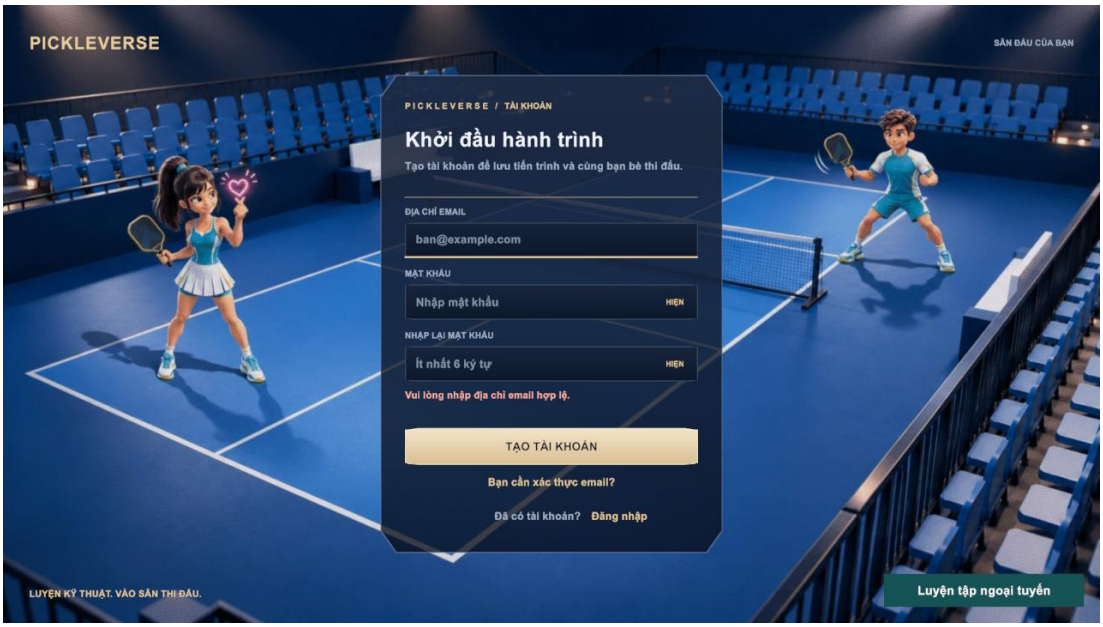


Figure 21. Register screen with an inline validation message

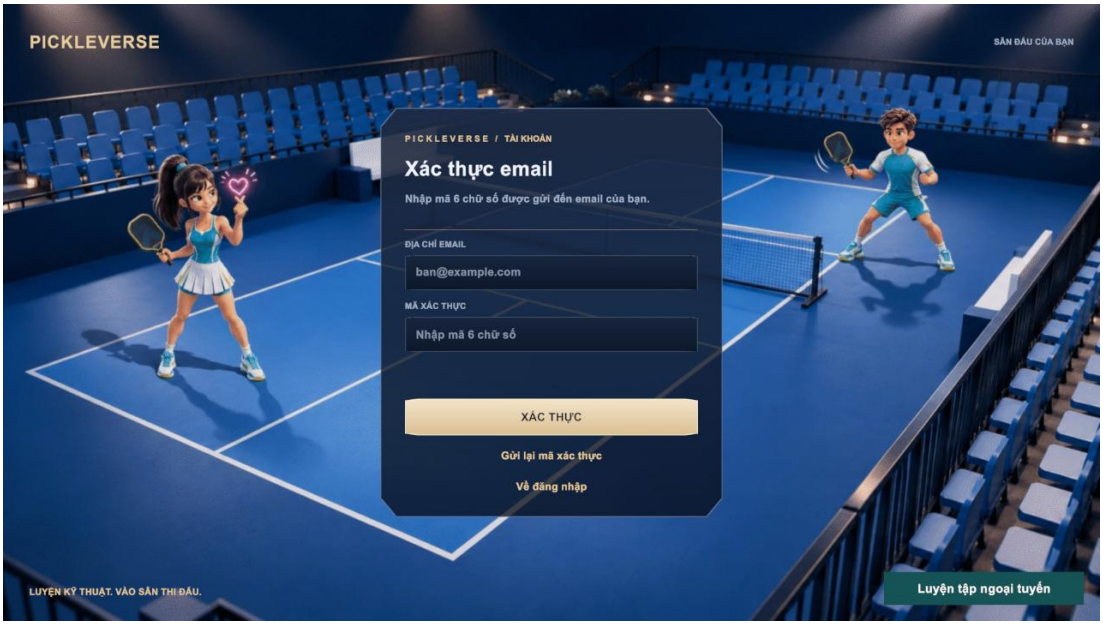


Figure 22. E-mail verification screen

Function details

Table 8. FR-01 data fields

Field	Type	Required	Validation and constraints
Email	Text (≤254)	Yes	Valid e-mail format (MSG01, MSG02); not already registered (MSG06)
Password	Password	Yes	6–128 characters (MSG03, MSG04, MSG17); show/hide toggle
Confirm password	Password	Yes	Must equal Password (MSG05)
Verification code	Text (6 digits)	Yes	Exactly six digits (MSG08); valid for 10 minutes; at most 5 wrong attempts

Business rules: BR-01, BR-02, BR-03, BR-04.

Normal flow:

1. The visitor enters e-mail, password and confirmation and selects **Tạo tài khoản**.
2. The client validates the fields; the API creates an ACTIVE but unverified account with the Player role and a player profile.
3. The API sends the OTP e-mail and the client shows MSG07 and the verification page.
4. The visitor enters the six-digit code and selects **Xác thực**.
5. The API checks the code, marks the e-mail as verified and the client returns to sign-in.

Alternative and exception flows:

- E1 – The e-mail is already registered: MSG06; an unverified owner can use **Gửi lại mã xác thực**.
- E2 – Resend is requested within 60 seconds of the previous code: HTTP 429 and MSG10.
- E3 – Wrong or expired code: the attempt is counted; after 5 wrong attempts the code is invalid and a new code must be requested.
- E4 – The e-mail cannot be delivered (SMTP not available): MSG22; an administrator can verify the account manually (FR-29).
- E5 – The e-mail is already verified: the client asks the visitor to sign in.

FR-02 Sign In and Sign Out

Related use cases: UC-03 Sign In, UC-05 Sign Out

Function trigger: Login screen → **Đăng nhập**; Settings → Overview → **Đăng xuất**. Token refresh runs automatically before the access token expires.

Function description: A visitor signs in with e-mail and password. The API verifies the BCrypt hash, issues a JWT access token (2 hours) and a refresh token (14 days, stored as a SHA-256 hash) and returns the profile summary. The client loads the profile, wallet, catalogue and settings on the Loading screen and opens the Menu. A restricted account receives a token limited to the appeal scope and is taken to Decisions and Appeals. Signing out revokes the refresh token on the server and clears the session on the device.

Screen layout

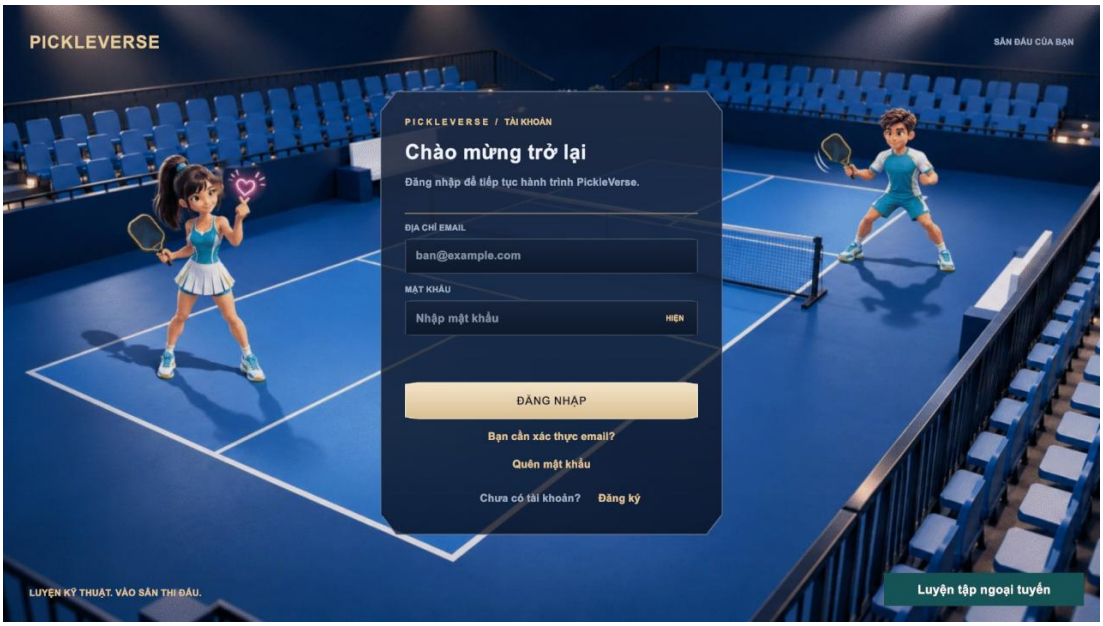


Figure 23. Login screen of the game client

Function details

Table 9. FR-02 data fields

Field	Type	Required	Validation and constraints
Email	Text	Yes	Not empty (MSG01), valid format (MSG02)
Password	Password	Yes	Not empty (MSG03)

Business rules: BR-04, BR-05.

Normal flow:

1. The visitor enters e-mail and password and selects **Đăng nhập**.
2. The API validates the credentials and returns tokens and the profile; the client shows MSG13 and the Loading screen.
3. On the first sign-in the client asks for the experience level (FR-04); otherwise it opens the Menu.
4. To sign out, the player opens Settings → Overview and selects **Đăng xuất**; the client returns to the Login screen.

Alternative and exception flows:

- E1 – Wrong e-mail or password: MSG12 (the message does not reveal which field is wrong).
- E2 – The e-mail is not verified: MSG11 and a link to the verification page.
- E3 – The account is suspended or banned: MSG14 and the Decisions and Appeals popup (FR-28).
- E4 – A system bot account tries to sign in: MSG65.
- E5 – The refresh token is expired or revoked: MSG18 and the Login screen.
- A1 – No network and a cached profile exists: the visitor can select **Luyện tập ngoại tuyến** (FR-09).

FR-03 Reset Password

Related use cases: UC-04 Reset Password

Function trigger: Login screen → **Quên mật khẩu**.

Function description: A visitor who has forgotten the password requests a reset code. To prevent account enumeration the API always answers with the same neutral message; a code is sent only when the address belongs to a verified account. The visitor enters the code and a new password; the API stores the new hash and revokes every refresh token of the account.

Screen layout

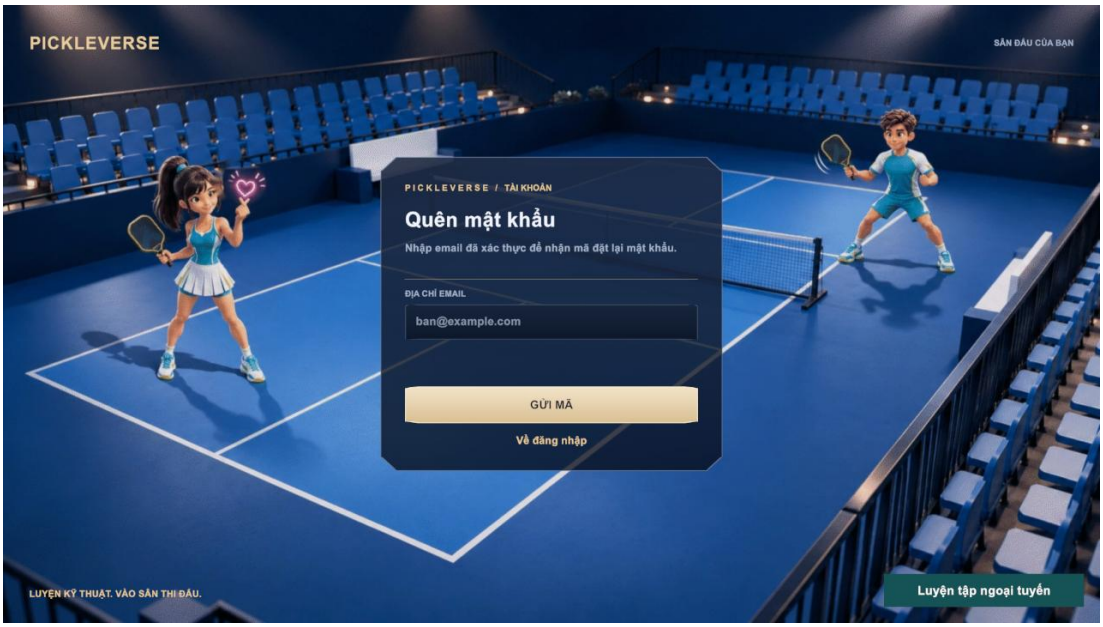


Figure 24. Forgot password screen

Function details

Table 10. FR-03 data fields

Field	Type	Required	Validation and constraints
Email	Text	Yes	Valid format (MSG02)
Reset code	Text (6 digits)	Yes	Valid for 10 minutes; at most 5 attempts (MSG16)
New password / confirmation	Password	Yes	6–128 characters; equal (MSG04, MSG05, MSG17)

Business rules: BR-02, BR-03, BR-05.

Normal flow:

1. The visitor enters the e-mail address and selects **Gửi mã**; the client shows MSG15.
2. The visitor enters the code, the new password and its confirmation.
3. The API changes the password, revokes all sessions and the client returns to sign-in.

Alternative and exception flows:

- E1 – Wrong or expired code: MSG16.
- E2 – Resend within 60 seconds: MSG10.

FR-04 Onboarding and Player Profile

Related use cases: UC-06 Choose Experience Level, UC-07 Update Player Name, UC-08 Select Character

Function trigger: First sign-in (experience popup); Menu → name tag or profile card edit icon (name); Menu → **Nam** / **Nữ** selector (character).

Function description: At the first sign-in the player answers MSG21. **Người mới** (Beginner) opens the tutorial; **Đã thành thạo** (Experienced) goes to the Menu. The answer is saved once. The player can later change the display name and choose the male or female character; the character is used in the menu, training and matches.

Screen layout



Figure 25. Menu (lobby) of a signed-in player

Function details

Table 11. FR-04 data fields

Field	Type	Required	Validation and constraints
Experience level	Choice	Yes (once)	Beginner or Experienced; cannot be changed afterwards
Display name	Text	Yes	3–18 characters, no control characters, unique (MSG19, MSG20)
Character	Choice	Yes	Male or Female; default Male

Business rules: BR-06, BR-07, BR-08.

Normal flow:

1. The player answers the experience question; the choice is saved.
2. The player edits the name and confirms; the API checks length and uniqueness and saves it.
3. The player selects **Nam** or **Nữ**; the menu character changes immediately and the choice is saved.

Alternative and exception flows:

- E1 – Name outside 3–18 characters: MSG19.
- E2 – Name already used: MSG20.
- E3 – The player is in a room, searching or in a match: profile changes are disabled until the activity ends.

FR-05 Configure Game Settings

Related use cases: UC-09 Configure Game Settings

Function trigger: Menu → **Cài đặt**, or Esc during a match or drill.

Function description: The player adjusts audio (music and effects volume), graphics (the quality options of the Graphics tab) and controls (keyboard bindings for PC and the touch layout for phones). Changes are previewed and stored on the device when the player selects **Áp dụng và lưu**.

Screen layout

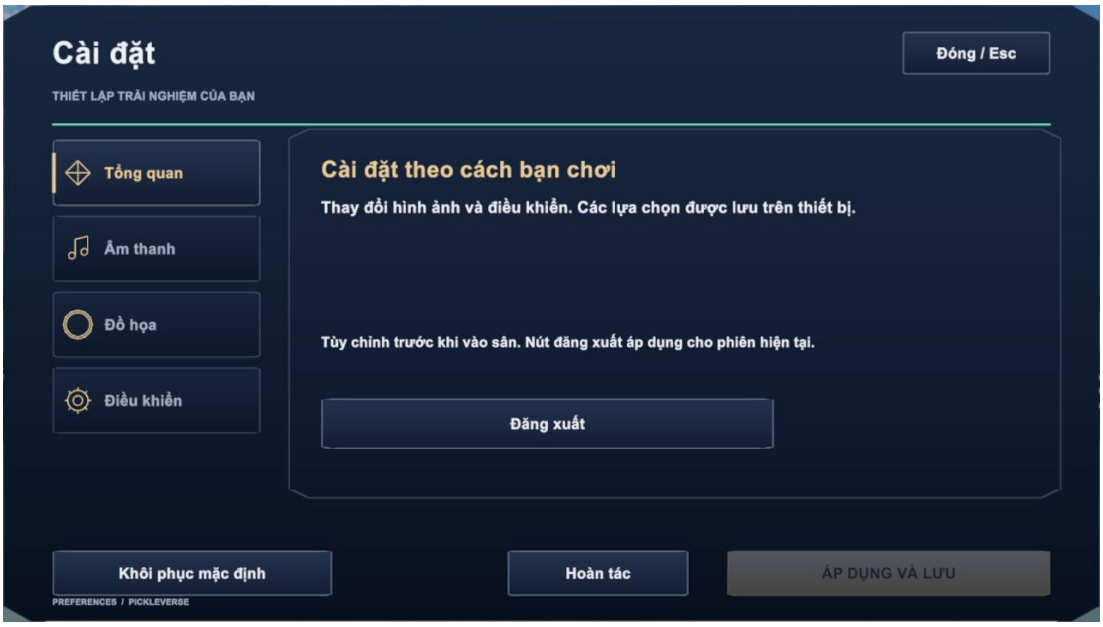


Figure 26. Settings – Overview tab

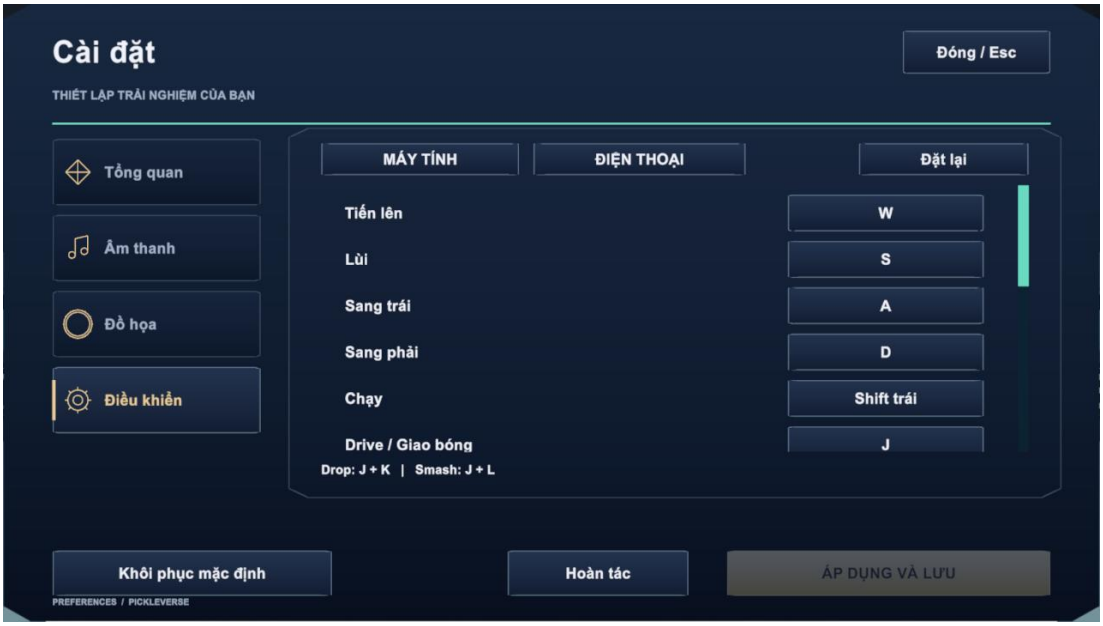


Figure 27. Settings – Controls tab with key bindings

Function details

Table 12. FR-05 data fields

Field	Type	Required	Validation and constraints
Music / effects volume	Slider 0–100	No	Default from the build
Graphics options	Choice	No	Applied after Apply and save
Key binding	Key	No	One key per action; Restore defaults resets all (defaults: W/A/S/D move, Left Shift run, J drive and serve, K dink, L volley, U power, I cross)

Business rules: BR-17.

Normal flow:

1. The player changes values; **Áp dụng và lưu** becomes active.
2. The player saves; settings apply immediately and persist on the device.

Alternative and exception flows:

- A1 – **Hoàn tác** restores the last saved values; **Khôi phục mặc định** restores factory values.

3.3 Tutorial and Technique Training (FE-02)

FR-06 Complete Tutorial

Related use cases: UC-10 Complete Tutorial

Function trigger: Automatically after choosing Beginner (FR-04); Mode Selection → **Tutorial / Hướng dẫn cơ bản**.

Function description: The tutorial teaches the court, the rules and the controls in 17 short steps (about 2–3 minutes). Each step shows a title, an explanation and a demonstration on the court, for example the court dimensions, a ball that does not cross the net, the two-bounce rule and the non-volley zone. Completion is saved to the profile.

Screen layout

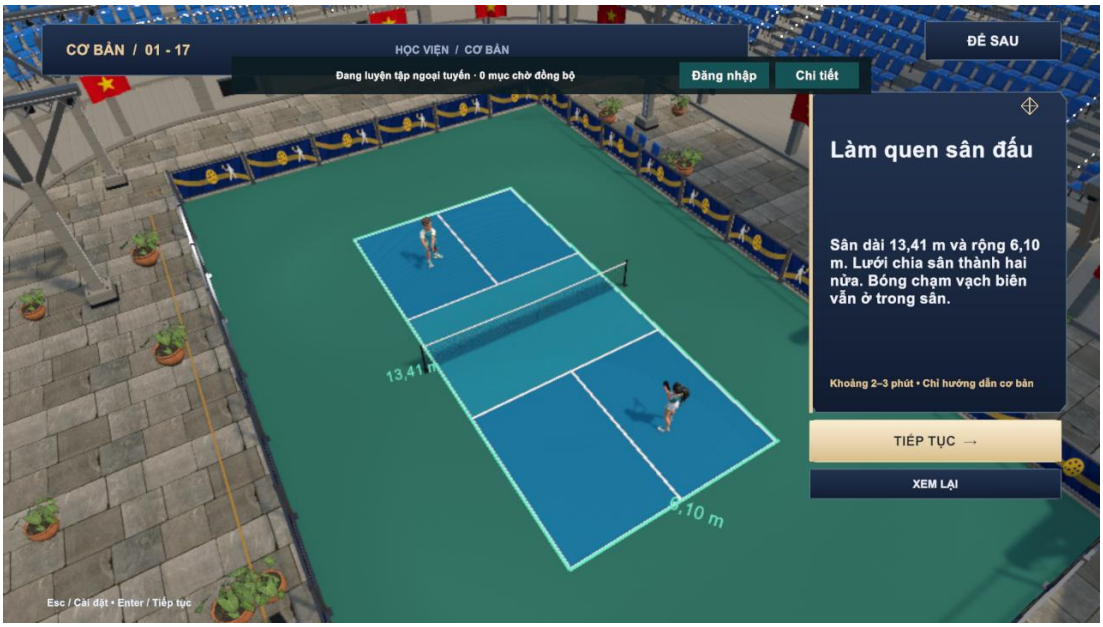


Figure 28. Tutorial step 1 – getting to know the court

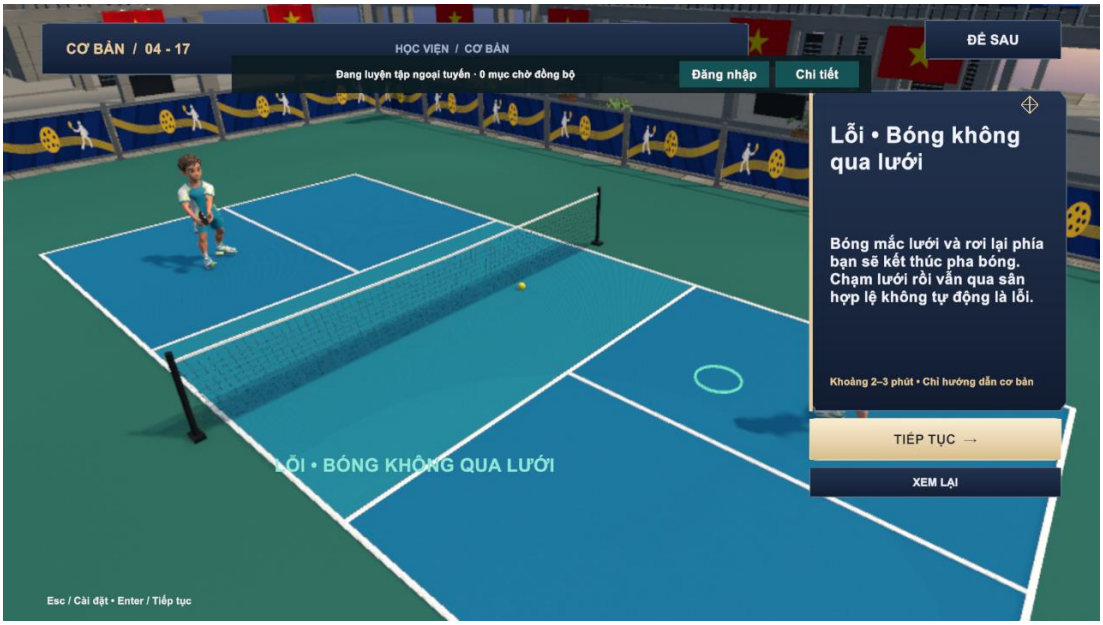


Figure 29. Tutorial step 4 – fault: the ball does not cross the net

Function details

Business rules: BR-09 to BR-14.

Normal flow:

1. The tutorial opens at step 1.
2. The player reads each step and selects **Tiếp tục** (Enter) to continue or **Xem lại** to replay the demonstration.
3. After step 17 the completion is saved and the Menu opens.

Alternative and exception flows:

- A1 – **Để sau** leaves the tutorial; it can be reopened from Mode Selection.

FR-07 Browse Lessons

Related use cases: UC-11 Browse Lessons

Function trigger: Mode Selection → **Luyện tập**; Training Progress → **Tiếp tục luyện**.

Function description: The academy lists the 15 published lessons grouped as Basic, Control, Advanced and Doubles. Each row shows the group, the first-pass reward, the best stars and whether the lesson is unlocked. Selecting a lesson opens its briefing: coach explanation, target, reward and the star rules.

Screen layout

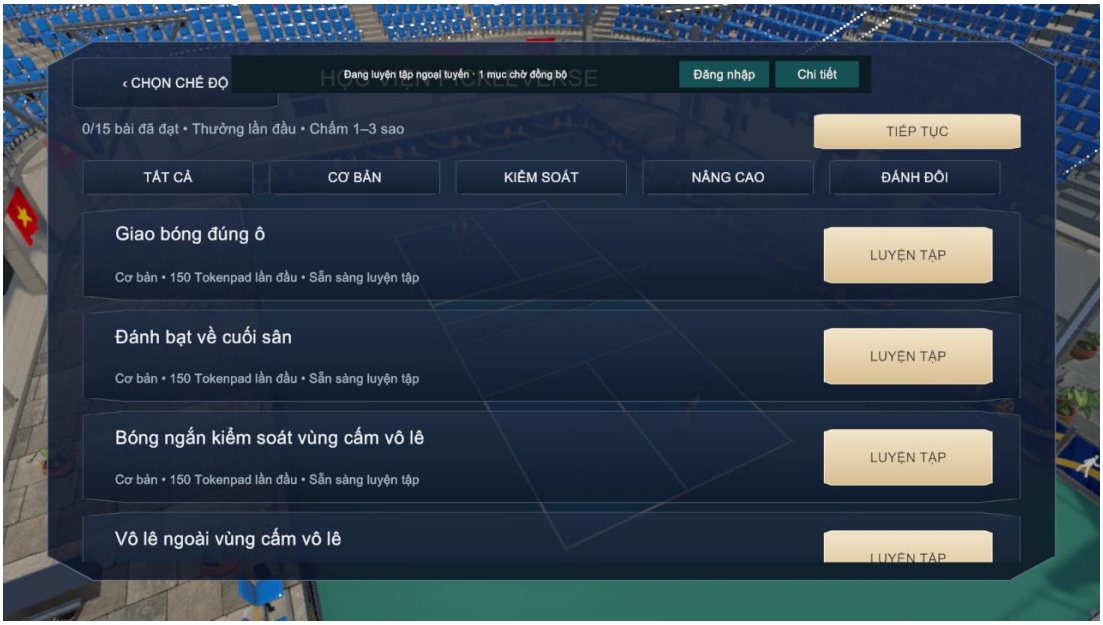


Figure 30. Lesson catalogue with group tabs

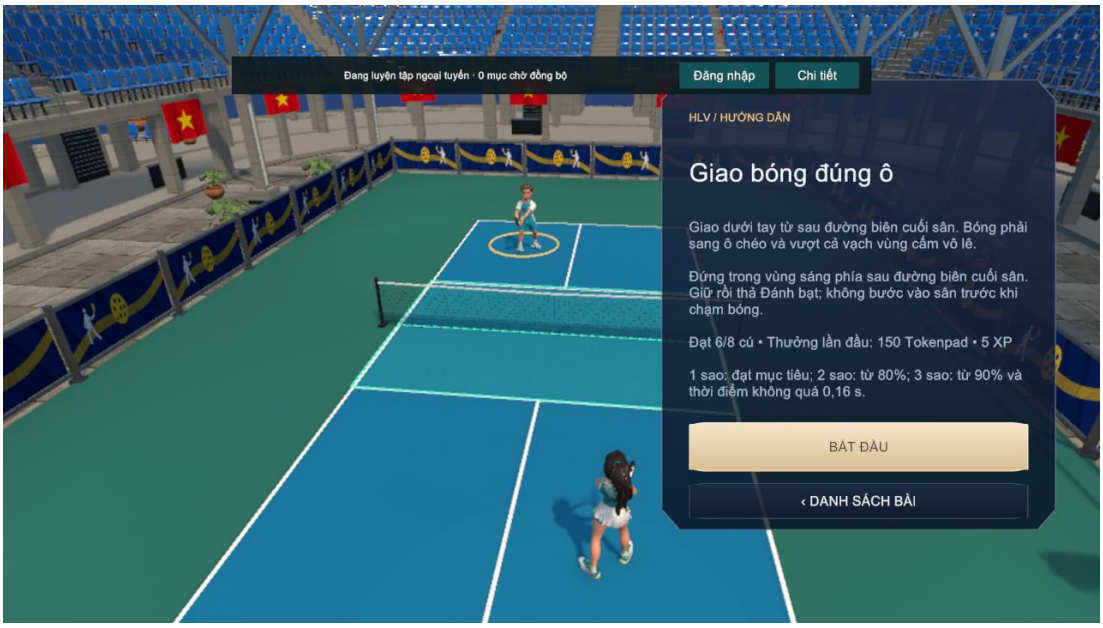


Figure 31. Lesson briefing for "Giao bóng đúng ô" (serve)

Function details

Business rules: BR-18, BR-19, BR-20.

Normal flow:

1. The player opens the academy; the catalogue loads from the server (or the cached catalogue when offline).
2. The player filters by group and selects **Luyện tập** on a lesson.
3. The briefing opens; **Bắt đầu** starts the drill (FR-08).

Alternative and exception flows:

- E1 – The lesson is locked: the start button is disabled and MSG23 is shown.

FR-08 Practise Drill

Related use cases: UC-12 Practise Drill

Function trigger: Lesson briefing → **Bắt đầu**; Drill Result → **Luyện lại**.

Function description: The player performs the drill on the training court, for example eight serves into the highlighted service court. For each shot the game measures whether the ball lands in the target area and the timing error between the key release and the ideal contact. In-drill feedback tells the player whether the release was early or late. At the end the client sends the attempt and its telemetry to the API, which validates it, applies the stars and the first-pass reward and returns a receipt.

Screen layout



Figure 32. Practice HUD of the volley drill: coach panel, feedback after a late hit ("Chưa đúng • Thử lại") and the shot cue L under the player

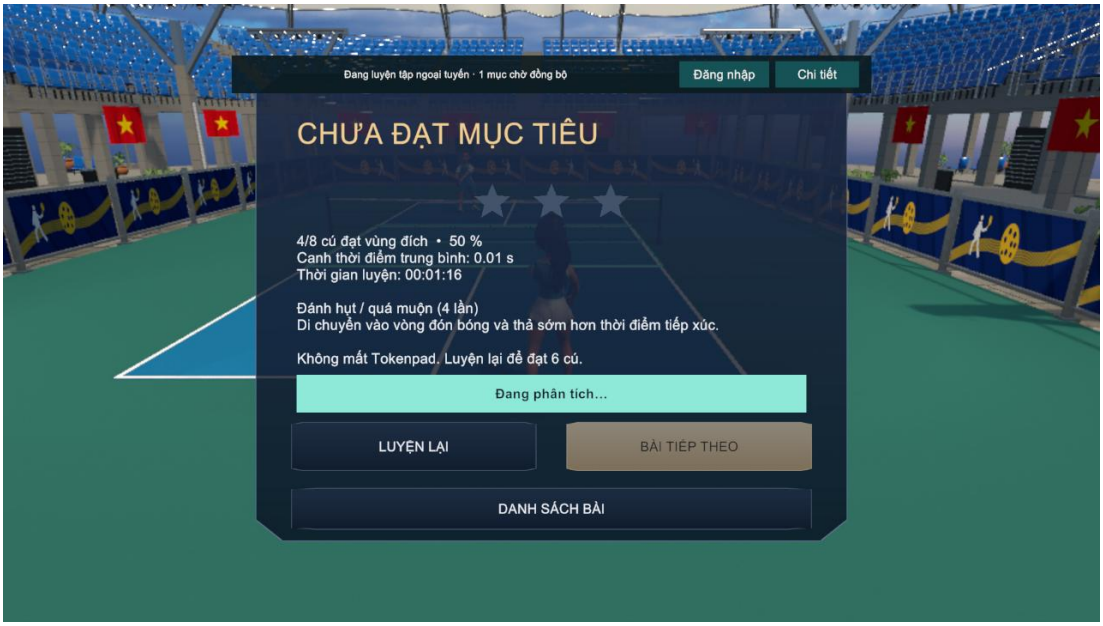


Figure 33. Drill result – target not reached

Function details

Table 13. FR-08 data fields

Field	Type	Required	Validation and constraints
Shots / target	Integer	System	Fixed per lesson (for example 6 of 8 for the serve lesson)
Hits in target	Integer	System	Counted by the rules engine
Mean timing error	Seconds	System	Absolute difference between release and ideal contact
Stars	0–3	System	Computed by BR-18; the best result is kept

Business rules: BR-16, BR-17, BR-18, BR-20.

Normal flow:

1. The drill starts and the HUD shows lesson, shot counter and key hints.
2. The player hits the required shots; feedback (MSG26, MSG27) appears after each shot.
3. After the last shot the Drill Result popup shows stars, accuracy, mean timing, time, the most frequent error and a tip.
4. The API stores the attempt; on the first pass it grants the lesson reward and unlocks the next lesson.
5. The coaching analysis appears when ready (FR-19).

Alternative and exception flows:

- A1 – Target not reached: MSG24 and MSG25; no currency is lost and **Bài tiếp theo** stays disabled.
- A2 – The device is offline: the attempt is queued for synchronisation (FR-09).
- E1 – The attempt is rejected by the server (for example inconsistent telemetry): receipt REJECTED and no reward.

FR-09 Offline Training and Synchronisation

Related use cases: UC-13 Practise Drill Offline, UC-14 Synchronise Offline Results

Function trigger: Login → **Luyện tập ngoại tuyến** when a profile is cached; automatic synchronisation when the connection returns, or banner → **Đăng nhập**.

Function description: A player who has signed in on the device at least once can train without a connection. The client uses the cached catalogue and profile, stores every finished attempt with a unique request ID in a local queue and shows the offline banner MSG28. When the player is online again, the queue is sent; the API applies each request at most once and answers APPLIED, DUPLICATE or REJECTED. Online-only features (bot and online matches, shop) are disabled offline.

Screen layout

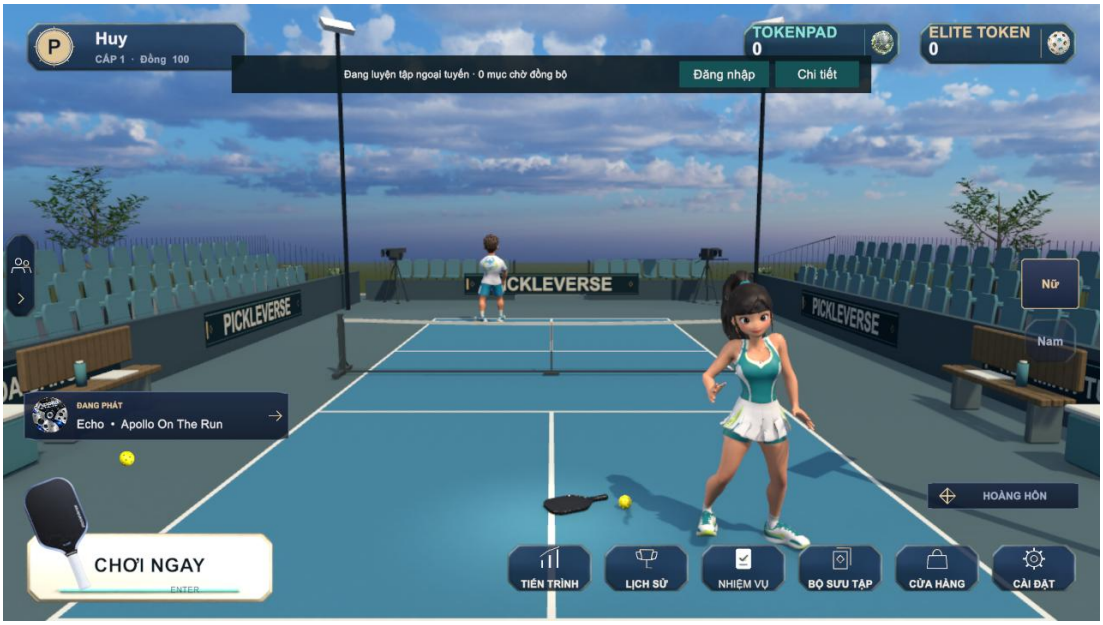


Figure 34. Menu in offline training mode with the offline banner

Function details

Business rules: BR-21.

Normal flow:

1. The player starts offline training from the Login screen.
2. Drills work as in FR-08; each result is added to the queue and the banner shows the pending count.
3. When connected, the client sends the queue; applied items are removed and rewards are added to the profile.

Alternative and exception flows:

- A1 – The same request is sent twice: receipt DUPLICATE; no second reward.
- E1 – No cached profile on the device: the offline button is not shown.
- E2 – A queued item is invalid: receipt REJECTED; the item is removed and the player is informed through **Chi tiết**.

FR-10 View Training Progress

Related use cases: UC-15 View Training Progress

Function trigger: Menu → **Tiến trình**.

Function description: The progress popup summarises the player's learning: passed lessons out of 15, the next lesson, the latest post-session feedback, total practice time, the most frequent error and the most retried lesson. For every lesson it shows the stars, the best accuracy and the four most recent accuracy results ("Gần đây"), so that the player can see the trend, with a Retry shortcut. Technique mastery is updated from the same attempts. Tabs give access to daily and weekly missions and achievements (FR-21).

Screen layout

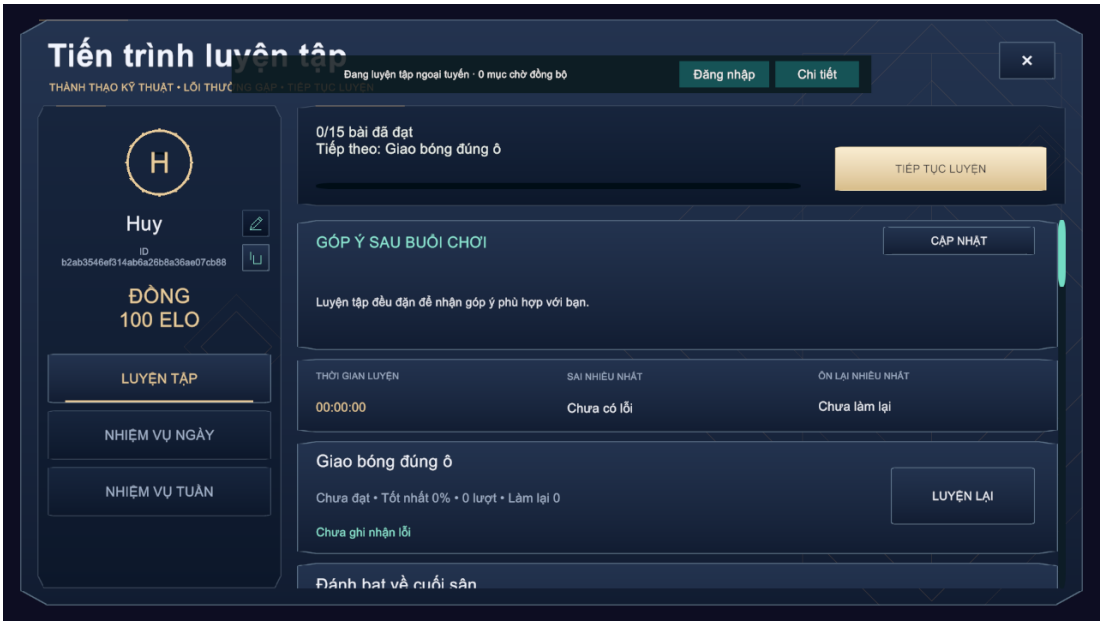


Figure 35. Training progress popup

Function details

Business rules: BR-18.

Normal flow:

1. The player opens Progress; the data is loaded from the profile.
2. The player selects **Luyện lại** on a lesson to open its briefing.

3.4 Core Gameplay and Bot Practice (FE-03)

FR-11 Control the Player and Hit Shots

Related use cases: UC-12, UC-16, UC-25

Function trigger: Any rally in a drill, bot match or online match.

Function description: On PC the player moves with W/A/S/D, runs with Left Shift and hits by holding and releasing shot keys. One of seven shot kinds is chosen from the key combination when the first key is released; U adds power and I aims cross-court. On phones the left joystick moves the player, the CHẠY button is held to run, and the DRIVE, DINK and VOLLEY buttons are held and released; dragging a held button up turns DRIVE into SMASH, DINK into DROP and VOLLEY into RESET, dragging left aims cross-court (CHÉO), dragging right adds power (LỰC) and dragging down cancels (HỦY); the serve uses DRIVE. The paddle contact is accepted only inside the timing window, and the quality of the contact (Perfect, Good, early or late) affects the ball speed and accuracy. The ball flight, bounce and net contact are simulated at 120 Hz. The camera stays behind the player's baseline and keeps the player above the shot cue (PC) or the touch controls (phone).

Screen layout



Figure 36. Bot match: the player is serving; the shot cue under the player shows the key combination J + I (cross-court serve)

Table 14. FR-11 screen layout

Area	Elements and behaviour
Score panel (top left)	Mode and difficulty, game number, games and points of each side, serving side and serve timer
Rule hint (left)	What the player must do now, for example "Bạn giao • giữ rồi thả J" and the score to call
Controls (top right)	Settings (Esc) and help (F1)
Court	Player character, opponent, ball, landing marker and fault banner
Shot cue (bottom centre)	Key combination and name of the shot being prepared; on phones the joystick and shot buttons take this area

Function details

Business rules: BR-16, BR-17.

Normal flow:

1. The player moves to the ball and holds a shot key (J, K or L, alone or combined).
2. The player releases the key close to the contact moment; the game selects the shot kind and checks the timing.
3. The ball is hit towards the target computed from the shot kind and modifiers.

Alternative and exception flows:

- A1 – Release outside ±65 ms of the ideal contact: no contact (miss) and feedback MSG26 or MSG27.
- A2 – Invalid key combination: the input is ignored until all keys are released.

FR-12 Apply Singles Match Rules

Related use cases: UC-16, UC-25

Function trigger: Every rally of a bot or online match.

Function description: The rules engine decides legal serves, faults, points, games and the match winner according to the official singles rules simplified for the game. In online matches the same engine runs on the dedicated server, which is the only authority for the score.

Screen layout

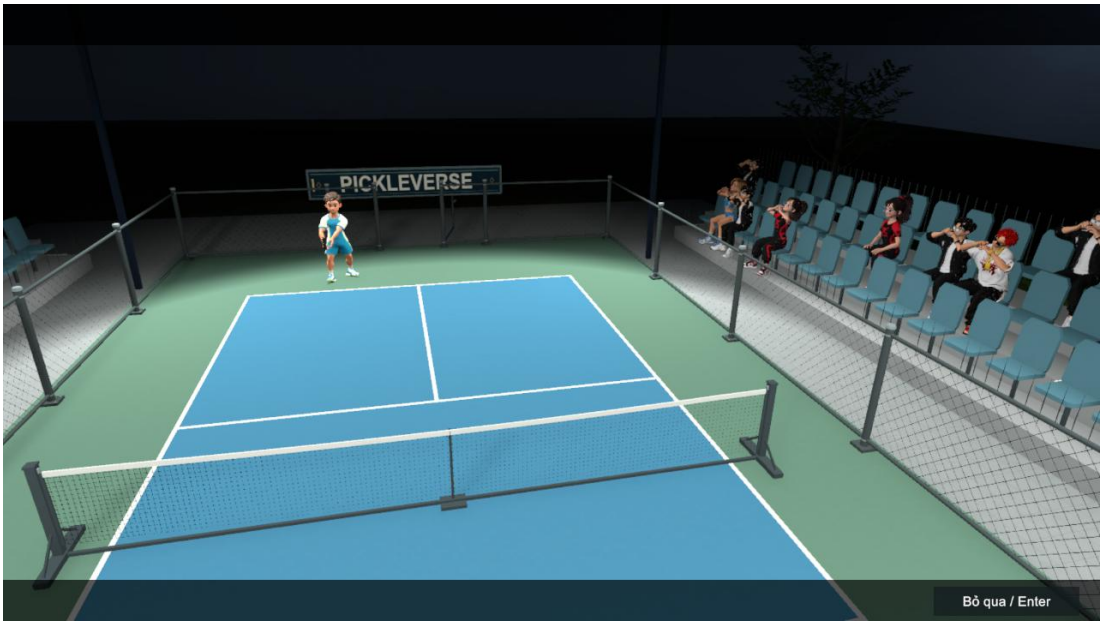


Figure 37. Bot match introduction

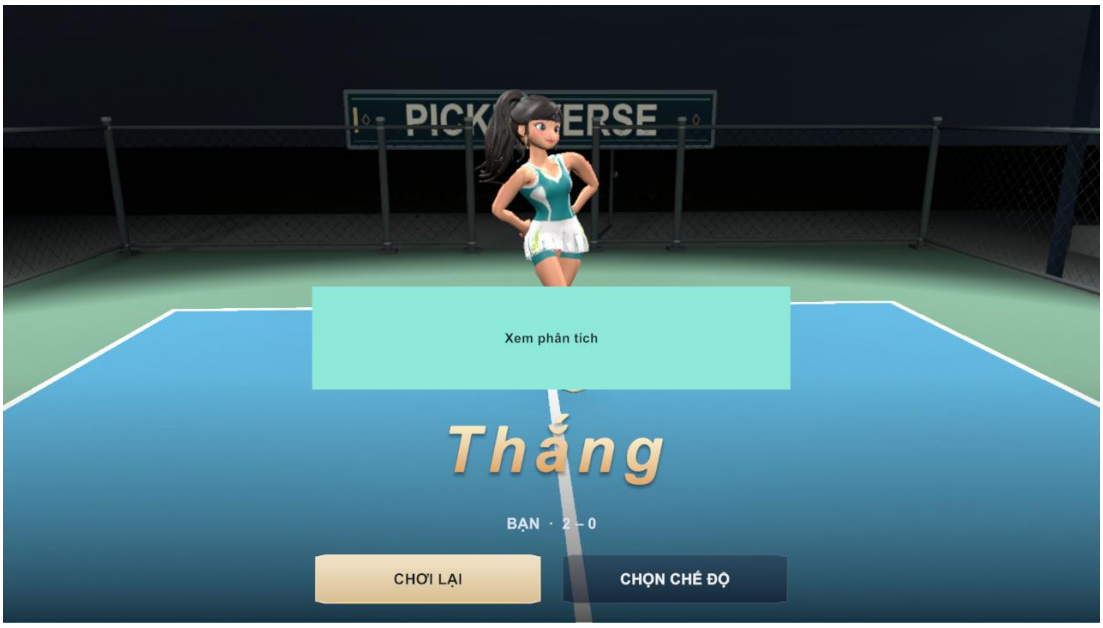


Figure 38. Match result – victory 2–0

Function details

Business rules: BR-09 to BR-15.

Normal flow:

1. The server of the rally serves underhand from behind the baseline into the diagonal service court within 10 seconds.
2. The return and the third shot must bounce; afterwards volleys are allowed outside the non-volley zone.
3. When a fault occurs, the rally ends, a fault banner (MSG30) is shown and the point or side-out is applied.
4. A game ends at 11 points with a 2-point lead; the first side to win two games wins the match (MSG31).

Alternative and exception flows:

- A1 – In the third game the players change ends when a side reaches 6 points.
- E1 – The serve timer expires: fault ServeTimeout and MSG29.

FR-13 Set Up and Play a Bot Match

Related use cases: UC-16 Play Bot Match

Function trigger: Menu → **Chơi ngay** → Mode Selection → **Đấu với bot** → Bot Match Setup → **Vào sân**.

Function description: The player chooses the difficulty, the time of day and the court, then plays a singles match against the computer. Easy, Medium and Hard use fixed reaction and accuracy settings; the adaptive option uses a behaviour-tree bot whose tier follows the player's Elo. After the match the result popup shows the winner, scores and rewards, and the coaching analysis can be opened.

Screen layout

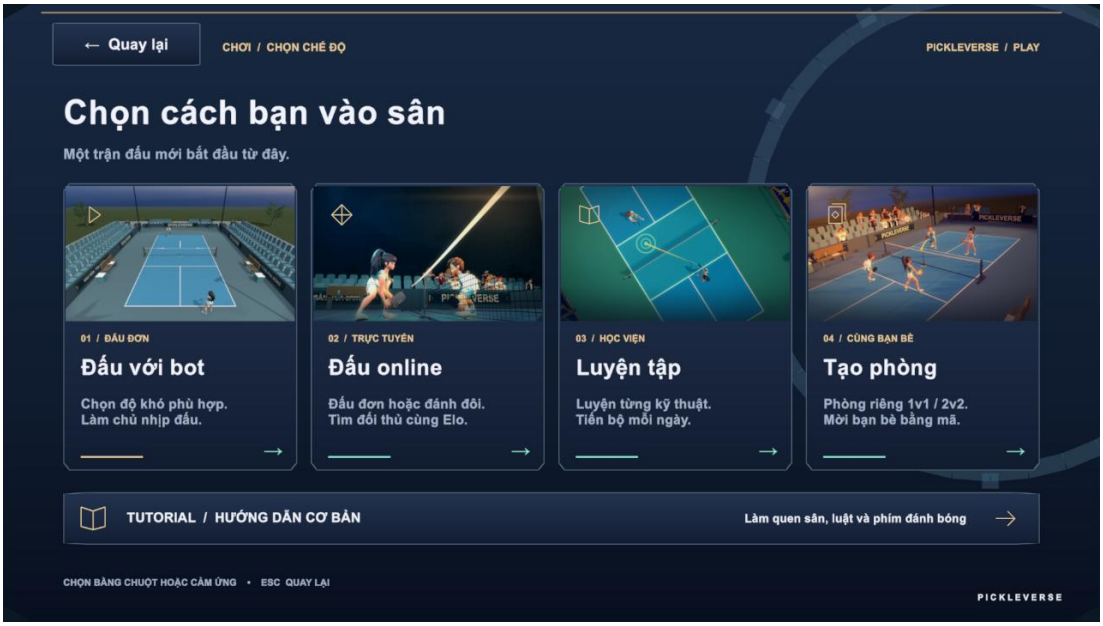


Figure 39. Mode selection

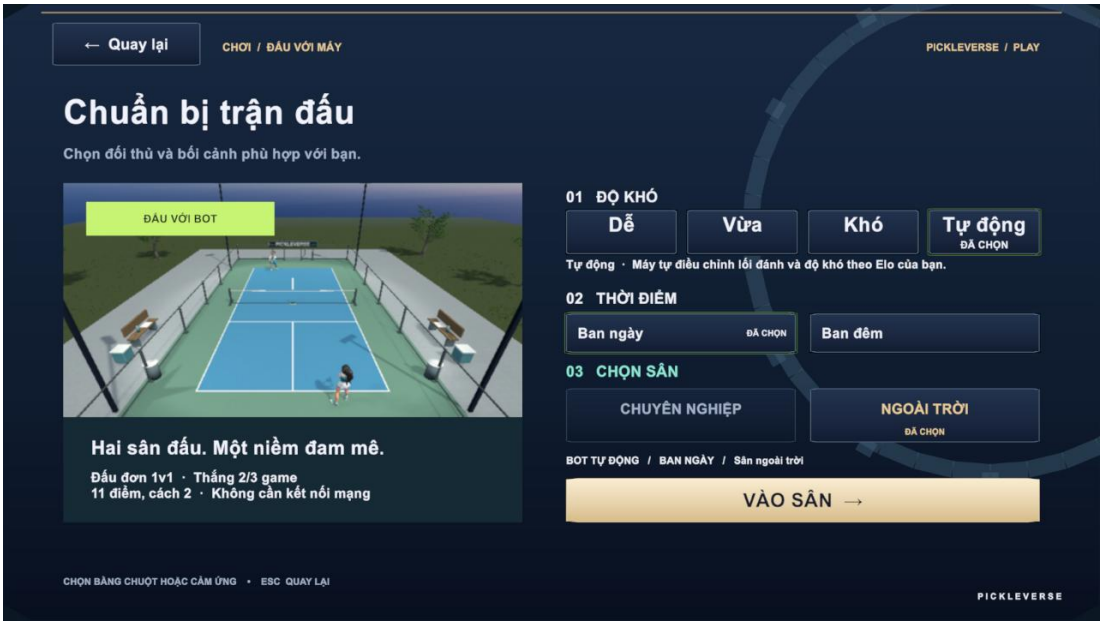


Figure 40. Bot match setup

This function uses the screen already shown in Figure 38; no separate screenshot is repeated here.

Function details

Table 15. FR-13 data fields

Field	Type	Required	Validation and constraints
Difficulty	Choice	Yes	Easy, Medium, Hard or Adaptive
Time of day	Choice	Yes	Day or night
Court	Choice	Yes	Professional or outdoor

Business rules: BR-10 to BR-16, BR-22, BR-31, BR-32.

Normal flow:

1. The player selects the options and **Vào sân**; the intro cinematic plays.
2. The match is played with FR-11 and FR-12.
3. At the end the API settles the result: rewards and XP are added; Elo does not change in a practice bot match.
4. The player selects **Chơi lại**, **Chọn chế độ** or **Xem phân tích**.

Alternative and exception flows:

- A1 – Ranked Bot match assigned by matchmaking (FR-14): Elo changes according to BR-23.
- E1 – The device is offline: the bot match card is disabled.

3.5 Online Matches (FE-04)

FR-14 Find an Online Match

Related use cases: UC-19 Find Online Match, UC-20 Confirm Match Readiness

Function trigger: Mode Selection → **Đấu online** → Online Lobby → **Đấu đơn 1v1** (ranked singles) or **Đấu đôi 2v2** (quick doubles) → **Tìm trận**.

Function description: Ranked singles: the player joins the ranked queue; the matchmaking worker pairs two searching players whose Elo is within the pairing window, and if no human opponent is found within 180 seconds a ranked bot opponent is assigned. **Quick doubles:** a player searches alone or the captain of a party of two (formed by room code or invitation) searches for both; the API fills two teams of two from solo players and parties in the same region, without Elo pairing and without a bot; the

match is unrated. In both modes every player must accept within 10 seconds. When all players accept, the API allocates the dedicated server and issues signed tickets, and the clients load the multiplayer court.

Screen layout

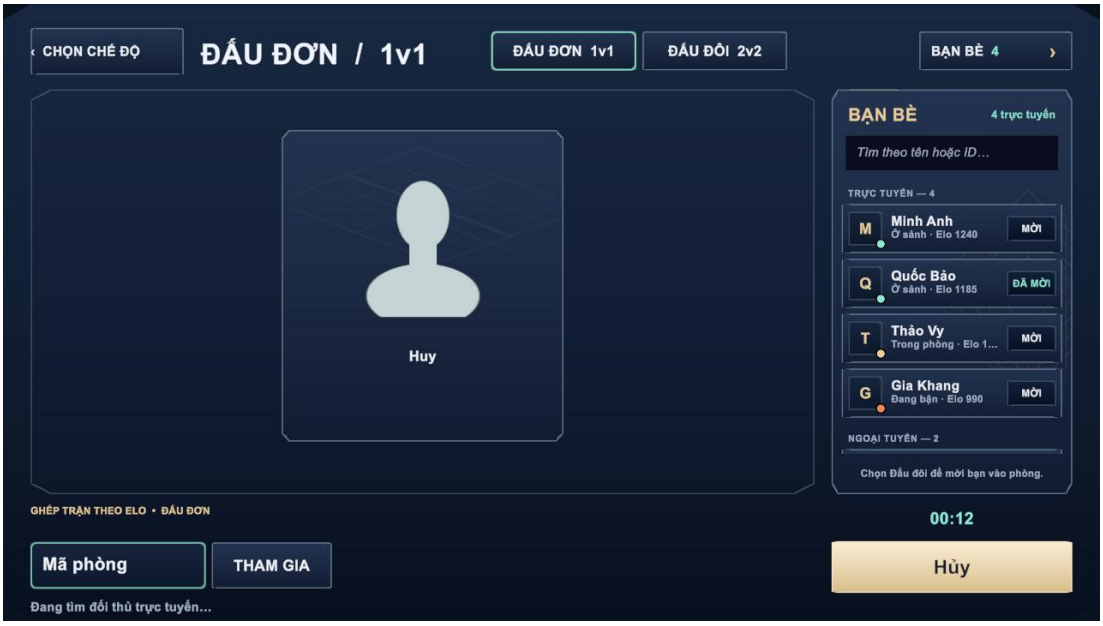


Figure 41. Searching for a singles opponent with the elapsed time (layout preview rendered in the Unity Editor)

Function details

Business rules: BR-22 to BR-27.

Normal flow:

1. The player selects **Tìm trận**; MSG32 and a timer are shown.
2. A pair (or two full teams for doubles) is found; the Ready Check shows MSG33 with a 10-second countdown.
3. All players accept; the server allocates the match and the court loads (FR-16).

Alternative and exception flows:

- A1 – Ranked singles only: no opponent within 180 seconds: a ranked bot is assigned and the match starts on the dedicated server.
- A2 – Quick doubles: no two full teams are found before the search lease ends: MSG34; there is no bot fallback.
- A3 – The player selects **Hủy** while searching: the ticket is cancelled.
- E1 – A player declines or the countdown ends: the offer closes; the decliner sees MSG35 and the other player returns to the queue.
- E2 – The dedicated server is busy: MSG36 and the request waits for the slot.
- E3 – The player already has a search or an unfinished match: MSG41.
- E4 – Search expired: MSG34.

FR-15 Private Rooms and Invitations

Related use cases: UC-21 Create Private Room, UC-22 Join Room by Code, UC-23 Invite Friend to Room, UC-24 Respond to Room Invitation

Function trigger: Mode Selection → **Tạo phòng**; Online Lobby → **Mã phòng** + **Tham gia bằng mã**; friend list → invite.

Function description: A player creates a singles (2 players) or doubles (up to 4 players) room. Other players join with the room code or accept an invitation sent to an online friend. The collapsible friends column of the lobby lists the friends who can be invited; only friends who are in the lobby receive invitations. The host can change teams, remove a guest and start the match. Custom matches are unrated and give no Elo. A doubles room of two players can also be used as a party for quick doubles (FR-14).

Screen layout

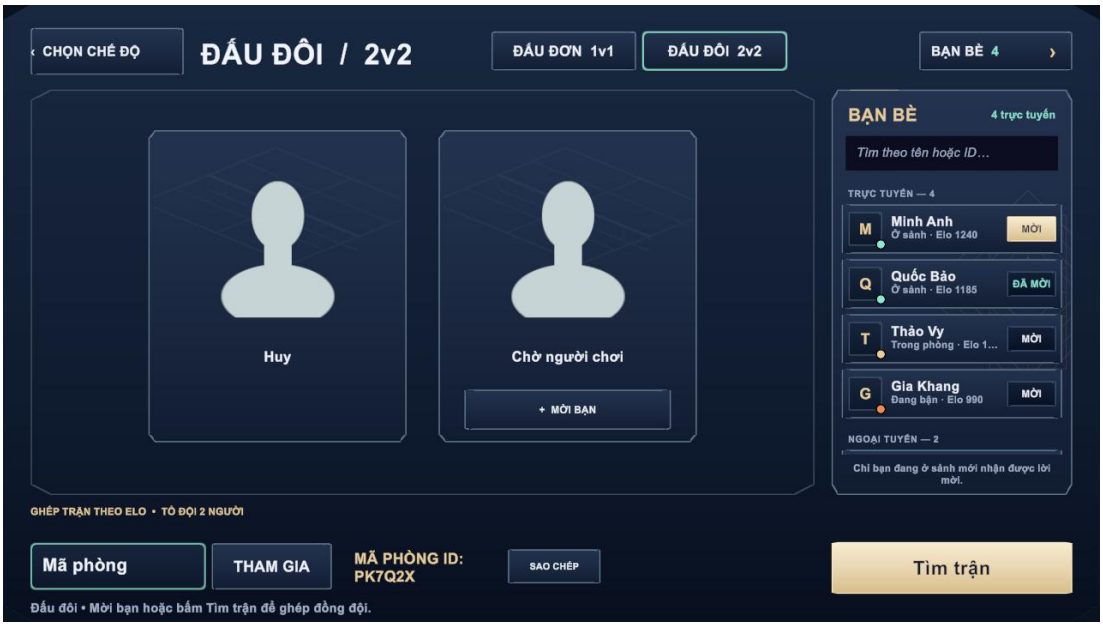


Figure 42. Online lobby – doubles room with the room code and the friends column (layout preview rendered in the Unity Editor with sample friend data)

Function details

Table 16. FR-15 data fields

Field	Type	Required	Validation and constraints
Room code	Text	Yes (to join)	Code of an open room on Photon; wrong code → MSG39
Mode	Choice	Yes	Singles 1v1 or Doubles 2v2

Business rules: BR-22, BR-30, BR-38.

Normal flow:

1. The host creates the room; the lobby shows the room code and roster slots.
2. Guests join by code or by accepting an invitation.
3. The host starts the match; the dedicated server is allocated as in FR-14.

Alternative and exception flows:

- E1 – Room full: MSG37.
- E2 – Room closed or the player was removed: MSG38.
- E3 – Invited player is busy or offline: MSG40.
- E4 – Blocked players cannot invite each other (BR-38).

FR-16 Play an Online Match

Related use cases: UC-25 Play Online Match

Function trigger: Automatically after the ready check (FR-14) or when the host starts a room (FR-15).

Function description: Each client connects to the dedicated server with its signed ticket through Photon Fusion. The server simulates the ball and applies the rules (FR-12); clients send only input and render server state with interpolation. The server sends heartbeats to the API every 8 seconds, records telemetry and the replay, and reports the final result.

Screen layout



Figure 43. Multiplayer court scene

Function details

Business rules: BR-12 to BR-17, BR-28, BR-29.

Normal flow:

1. The court loads and the server starts the match after all players are connected.
2. Players play rallies; scores are shown from the server state.
3. The server ends the match and the result popup appears (FR-18).

Alternative and exception flows:

- E1 – Photon cannot be reached: MSG45.
- E2 – The connection to the match server is lost: MSG42 and FR-17 starts.

FR-17 Reconnect to or Leave a Match

Related use cases: UC-26 Reconnect to Match, UC-27 Leave Match

Function trigger: Connection loss during an online match; pause menu → leave.

Function description: If a player's connection drops, the client shows a reconnecting overlay and retries for up to 30 seconds with the same identity. The dedicated server pauses the match and keeps the player's slot for 30 seconds. A player who reconnects in time continues with the server score. A player who leaves, or who does not return within 30 seconds, is recorded as having forfeited and the server ends the match as interrupted; interrupted matches change no Elo and give no reward. If the dedicated server itself is lost, its lease expires and the API ends the match as interrupted (BR-28).

Screen layout

Table 17. FR-17 screen layout

Area	Elements and behaviour
Reconnecting overlay	Status text and countdown; the court stays visible in the background
Leave dialog	Warning about the consequence and Confirm / Cancel buttons

Function details

Business rules: BR-28, BR-29.

Normal flow:

1. The client detects the loss and starts reconnecting.
2. The player rejoins with the same ticket identity and control is restored.

Alternative and exception flows:

- E1 – Reconnection fails within 30 seconds: MSG43; the result is shown when the server finishes processing.
- A1 – The player confirms leaving: MSG44 and the Menu opens.

FR-18 Record the Match Result and Update Elo

Related use cases: UC-16, UC-25 (non-screen settlement)

Function trigger: End of a bot match (reported by the client) or of an online match (reported by the dedicated server).

Function description: The API settles each match exactly once in a single transaction: it stores the game scores, marks the match FINISHED, applies Elo for ranked matches, grants Tokenpad and XP, and updates missions, achievements and streaks. The result popup and the history (FR-24) show the outcome.

Screen layout

This function uses the screen already shown in Figure 38; no separate screenshot is repeated here.

Function details

Business rules: BR-22, BR-23, BR-29, BR-32 to BR-35.

Normal flow:

1. The API receives the result with the match identifier.
2. It validates that the match is still PLAYING and that the reporter is allowed to report it.
3. It saves scores, Elo, rewards and progress, then returns the summary to the clients.

Alternative and exception flows:

- A1 – The result is sent again: the stored result is returned and nothing is applied twice.
- E1 – The match was stopped by an administrator before the result: no Elo or reward (BR-48).

3.6 AI Coaching (FE-05)

FR-19 View Coaching Summary and Drill Recommendations

Related use cases: UC-17 View Coaching Summary, UC-18 View Drill Recommendation

Function trigger: Drill Result → analysis panel; Match Result → **Xem phân tích**.

Function description: The client uploads the shot telemetry of the drill or match (for online matches the dedicated server uploads it). The hybrid AI worker sends the session to the AI coaching service, which detects common errors with rules (for example late contact, dink too high, entering the non-volley zone after a volley), writes a short summary and recommends at most three drills. When the gated tier model is approved, it also estimates the player's tier. The coaching popup shows the counts of shots, contacts and serves, the main errors with their frequency, and the recommended drills.

Screen layout

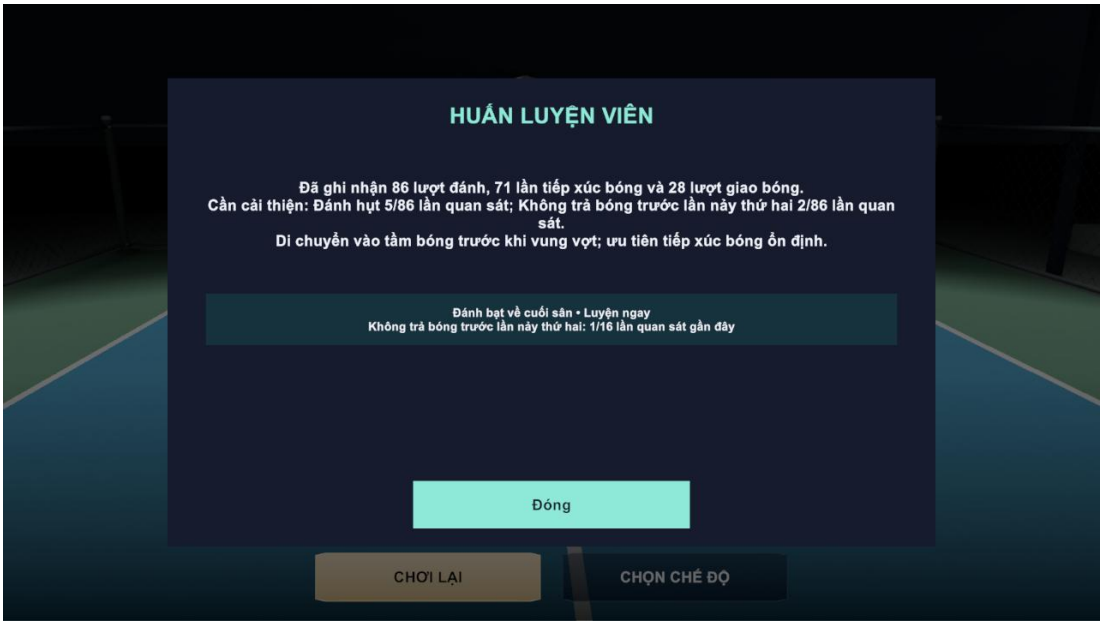


Figure 44. Coaching summary after a bot match

Function details

Business rules: BR-50, BR-51.

Normal flow:

1. The player opens the analysis; the client polls the analysis job.
2. The summary and recommendations appear when the job is completed.
3. **Luyện ngay** opens the recommended lesson briefing.

Alternative and exception flows:

- A1 – The analysis is still running: the panel shows "Đang phân tích…".
- E1 – Too few shot opportunities: no finding is reported for that error.
- E2 – AI service unavailable: the job is retried by the worker; the player can open the analysis later.

3.7 Progress, Rewards and Shop (FE-06)

FR-20 Earn Experience, Levels and Progression Rewards

Related use cases: UC-15, UC-16 (non-screen progression)

Function trigger: Settlement of a drill (FR-08) or match (FR-18); claiming mission rewards (FR-21).

Function description: Players earn XP from matches, lessons and missions. Reaching the XP threshold raises the level, which is shown in the name tag and history card. Two items are granted automatically once: the THOT paddle after 10 passed lessons and the GEMSTONE paddle at level 25.

Screen layout

This function uses the screen already shown in Figure 25; no separate screenshot is repeated here.

Function details

Business rules: BR-20, BR-32, BR-33, BR-37.

Normal flow:

1. XP is added by the settlement.
2. If XP reaches level × 200, the level increases and the remaining XP carries over.
3. Progression grants are checked and recorded in the reward log so that each is given once.

FR-21 Missions, Streak and Achievements

Related use cases: UC-37 to UC-40

Function trigger: Menu → **Nhiệm vụ**; Progress → **Nhiệm vụ ngày / tuần**.

Function description: Seven daily and ten weekly missions track matches, wins, points, valid returns and serves, bot matches and human matches. Completing all daily or weekly missions unlocks a set bonus. The daily streak counts consecutive days with at least one match; it has no reward of its own but is rewarded through the weekly mission for a 3-day streak (350 Tokenpad, 20 XP) and the achievement for a 5-day streak (500 Tokenpad, 30 XP). Ten achievements reward long-term goals such as 5, 25 and 100 wins.

Screen layout

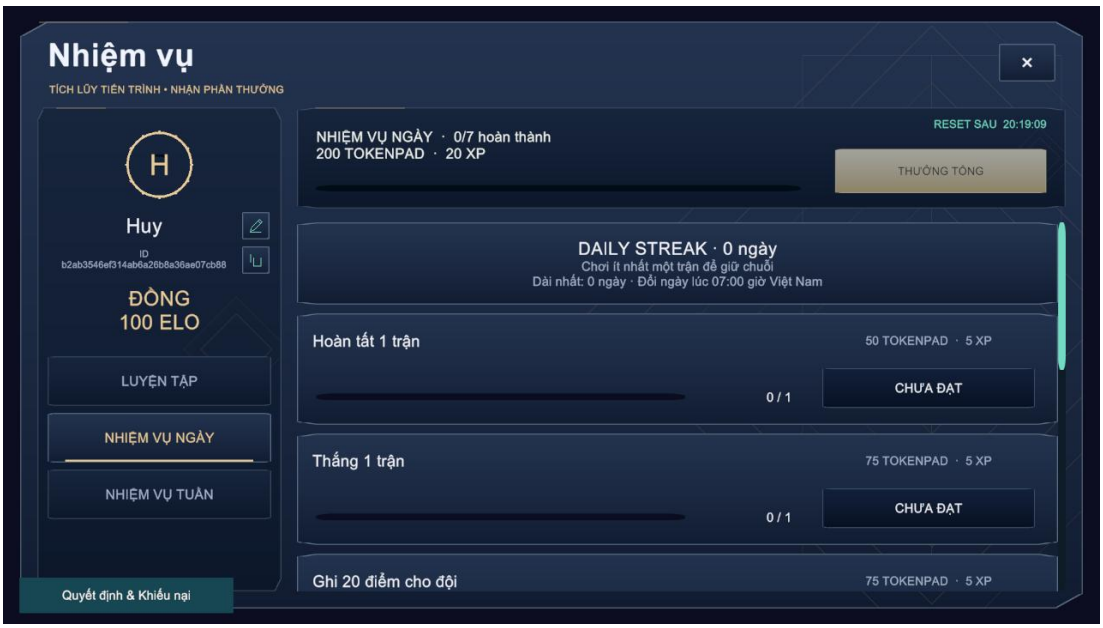


Figure 45. Missions popup with daily streak and reset countdown

Function details

Business rules: BR-34, BR-35.

Normal flow:

1. The player opens Missions; progress and the reset countdown are shown.
2. A completed mission shows **NHẬN THƯỞNG** (or **NHẬN THƯỞNG TỔNG** for the set bonus); the player claims it and the reward is added.
3. Achievement rewards are claimed in the same way (MSG49).

Alternative and exception flows:

- A1 – A reward is claimed twice: the second claim adds nothing.
- A2 – Missions renew at 07:00 Vietnam time; unclaimed progress of the old cycle is closed.

FR-22 Browse Shop and Purchase Cosmetic

Related use cases: UC-41 Browse Shop, UC-42 Purchase Cosmetic

Function trigger: Menu → **Cửa hàng**.

Function description: The shop lists cosmetic paddles with preview, description and price. An item can have a Tokenpad price, an Elite Token price or both. The purchase debits the wallet and adds the item to the collection in one transaction.

Screen layout



Figure 46. Shop with featured GEMSTONE and THOT paddles

Function details

Business rules: BR-36.

Normal flow:

1. The player selects an item and its currency and confirms the purchase.
2. The API checks ownership and balance, debits the price and grants the item.
3. The wallet and the collection are refreshed.

Alternative and exception flows:

- E1 – Not enough balance: MSG46.
- E2 – Item already owned: MSG47.
- E3 – Transaction failure: no balance is changed.

FR-23 View Collection and Equip Cosmetic

Related use cases: UC-43 View Collection, UC-44 Equip Cosmetic

Function trigger: Menu → **Bộ sưu tập**.

Function description: The collection shows all paddles with their status: in use, owned or not owned (preview only). The player equips an owned paddle, which is then used in matches and replays.

Screen layout

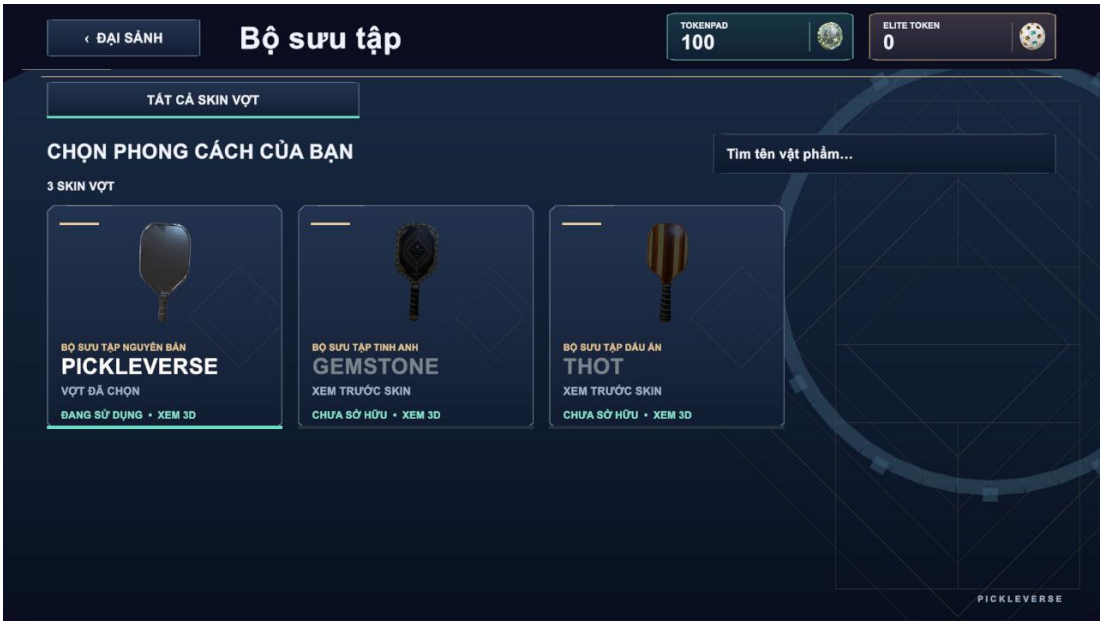


Figure 47. Collection of paddle skins

Function details

Business rules: BR-36.

Normal flow:

1. The player selects an owned paddle and **TRANG BỊ**.
2. The API marks it as the only equipped item; the menu character shows it.

Alternative and exception flows:

- E1 – The item is not owned: MSG48.

3.8 Social, History and Ranking (FE-07)

FR-24 View Match History and Watch Replay

Related use cases: UC-28 View Match History, UC-29 Watch Match Replay

Function trigger: Menu → **Lịch sử**; history entry → replay button.

Function description: The history popup lists the player's last 20 matches in Vietnam time with opponent, mode, result (win, loss or interrupted), game score and Elo change, plus the level card. A match with a stored replay can be watched in the Replay Viewer with pause, seek and speeds from 0.5× to 2×. Only participants can open a replay.

Screen layout



Figure 48. Match history with interrupted ranked matches

Function details

Business rules: BR-29, BR-40.

Normal flow:

1. The player opens History; the list loads from the API.
2. The player selects a replay; the manifest and chunks are downloaded and played.

Alternative and exception flows:

- A1 – No replay stored: the entry shows "Chưa có replay".
- E1 – A player who did not take part requests the replay: HTTP 403.

FR-25 Manage Friends and Blocked Players

Related use cases: UC-32 to UC-36

Function trigger: Menu → friend drawer → **Bạn bè**.

Function description: The player searches other players by name or ID, sends friend requests, accepts or declines received requests, removes friends and blocks or unblocks players. Online friends can be invited to a room (FR-15).

Screen layout



Figure 49. Friends drawer

Function details

Business rules: BR-38.

Normal flow:

1. The player searches by name or ID and sends a request.
2. The receiver accepts or declines it under the requests list.
3. The player can remove a friend or block a player from the list.

Alternative and exception flows:

- E1 – No match for the search: MSG51.
- E2 – The other player has already sent a request: MSG50.
- E3 – A request to oneself or to a blocked player is rejected.

FR-26 View Leaderboard

Related use cases: UC-31 View Leaderboard

Function trigger: Menu → friend drawer → **Bảng xếp hạng**.

Function description: The leaderboard lists players by Elo in descending order with rank tier, and highlights the player's own position, which can be located with **Vị trí của tôi**.

Screen layout



Figure 50. Elo leaderboard

Function details

Business rules: BR-27, BR-39.

Normal flow:

1. The player opens the leaderboard; the cached list is loaded.
2. The player searches by name or jumps to the own position.

3.9 Reports, Restrictions and Appeals (FE-08)

FR-27 Report a Player

Related use cases: UC-30 Report Player

Function trigger: Match History → **Tố cáo** on a finished match.

Function description: A player reports an opponent from one of their own matches with a reason. The report is stored as PENDING for administrators (FR-30).

Screen layout

This function uses the screen already shown in Figure 48; no separate screenshot is repeated here.

Function details

Table 18. FR-27 data fields

Field	Type	Required	Validation and constraints
Reported player	Choice	Yes	A participant of the match other than the reporter (MSG52)
Reason	Text (≤254)	Yes	Not empty

Business rules: BR-41, BR-44.

Normal flow:

1. The player selects **Tố cáo**, the opponent and enters the reason.
2. The API stores the report; the button changes to the reported state.

Alternative and exception flows:

- E1 – The player has already reported this player in this match: the duplicate is rejected.
- E2 – No eligible player: MSG52.

FR-28 View Sanction Decision and Submit Appeal

Related use cases: UC-45 View Sanction Decision, UC-46 Submit Appeal

Function trigger: Sign-in of a restricted account (automatic); Menu → **Quyết định & Khiếu nại**.

Function description: A restricted player sees each sanction with its type (suspended or banned), reason and expiry, and can submit one appeal per sanction. The appeal status (PENDING, CHECKING, ACCEPTED, DECLINED) and the administrator's decision reason are shown when available.

Screen layout

Table 19. FR-28 screen layout

Area	Elements and behaviour
Decision list	Restriction type, reason, start and expiry ("Không thời hạn" for an indefinite ban)
Appeal form	Statement field (10–4000 characters) and Send button; hidden when an appeal already exists
Appeal status	Status badge and decision reason

Function details

Table 20. FR-28 data fields

Field	Type	Required	Validation and constraints
Statement	Text	Yes	10–4000 characters (MSG53)

Business rules: BR-42, BR-43, BR-44.

Normal flow:

1. The restricted player opens the popup and reads the decision.
2. The player writes the statement and sends it; the status becomes PENDING.

Alternative and exception flows:

- E1 – Statement shorter than 10 characters: MSG53.
- E2 – No sanction to appeal: MSG54.
- E3 – An appeal already exists for the sanction: the form is not shown.

3.10 Administration Web Application (FE-09)

FR-29 Admin Sign-in and User Management

Related use cases: UC-47 to UC-53

Function trigger: Browser → admin web URL → Login; sidebar → **Quản lý người dùng** (/users).

Function description: An administrator signs in with an Admin account; other roles are rejected. On the Users page the administrator searches and filters accounts, assigns the Player or Admin role, verifies an e-mail manually when delivery failed, applies or revokes a sanction with a reason, and resets selected progress groups of a player after previewing the impact. Every change is written to the audit log.

Screen layout

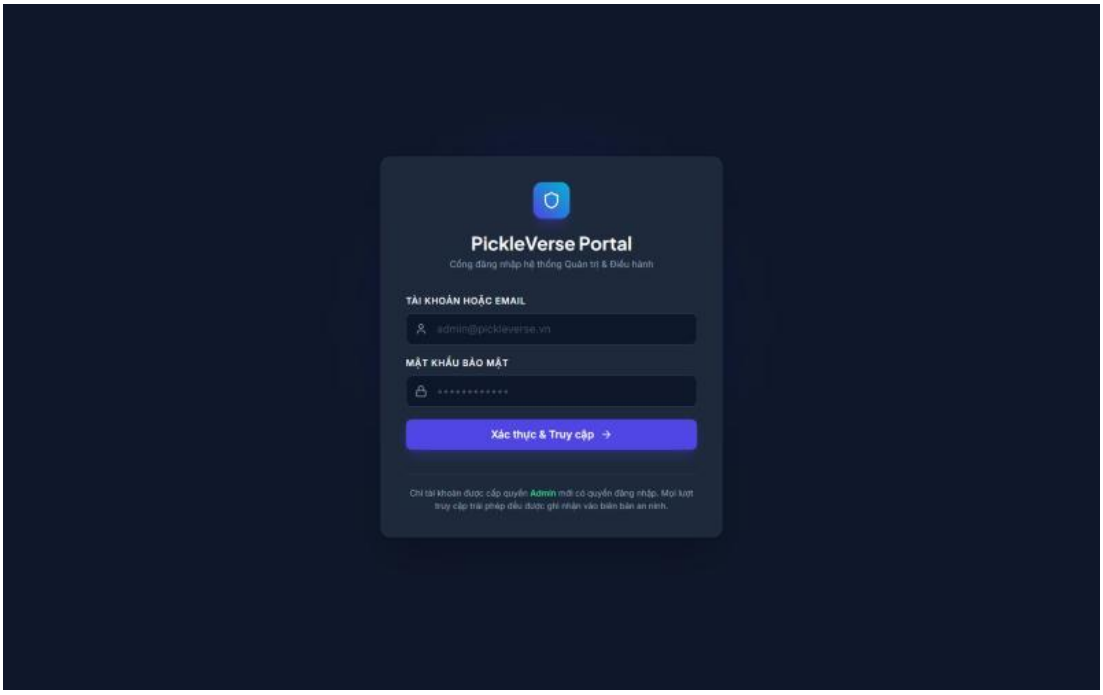


Figure 51. Admin web login page

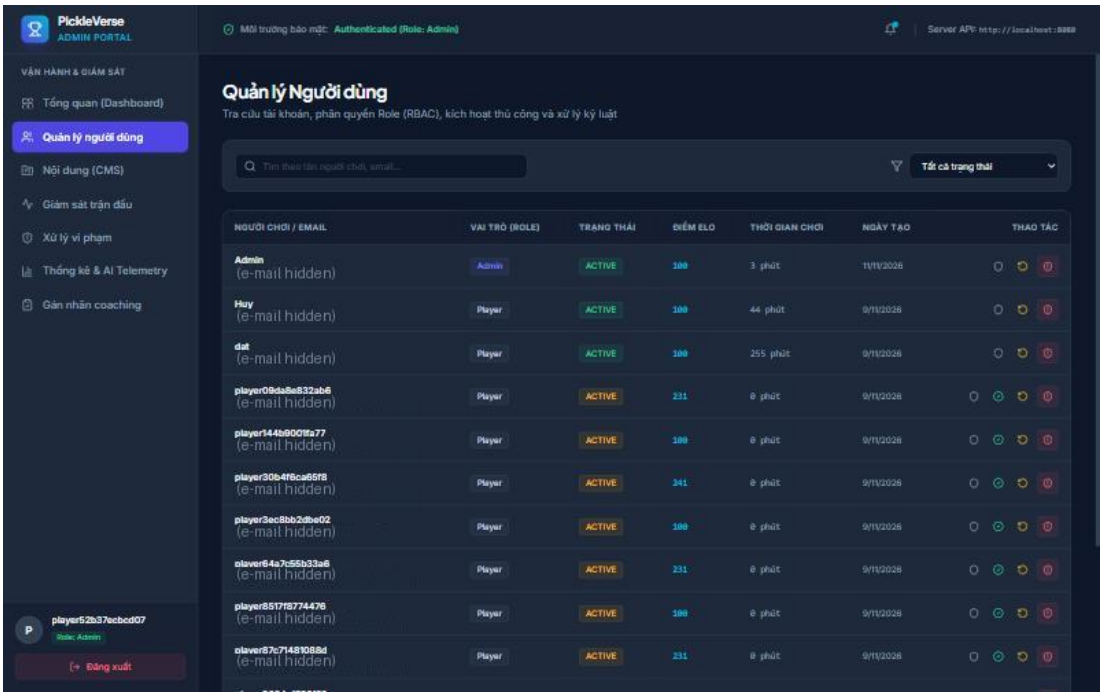


Figure 52. Users page of the admin web application

Function details

Table 21. FR-29 data fields

Field	Type	Required	Validation and constraints
Search text	Text	No	Matches name, e-mail or ID
Restriction	Choice	Yes (sanction)	SUSPENDED or BANNED
Expiry	Date-time (UTC)	Suspend: yes	Must be in the future (MSG56)
Reason	Text	Yes	3–2000 characters (MSG55)

Field	Type	Required	Validation and constraints
Reset groups	Multi-choice	Yes (reset)	At least one of LEARNING, RANKING, MISSIONS, ECONOMY

Business rules: BR-42, BR-45, BR-46.

Normal flow:

1. The administrator signs in and opens Users.
2. The administrator searches, selects an account and an action.
3. The administrator confirms; the API applies the change, writes the audit record and the table refreshes.

Alternative and exception flows:

- E1 – Non-admin account at login or on any admin API: HTTP 403.
- E2 – Sanction on the own account or a system account: MSG57.
- E3 – Reset while the player is searching or in a match: MSG58.
- E4 – Invalid reason or expiry: MSG55, MSG56.

FR-30 Review Player Reports and Appeals

Related use cases: UC-54 Review Player Report, UC-55 Review Appeal, UC-51 Apply Sanction

Function trigger: Sidebar → **Xử lý vi phạm** (/moderation).

Function description: The administrator opens a pending report, marks it CHECKING, examines the match evidence and replay, and accepts or declines it with a reason; accepting may apply a sanction to the reported player. Appeals are handled in the same way; accepting an appeal revokes only the sanction it concerns.

Screen layout

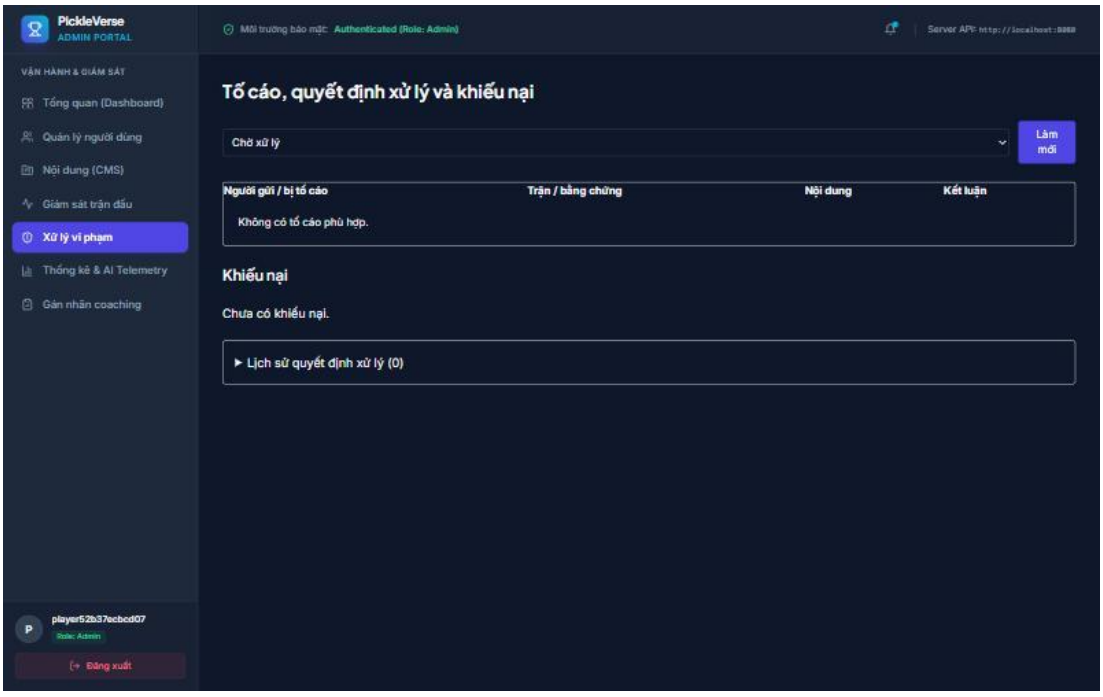


Figure 53. Moderation page

Function details

Table 22. FR-30 data fields

Field	Type	Required	Validation and constraints
Decision	Choice	Yes	ACCEPTED or DECLINED

Field	Type	Required	Validation and constraints
Decision reason	Text	Yes	3–2000 characters (MSG55)

Business rules: BR-42, BR-43, BR-44, BR-45.

Normal flow:

1. The administrator selects a report and reviews evidence.
2. The administrator enters the decision and reason and saves.
3. The report status changes; a sanction, if chosen, is applied and the player is restricted.

Alternative and exception flows:

- E1 – The report or appeal was changed by another administrator: the latest version is reloaded.
- E2 – A non-admin token: MSG63 (HTTP 403).

FR-31 Manage Training Content, Items and Missions

Related use cases: UC-56 to UC-61

Function trigger: Sidebar → **Nội dung (CMS)** (/cms).

Function description: The administrator creates and edits lessons as drafts, publishes a saved draft so that new attempts use it, and unpublishes a lesson to hide it from players; attempts already in progress keep their version. The administrator also maintains technique reference values, adds shop items and creates missions.

Screen layout

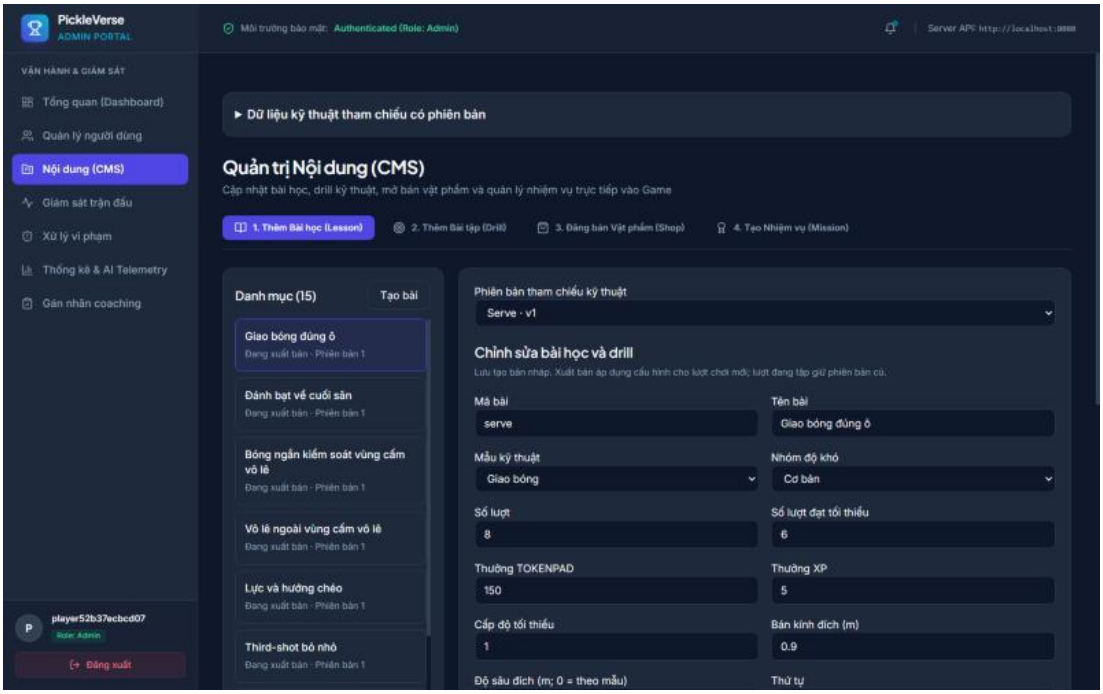


Figure 54. Content management page

Function details

Table 23. FR-31 data fields

Field	Type	Required	Validation and constraints
Lesson code	Text	Yes	Unique (MSG59)
Target	Integer	Yes	1 ≤ target ≤ number of shots
Prerequisite	Lesson	No	Must not create an unlock cycle
Price	Integer	Yes (item)	≥ 0, with a currency

Business rules: BR-19, BR-45, BR-47.

Normal flow:

1. The administrator edits a lesson and selects **Lưu bản nháp**.
2. The administrator selects **Xuất bản bản đã lưu**; the lesson becomes PUBLISHED.
3. **Gỡ xuất bản** sets it to UNPUBLISHED.

Alternative and exception flows:

- E1 – Duplicate lesson code: MSG59.
- E2 – The content changed since it was loaded: MSG60.
- E3 – The prerequisite creates a cycle: the save is rejected.

FR-32 Monitor Matches and View Analytics

Related use cases: UC-62 to UC-66

Function trigger: Sidebar → **Tổng quan (Dashboard)**, **Giám sát trận đấu** (/matches), **Thống kê & AI Telemetry** (/analytics).

Function description: The dashboard summarises today's activity and pending work. The Matches page shows live matches with the score and connection state reported by the dedicated server and lets the administrator stop a live match with a reason. The Analytics page reports daily active players, retention cohorts, popular lessons with pass rate (passes divided by started attempts), the technique-error heat map and matchmaking health for a chosen date range.

Screen layout

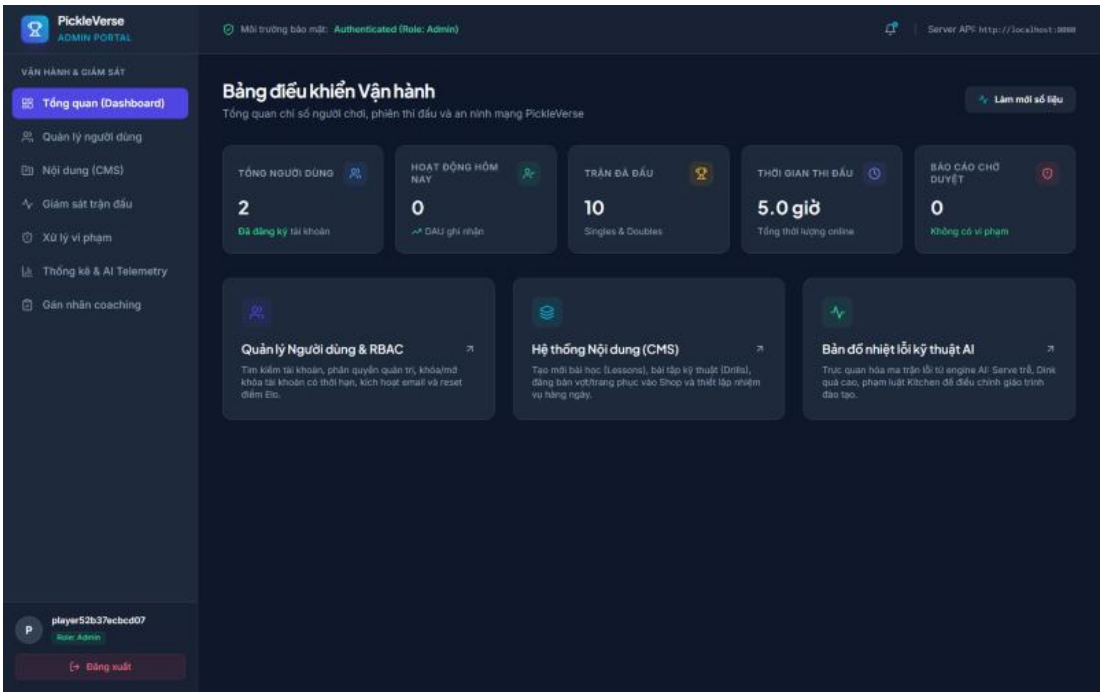


Figure 55. Admin dashboard

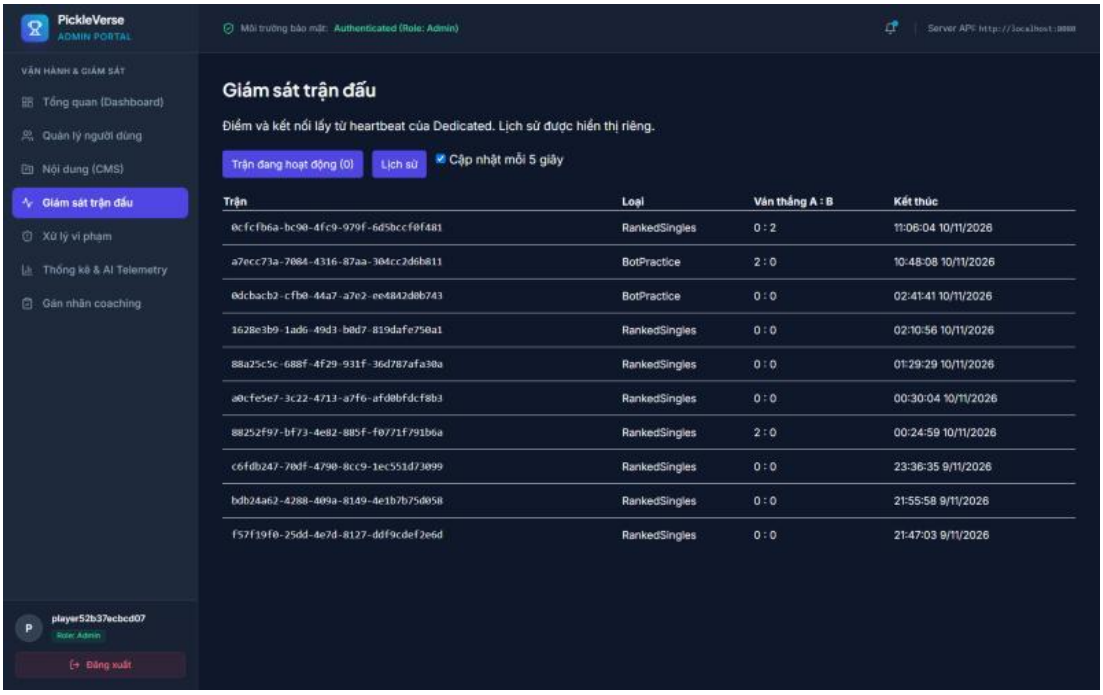


Figure 56. Live matches page

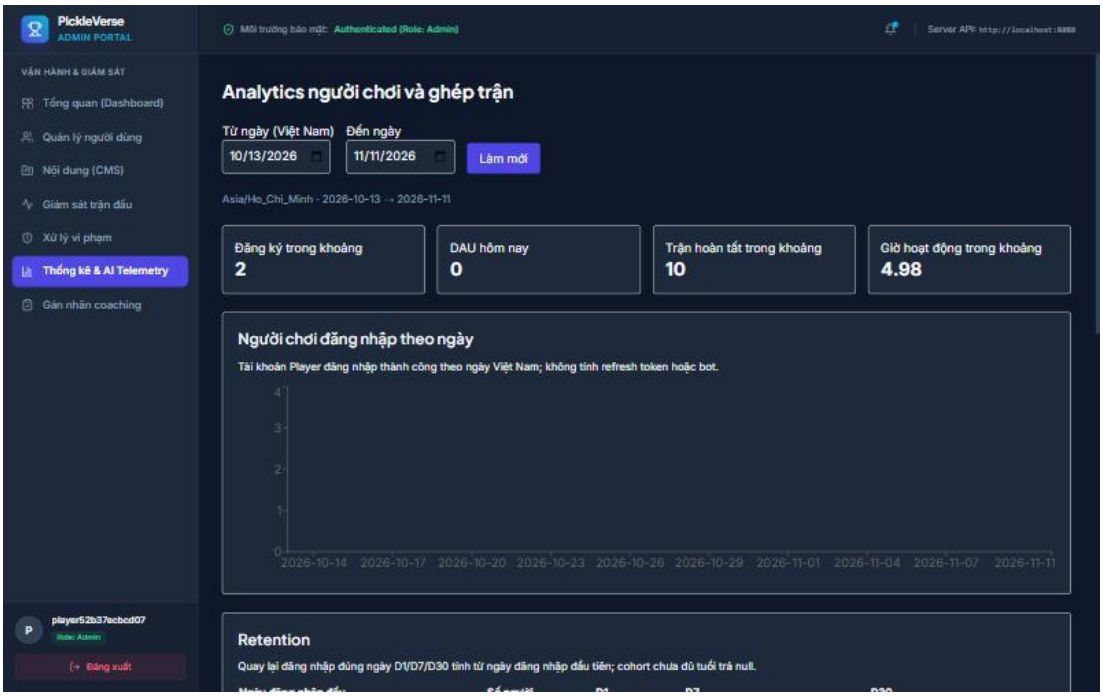


Figure 57. Analytics page

Function details

Table 24. FR-32 data fields

Field	Type	Required	Validation and constraints
Stop reason	Text	Yes	3–2000 characters
From / To date	Date	Yes	From ≤ To; at most 366 days (MSG62)

Business rules: BR-45, BR-48, BR-49.

Normal flow:

1. The administrator opens Matches; the list refreshes every 5 seconds.
2. To stop a match the administrator enters a reason and confirms MSG61; the server ends the match as interrupted.
3. The administrator opens Analytics, sets the dates and reads the reports.

Alternative and exception flows:

- E1 – The match already finished: the stop is ignored and the final result is kept.
- E2 – Invalid date range: MSG62.

3.11 Expert Review (FE-10)

FR-33 Review Coaching Sessions and Export Dataset

Related use cases: UC-67 Review Coaching Session, UC-68 Export Labelled Dataset

Function trigger: Sidebar → **Gán nhãn coaching** (/coaching).

Function description: An administrator acting as a pickleball expert inspects the shot-level evidence of a real training session, assigns the player level and confirms or rejects each error, and assigns the data group. All sessions of one player must use the same data group. The labelled sessions are exported to calibrate the rule thresholds and to evaluate the tier model against NFR-06.

Screen layout

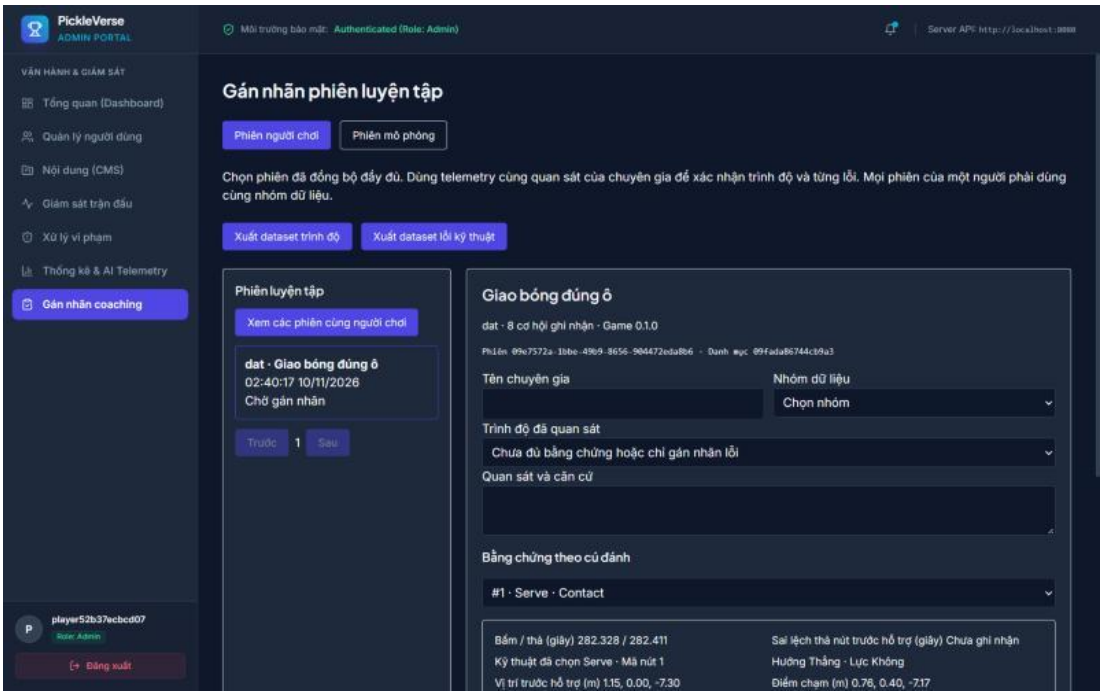


Figure 58. Expert review page

Function details

Table 25. FR-33 data fields

Field	Type	Required	Validation and constraints
Level	Choice	Yes	Beginner, intermediate or advanced
Error labels	Per error code	At least one	Present or absent with evidence (MSG64)
Data group	Choice	Yes	Training, calibration or holdout; consistent for a player

Business rules: BR-45, BR-50.

Normal flow:

1. The administrator selects a session and reviews the evidence.

2. The administrator sets the labels and saves the review.
3. The administrator exports the labelled dataset.

Alternative and exception flows:

- E1 – Missing confirmation, evidence or labels: MSG64.
- E2 – The session already has a newer review: the evidence must be reloaded.
- E3 – The player's sessions are in different data groups: HTTP 409.

4. Non-Functional Requirements

4.1 External Interfaces

4.1.1 User Interfaces

- The game interface is in Vietnamese with the project's Vietnamese fonts; it scales for 16:9 (1920×1080, 1280×720) and 20:9 phone screens; every screen can be operated with mouse and keyboard on PC and with touch on Android; Esc closes overlays and opens Settings, and Enter confirms.
- In-match information (score, serving side, timers, fault banners) is placed at the screen edges so that it does not cover the court.
- The administration web application is a responsive single-page application for desktop browsers with a fixed sidebar, tables with paging and confirmation dialogs for every destructive action.

4.1.2 Hardware Interfaces

- PC: keyboard and mouse; game controller input is not supported in release 1.0.
- Android: multi-touch screen for the virtual joystick and shot buttons.
- Server: Linux host with Docker; the dedicated game-server container is limited to 2 CPUs and 2 GB RAM by default.

4.1.3 Software Interfaces

Table 26. Software interfaces

Interface	Description
Web API (REST/JSON)	129 endpoints under /api (Report 4, §3.9), JSON bodies wrapped in the ApiResponse<T> envelope {success, message, data}; error responses carry {success, message, code, errors}; JWT bearer authentication; Swagger/OpenAPI documentation
AI coaching service	Internal HTTP API POST /v1/analyze-session protected by the X-AI-Key header; called only by the Web API worker
Dedicated server ↔ Web API	HTTP calls for lease, heartbeat, telemetry, replay and result, authenticated with a service key; match tickets signed with HMAC-SHA256
Photon Fusion 2.1.3	Real-time session service (protocol version 18) used by clients and the dedicated server
PostgreSQL 16 / Redis 7.4	Persistent storage through Npgsql Entity Framework Core; Redis for matchmaking queue, leases, presence and caches
S3-compatible storage	Replay manifests and chunks through the S3 API

4.1.4 Communication Interfaces

- Client and admin web to Web API: HTTP/1.1 with JSON over TLS in production (port 443); the staging environment serves the API and the admin web over HTTPS on port 443 since 13/11/2026 (HTTP on port 8080 until then).
- Game traffic: UDP through Photon Cloud and to the dedicated server on UDP port 3669.
- E-mail: SMTP with STARTTLS on port 587 to Gmail.
- All timestamps are exchanged in UTC (ISO 8601) and displayed in Vietnam time (UTC+7).

4.2 Quality Attributes

Each quality attribute below is measurable and is verified by the test case named in the last column (Report 5).

4.2.1 Usability

Table 27. Usability requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-01	A new player can learn the game without outside help.	At least 8 of 10 first-time testers complete the 17-step tutorial in 5 minutes or less and then pass the first lesson (serve) within three attempts, without help from the team.	Moderated usability test with 10 students who have not played PickleVerse (Report 5, TC-N01)

4.2.2 Reliability

Table 28. Reliability requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-07	Online services are available when players want to play.	The API health endpoint /health/ready is available ≥ 99.5% of the time measured every 5 minutes by an external probe over the 14 days before the final defense (at most 100 minutes of downtime).	Uptime probe report (TC-N07)
NFR-08	No result or reward is lost or duplicated.	Every match result, training receipt and purchase is applied exactly once under retries; after a container failure the service is restored from Docker Hub images within 1 hour (MTTR) with at most 24 hours of data loss from the daily database backup. No open Critical defect is allowed at release (Critical = loss of player data or inability to sign in or play).	Retry tests on settlement and sync APIs; restore drill (TC-N08)
NFR-11	Training is possible without a connection.	After one online sign-in, all published lessons can be practised offline, and 100% of offline attempts are synchronised exactly once (no duplicate reward) when the device reconnects.	Offline–online test with 20 queued attempts (TC-N11)

4.2.3 Performance

Table 29. Performance requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-02	Online play is responsive for players in Vietnam.	Round-trip time between the client and the staging game server is at most 100 ms at the 95th percentile over 10 online matches on home broadband in Ho Chi Minh City.	Fusion RTT statistics logged by the client (TC-N02)
NFR-03	Matches are smooth on mid-range devices.	Average frame rate ≥ 60 FPS (1% low ≥ 45 FPS) during a bot match on a reference PC (Intel Core i5 10th generation, GTX 1650, 16 GB RAM, 1920×1080, default quality); average ≥ 30 FPS on a reference Android phone (Snapdragon 7-series, 6 GB RAM).	Unity Profiler capture of a full bot match on each device (TC-N03)
NFR-04	The Web API supports the target player population.	With 2,000 concurrent virtual users performing a read-mostly mix (profile, missions, leaderboard, history, catalogue) the API keeps the 95th-percentile response time ≤ 500 ms and the error rate < 1% for 10 minutes. The release capacity of online matches equals the number of dedicated server slots (one in the staging deployment, LI-07 in Report 1).	k6 load test against staging (TC-N04)
NFR-05	Coaching feedback arrives quickly.	The AI coaching service returns the analysis of one drill or match in ≤ 1 second at the 95th percentile once the telemetry has been received; the coaching popup shows the result within 5 seconds of the end of the drill when online.	Timing of 50 analysis requests from the worker log (TC-N05)

4.2.4 Accuracy of AI Coaching

Table 30. Accuracy of AI Coaching requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-06	Coaching findings are trustworthy.	Each rule-based finding reaches precision ≥ 0.80 and recall ≥ 0.70 on the expert-labelled holdout sessions. The tier model is shown to players only when it is approved: real expert data, at least 10 holdout players per tier, macro-F1 ≥ 0.70 and at least 0.05 above the majority baseline, prediction confidence ≥ 0.65 and at least 20 shot opportunities.	Calibration and training reports of the AI service (TC-N06)

4.2.5 Security

Table 31. Security requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-09	Accounts and match results are protected.	Passwords are stored with BCrypt; access tokens are signed JWTs valid for 2 hours and refresh tokens are stored as SHA-256 hashes; OTPs and dedicated-server tickets are protected with HMAC; authentication endpoints are rate-limited; every admin endpoint requires the Admin role; secrets are supplied by environment variables and never committed; production traffic uses HTTPS (TLS 1.2+); match outcomes are decided only by the dedicated server.	Security checklist, API negative tests and secret scan (TC-N09)

4.2.6 Compatibility

Table 32. Compatibility requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-10	The game runs on the target platforms.	The client runs on Windows 10/11 64-bit and on Android 8.0 (API 26) or later on ARM64, at 16:9 and 20:9 aspect ratios with keyboard or touch input; the admin web supports the latest Chrome and Edge at 1366×768 or larger.	Installation and smoke test on each platform (TC-N10)

4.2.7 Maintainability and Privacy

Table 33. Maintainability and Privacy requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-12	The system can be changed safely and protects personal data.	The backend keeps the layered dependency rule (API → Services → Repositories → Entities) with no reverse reference; core rule classes are covered by automated unit tests that run before release; other players can see only the display name, player ID, rank and Elo, never the e-mail address, which is visible only to administrators.	Architecture check of project references, test run report and API response review (TC-N12)

5. Requirement Appendix

5.1 Business Rules

Table 34. Business rules

ID	Group	Rule Definition
BR-01	Account	An e-mail address can be used by only one account.
BR-02	Account	A password has 6 to 128 characters and is stored only as a BCrypt hash.
BR-03	Account	A verification or reset code has six digits, is valid for 10 minutes, can be re-sent only after 60 seconds and allows at most 5 wrong attempts.
BR-04	Account	Only an account with a verified e-mail can sign in. System bot accounts can never sign in. A suspended or banned account can sign in only with the appeal scope.
BR-05	Account	An access token is valid for 2 hours and a refresh token for 14 days; a refresh token is replaced each time it is used. Resetting the password or signing out revokes the refresh tokens concerned.

ID	Group	Rule Definition
BR-06	Account	A display name has 3 to 18 characters, contains no control characters and is unique among players.
BR-07	Account	The experience level (Beginner or Experienced) is chosen once, at the first sign-in.
BR-08	Account	The display name and character cannot be changed while the player is in a room, searching for a match or playing a match.
BR-09	Gameplay	The court is 13.41 m × 6.10 m; the non-volley zone extends 2.13 m from the net on each side. A ball touching a line is in.
BR-10	Gameplay	A singles match is the best of three games. A game is won by the first side to reach 11 points with a lead of at least 2. Only the serving side scores a point; a fault by the server gives the serve to the opponent (side-out).
BR-11	Gameplay	The serve is hit underhand, with both feet behind the baseline, into the diagonal service court and beyond the non-volley zone, within 10 seconds after the server is ready. The server serves from the right court when the server's score is even and from the left when it is odd.
BR-12	Gameplay	Two-bounce rule: the return of serve and the third shot must bounce once before they are hit.
BR-13	Gameplay	A volley may not be hit while standing in the non-volley zone, and the player's momentum after a volley may not carry them into the zone.
BR-14	Gameplay	The rally ends with a fault of one of 17 types: serve foot, serve motion, serve timeout, short serve, serve to the wrong court, missed serve, out, ball into own court, two-bounce rule, double bounce, double hit, kitchen volley, kitchen momentum, net touch, body hit, net plane and permanent object.
BR-15	Gameplay	In the third game the players change ends when a side first reaches 6 points. Time-outs are not offered in release 1.0.
BR-16	Gameplay	A paddle contact is accepted only when the key is released within ±65 ms of the ideal contact time. A contact is Perfect when the contact error is ≤ 0.145 m and the timing error is within ±30 ms.
BR-17	Gameplay	Shot keys: J drive (and serve), K dink, L volley, J+K drop, J+L smash, K+L reset; U adds power and I aims cross-court. The shot is decided once, when the first base key is released.
BR-18	Training	Drill score: 0 stars below the target; 1 star when the target is reached; 2 stars at ≥ 80% hits; 3 stars at ≥ 90% hits with a mean timing error ≤ 0.16 s. The best result of each lesson is kept.
BR-19	Training	A lesson is unlocked when its prerequisite lesson has been passed; a locked lesson cannot be started. Prerequisites cannot form a cycle.
BR-20	Training	The first pass of a lesson grants 150, 225, 300 or 450 Tokenpad and 5, 7, 10 or 15 XP according to the lesson group; later passes and failed attempts grant no currency and cost nothing.
BR-21	Training	Offline training is allowed only for a profile cached by a previous online sign-in. Each attempt carries a unique request ID and is applied at most once (receipt APPLIED, DUPLICATE or REJECTED).
BR-22	Match	Match kinds are Bot Practice, Ranked Bot, Ranked Singles, Matched Doubles and Custom. Only Ranked Singles and Ranked Bot change Elo.
BR-23	Match	A new player starts at 100 Elo, which is also the minimum. After a ranked match Elo changes by round(32 × (S − E)), where S is 1 for a win and 0 for a loss and E = 1 / (1 + 10^((R_opponent − R_own)/400)); the result is never below 100.
BR-24	Match	Ranked pairing uses a window of 120 Elo. During the first 30 seconds the window extends only upward (only downward for a player with 5 or more losses); after 30 seconds it extends 120 in both directions.
BR-25	Match	When a match is found, every player must accept within 10 seconds; a decline or a timeout cancels the offer.
BR-26	Match	Ranked singles only: if no human opponent is found within 180 seconds, a ranked bot is assigned with Elo = own + 120 (own − 100 for a player with 5 or more losses). Quick doubles has no bot fallback; its search ends when the search lease expires.
BR-27	Match	Rank tiers by Elo: Đồng < 500, Bạc < 1000, Vàng < 2000, Bạch Kim < 3500, Kim Cương < 5000, Cao Thủ < 7000, Huyền Thoại ≥ 7000.
BR-28	Match	A disconnected client tries to reconnect for 30 seconds, and the dedicated server keeps the player's slot for 30 seconds while the match is paused; a player who does not return is recorded as having forfeited and the match ends as interrupted. Clients send a heartbeat to the API every 8 seconds; in shared-mode matches (local development only) the API interrupts the match when a player has sent no heartbeat for 35 seconds. If the dedicated server stops renewing its lease, the lease monitor ends the match as interrupted.
BR-29	Match	A match result is saved exactly once. An interrupted or stopped match gives no Elo and no reward.
BR-30	Match	A singles room holds 2 players and a doubles room up to 4. The host starts a custom match and may start a doubles room before all four slots are filled; custom matches are unrated.
BR-31	Match	Bot levels Easy, Medium and Hard use fixed reaction, accuracy and error settings. The adaptive bot selects its tier from the player's Elo: below 500 beginner, below 2000 intermediate, otherwise advanced.

ID	Group	Rule Definition
BR-32	Economy	Match rewards: bot match win 100 / loss 50 Tokenpad; online match win 150 / loss 75 Tokenpad; every finished match gives 10 XP for a win and 5 XP for a loss.
BR-33	Economy	Reaching level n+1 requires n × 200 XP at level n; surplus XP carries over.
BR-34	Economy	There are 7 daily and 10 weekly missions. Daily missions renew at 07:00 Vietnam time (UTC+7) and weekly missions at the start of each mission week. Completing all daily missions gives 200 Tokenpad + 20 XP and all weekly missions 1,000 Tokenpad + 50 XP. The daily streak grows by one for each day with at least one finished match; streaks are rewarded by the weekly mission "3-day streak" (350 Tokenpad + 20 XP) and the achievement "5-day streak" (500 Tokenpad + 30 XP).
BR-35	Economy	Each mission, mission set and achievement reward can be claimed only once.
BR-36	Economy	A purchase requires enough balance in the chosen currency and is refused if the item is already owned; the debit and the grant are one transaction. Exactly one paddle is equipped at a time. Elite Token has no earning source in release 1.0.
BR-37	Economy	The THOT paddle is granted once after 10 passed lessons and the GEMSTONE paddle once at level 25, unless already owned.
BR-38	Social	A player cannot send a friend request to themself, send a duplicate request, or send one to a player who has already sent them a request. Blocking removes the friendship and prevents friend requests and room invitations in both directions.
BR-39	Social	The leaderboard is ordered by Elo descending and is cached for 15 seconds.
BR-40	Social	A match history and its replay can be read only by the players who took part and by administrators.
BR-41	Moderation	A player can report another participant of their own finished match once per match; self-reporting is not allowed; the reason has at most 254 characters.
BR-42	Moderation	A sanction is SUSPENDED (expiry required and in the future) or BANNED (expiry optional). The reason has 3 to 2,000 characters. An administrator cannot sanction their own account or a system account. An expired suspension ends automatically.
BR-43	Moderation	A restricted player can submit one appeal per sanction, with a statement of 10 to 4,000 characters. Accepting an appeal revokes only the sanction it concerns.
BR-44	Moderation	Reports and appeals move from PENDING to CHECKING to ACCEPTED or DECLINED; a decision requires a reason.
BR-45	Administration	Only the Admin role can use the administration application and API (otherwise HTTP 403). Every administrative change is recorded in the audit log with actor, action, target and time.
BR-46	Administration	Progress reset works on the groups LEARNING, RANKING, MISSIONS and ECONOMY after an impact preview, and is refused while the player is searching or in a match. It keeps the account, friends, match history, decisions and reward log; rewards already granted are not granted again.
BR-47	Administration	A lesson moves DRAFT → PUBLISHED → UNPUBLISHED (and can be published again). Publishing applies to new attempts only. A save is refused when the content changed after it was loaded.
BR-48	Administration	Stopping a live match requires a reason; when the stop is recorded before the final result the match is interrupted without Elo or reward.
BR-49	Administration	Analytics dates are calendar days in Vietnam time; the start date is not after the end date and a range covers at most 366 days.
BR-50	AI coaching	A coaching finding is reported only when the session contains at least 5 opportunities for that error; at most 3 drills are recommended; the tier estimate is shown only under the approval conditions of NFR-06.
BR-51	AI coaching	Telemetry and replays of dedicated-server matches are submitted only by the dedicated server; a client can submit telemetry only for its own drills and bot matches.

5.2 Common Requirements

- Language: all player-facing and administrator-facing text is Vietnamese; documents and source code identifiers are English.
- Date and time: stored in UTC and shown in Vietnam time (dd/MM/yyyy HH:mm); daily cycles (missions, streaks, analytics days) start at 07:00 or 00:00 Vietnam time as stated in the related rule.
- API responses: every JSON response uses the ApiResponse envelope; HTTP 400 validation, 401 not signed in, 403 forbidden, 404 not found, 409 conflict, 429 too many requests, 503 dependency unavailable; HTTP 500 responses never expose internal details.
- Idempotency: commands that change progress, wallet or match state carry a client request ID; repeating a request returns the first result.
- Concurrency: administrative edits use a version check and ask the user to reload when the data changed.

- Lists: account lists are paged (20 rows per page by default in the API); the game history shows the last 20 matches.
- Logging: the API, workers and dedicated server write structured logs without passwords, tokens or codes.

5.3 Application Messages List

Table 35. Application messages

#	Message code	Message Type	Context	Content
1	MSG01	In red, under the text box	E-mail field is empty	Vui lòng nhập email.
2	MSG02	In red, under the text box	E-mail format is invalid	Vui lòng nhập địa chỉ email hợp lệ.
3	MSG03	In red, under the text box	Password field is empty	Vui lòng nhập mật khẩu.
4	MSG04	In red, under the text box	Password shorter than 6 characters	Mật khẩu cần ít nhất 6 ký tự.
5	MSG05	In red, under the text box	Confirmation differs from the password	Mật khẩu xác nhận chưa khớp.
6	MSG06	In red, under the form	E-mail already registered	Email đã được đăng ký. Nếu chưa xác thực, hãy dùng chức năng gửi lại mã.
7	MSG07	In line, information	Registration succeeded	Đã đăng ký. Hãy nhập mã được gửi tới email để kích hoạt tài khoản.
8	MSG08	In red, under the text box	Code is not six digits	Mã xác thực gồm 6 chữ số.
9	MSG09	In line, information	A new code was sent	Mã xác thực đã được gửi. Hãy kiểm tra cả thư rác.
10	MSG10	In red, under the form	Code re-sent within 60 seconds (HTTP 429)	Vui lòng chờ ít nhất 60 giây trước khi gửi lại mã.
11	MSG11	In red, under the form	Sign-in with an unverified e-mail	Email chưa được xác thực. Hãy nhập OTP hoặc yêu cầu gửi lại mã.
12	MSG12	In red, under the form	Wrong e-mail or password	Email hoặc mật khẩu không chính xác.
13	MSG13	Toast message	Sign-in succeeded	Đăng nhập thành công.
14	MSG14	Popup	Sign-in of a suspended or banned account	Tài khoản đang bị hạn chế. Bạn có thể xem quyết định và gửi khiếu nại.
15	MSG15	In line, information	Password reset requested (same text whether or not the e-mail exists)	Nếu email thuộc tài khoản đã xác thực, mã đặt lại mật khẩu sẽ được gửi. Hãy kiểm tra cả thư rác.
16	MSG16	In red, under the form	Reset code wrong or expired	Mã đặt lại mật khẩu không hợp lệ hoặc đã hết hạn.
17	MSG17	In red, under the form	Password rule rejected by the API	Mật khẩu cần từ 6 đến 128 ký tự và xác nhận phải khớp.
18	MSG18	Popup	Refresh token expired or revoked	Phiên đăng nhập không hợp lệ hoặc đã hết hạn.
19	MSG19	In red, under the text box	Display name length invalid	Tên người chơi cần từ 3 đến 18 ký tự.
20	MSG20	In red, under the text box	Display name already used	Tên người chơi đã được sử dụng. Hãy chọn tên khác.
21	MSG21	Popup question	First sign-in experience choice	Bạn đã quen với pickleball chưa? – Người mới sẽ được hướng dẫn luật và điều khiển. Nếu đã thành thạo, bạn có thể vào sảnh ngay.
22	MSG22	In red, under the form	Verification e-mail could not be sent	Chưa gửi được email xác thực. Vui lòng thử gửi lại mã sau ít nhất 60 giây.
23	MSG23	In line	Starting a locked lesson	Bài tập chưa mở khóa.
24	MSG24	Popup title	Drill target not reached	CHƯA ĐẠT MỤC TIÊU
25	MSG25	In line	Drill failed	Không mất Tokenpad. Luyện lại để đạt {target} cú.
26	MSG26	HUD feedback	Key released late	Bạn thả nút hơi muộn. Hãy chuẩn bị vợt và thả nút sớm hơn.
27	MSG27	HUD feedback	Key released early	Bạn thả nút hơi sớm. Hãy đợi bóng tiến gần hơn.

#	Message code	Message Type	Context	Content
28	MSG28	Banner	Offline training mode	Đang luyện tập ngoại tuyến · {n} mục chờ đồng bộ
29	MSG29	HUD banner	Serve timer expired	Bạn đã hết 10 giây giao bóng. Hãy chuẩn bị và giao bóng trước khi hết thời gian.
30	MSG30	HUD banner	Rally ended by a fault (examples)	Lỗi • Bóng ra ngoài / Lỗi • Nảy hai lần / Lỗi • Vô lê trong vùng cấm vô lê / Lỗi • Bóng không qua lưới / Lỗi chân giao bóng
31	MSG31	Result title	Player wins the match	Bạn thắng trận!
32	MSG32	In line	Searching for an opponent	Đang tìm đối thủ trực tuyến…
33	MSG33	Popup	Match found, ready check	Đã tìm thấy trận • Xác nhận để tham gia.
34	MSG34	In line	Search expired	Hết thời gian tìm trận. Hãy thử lại.
35	MSG35	In line	Player declined the match	Bạn đã hủy tham gia. Bấm Tìm trận để tìm lại.
36	MSG36	In line	Dedicated server slot is busy	Máy chủ đang bận • Đang chờ lượt thi đấu…
37	MSG37	In line	Room is full	Phòng đã đủ người.
38	MSG38	In line	Room closed or player removed	Phòng đã đóng hoặc bạn đã được mời ra.
39	MSG39	In line	Joining a room failed (for example wrong code)	Không tham gia được phòng. Vui lòng thử lại.
40	MSG40	In line	Invited player cannot receive invitations	Người chơi chưa sẵn sàng nhận lời mời.
41	MSG41	In line	Player already searching or in an unfinished match	Người chơi đang tìm trận hoặc có trận chưa hoàn tất.
42	MSG42	Overlay	Connection to the match server lost	Mất kết nối máy chủ thi đấu.
43	MSG43	Popup	Reconnection failed	Không thể kết nối lại trận. Kết quả sẽ được cập nhật khi máy chủ xử lý xong.
44	MSG44	Toast message	Player left the match	Bạn đã rời trận.
45	MSG45	Popup	Photon Cloud not reachable	Không kết nối được Photon. Vui lòng kiểm tra mạng và thử lại.
46	MSG46	Popup	Balance too low for a purchase	Số dư tiền tệ không đủ để mua vật phẩm này.
47	MSG47	Popup	Item already owned	Bạn đã sở hữu vật phẩm này rồi.
48	MSG48	Popup	Equipping an item that is not owned	Bạn chưa sở hữu vật phẩm này.
49	MSG49	Toast message	Achievement reward claimed	Mở khóa thành công danh hiệu và phần thưởng: {reward}!
50	MSG50	In line	The other player already sent a request	Người chơi đã gửi lời mời cho bạn. Hãy chấp nhận hoặc từ chối.
51	MSG51	In line	No search result	Không tìm thấy tên hoặc ID phù hợp.
52	MSG52	In line	No player can be reported	Không có người chơi phù hợp để tố cáo
53	MSG53	In red, under the text box	Appeal statement too short	Nội dung khiếu nại cần ít nhất 10 ký tự.
54	MSG54	In line	No decision to appeal	Không có quyết định xử lý để khiếu nại.
55	MSG55	In red, under the text box	Reason length invalid (admin)	Lý do phải có từ 3 đến {max} ký tự.
56	MSG56	In red, under the text box	Suspension without a future expiry	Tạm ngưng cần thời hạn trong tương lai; khóa có thể không có thời hạn.
57	MSG57	Toast message	Admin tries to sanction own account	Không thể tự khóa tài khoản quản trị.
58	MSG58	In line, in the dialog	Reset while player searching or in a match	Người chơi đang tìm trận hoặc có trận chưa hoàn tất. Chưa thể reset.
59	MSG59	In red, under the text box	Lesson code already exists	Mã bài học đã tồn tại.
60	MSG60	Toast message	Content changed since it was loaded	Nội dung đã thay đổi. Hãy tải lại trước khi lưu.
61	MSG61	Warning in the dialog	Confirming a match stop	Trận sẽ bị gián đoạn và không tính Elo/thưởng nếu lệnh dừng được ghi trước kết quả cuối.
62	MSG62	In red, under the date fields	Invalid analytics range	Khoảng ngày thống kê không hợp lệ. / Biểu đồ hỗ trợ tối đa 366 ngày mỗi lần xem.
63	MSG63	Toast message	Non-admin calls a moderation function (HTTP 403)	Bạn không có quyền xử lý vi phạm.
64	MSG64	In red, under the form	Expert review incomplete	Hãy xác nhận phiên người thật, người đánh giá, bằng chứng và ít nhất một nhãn.
65	MSG65	In red, under the form	System account sign-in attempt	Tài khoản hệ thống không được đăng nhập.

5.4 Definitions and Acronyms

Table 36. Definitions and acronyms

Term	Definition
Dink	A soft shot that drops into the opponent's non-volley zone
Drill	A practice exercise with a fixed number of shots and a target
Elo	A rating system in which the rating change depends on the expected result against the opponent
Kitchen / NVZ	Non-volley zone: the area within 2.13 m of the net where volleys are not allowed
Rally	The exchange of shots from the serve until a fault
Side-out	Loss of serve to the opponent after a fault by the serving side
Telemetry	Shot-level data (timing, contact, landing, position) recorded by the game
Tokenpad / Elite Token	In-game currencies; Tokenpad is earned by playing
Dedicated server	A headless Unity server that runs an online match with authority over the result
Photon Fusion	Third-party networking library and cloud used for real-time sessions
API, JWT, OTP, HMAC	Application programming interface; JSON Web Token; one-time password; hash-based message authentication code
BR, FR, NFR, UC, MSG	Business rule; functional requirement; non-functional requirement; use case; message

5.5 Appendix A – Traceability to the Review 2 Commitment

The table below links the 18 features committed in the Review 2 form BM04 (07/11/2026) to the use cases and functional requirements of this specification, with the workflow and the member accountable for each feature at the reviews. Report 5 links each requirement to its test cases.

Table 37. Traceability from Review 2 features to requirements and owners

Review 2 feature	Name	Functional requirements	Use cases	Workflow	Owner (BM04)
F01	Account and first steps	FR-01, FR-02, FR-03, FR-04	UC-01 to UC-08	W01	Trần Nguyễn Hiếu Thuận
F02	Rules and controls	FR-05, FR-06, FR-11	UC-09, UC-10	W01	Trần Nguyễn Hiếu Thuận
F03	Guided drills	FR-07, FR-08	UC-11, UC-12	W01	Nguyễn Gia Huy
F04	Levels and items	FR-20, FR-23	UC-43, UC-44	W01	Nguyễn Văn Thành Đạt
F05	Adaptive bot	FR-12, FR-13	UC-16	W02	Nguyễn Gia Huy
F06	Coaching and tier	FR-19	UC-17, UC-18	W02	Nguyễn Văn Thành Đạt
F07	Quick match	FR-14	UC-19, UC-20	W03	Nguyễn Văn Thành Đạt
F08	Private rooms	FR-15	UC-21 to UC-24	W03	Nguyễn Văn Thành Đạt
F09	Dedicated play	FR-16, FR-17	UC-25 to UC-27	W03	Nguyễn Văn Thành Đạt
F10	Result and Elo	FR-18	UC-25	W03	Nguyễn Văn Thành Đạt
F11	Progress and replay	FR-10, FR-21, FR-24	UC-15, UC-28, UC-29, UC-39	W02	Nguyễn Gia Huy
F12	Friends and ranks	FR-25, FR-26	UC-31 to UC-36	W03	Trần Nguyễn Hiếu Thuận
F13	Daily and streak rewards	FR-21	UC-37 to UC-40	W01	Nguyễn Gia Huy
F14	Offline practice	FR-09	UC-13, UC-14	W01	Trần Nguyễn Hiếu Thuận
F15	Admin and accounts	FR-29	UC-47 to UC-53	W04	Huỳnh Trung Kiên
F16	Lesson publishing	FR-31	UC-56 to UC-61	W04	Huỳnh Trung Kiên
F17	Match care and reports	FR-27, FR-28, FR-30, FR-32	UC-30, UC-45, UC-46, UC-54, UC-55, UC-62 to UC-64	W04	Huỳnh Trung Kiên
F18	Admin charts	FR-32	UC-65, UC-66	W04	Huỳnh Trung Kiên

Review 2 feature	Name	Functional requirements	Use cases	Workflow	Owner (BM04)
–	Shop (not in the Review 2 list)	FR-22	UC-41, UC-42	–	–
–	Expert review (addition to the proposal)	FR-33	UC-67, UC-68	–	–

5.6 References

[1] USA Pickleball, "Official Rulebook 2024." https://usapickleball.org/what-is-pickleball/official-rules/ (accessed 11 Nov. 2026).

[2] K. E. Wiegers and J. Beatty, Software Requirements, 3rd ed. Microsoft Press, 2013.

[3] Photon Engine, "Fusion 2 Documentation." https://doc.photonengine.com/fusion/current (accessed 11 Nov. 2026).

[4] GFA26SE111, "Report 1 – Project Introduction" and "Report 4 – Software Design Document," PickleVerse, 2026.